FLOOD VULNERABILITY OF THE MUNICIPALITY OF LOS BAÑOS, LAGUNA USING HYBRID MACHINE LEARNING MODELS

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In partial fulfillment of the requirement for the degree
Bachelor of Science in Computer Science
Specialized in Intelligent System

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The Researchers

DEDICATION

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J.C.D.L

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ABSTRACT

Flooding is among the most persistent natural hazards in the Philippines, severely affecting low-lying and coastal municipalities including Los Baños, Laguna. This study developed a high-resolution flood vulnerability assessment framework by integrating the physically based Soil and Water Assessment Tool (SWAT) with optimized machine learning algorithms. A ten-year dataset (2014–2023) encompassing rainfall, temperature, elevation, slope, soil type, and land use was processed through SWAT to simulate hydrological responses across 145 Hydrological Response Units (HRUs). These outputs served as inputs for hybrid models, including Random Forest, XGBoost, Multi-Layer Perceptron (MLP), Long Short-Term Memory (LSTM), and Support Vector Classifier (SVC), each optimized using Particle Swarm Optimization (PSO), Bayesian Optimization (BO), or Genetic Algorithm (GA). Performance was evaluated through K-Fold cross-validation using precision, recall, F1-score, and AUC-ROC metrics. Results revealed that the Bayesian-optimized MLP model (MLP–BO + K-Fold) achieved the highest accuracy (Macro-F1 = 0.9884; Accuracy = 0.98), effectively delineating Bayog, Tadlac, Mayondon, Malinta, and Bambang as high-risk barangays. The integration of SWAT-derived hydrological parameters with advanced optimization-based learning significantly enhanced the spatial precision and reliability of flood vulnerability mapping. The proposed hybrid framework provides a scalable, data-driven tool for disaster preparedness, land-use planning, and climate adaptation, supporting the broader goals of building resilient and sustainable communities in the Philippines.

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DEFINITION OF TERMS

Algorithm	A set of rules or step-by-step instructions designed to perform a specific task or solve a problem. In this study, it refers to computational models used to assess flood vulnerability.
Calibration	The process of adjusting model parameters so that simulation outputs closely match observed data, enhancing model accuracy and reliability.
Cross-Validation (K-Fold Cross-Validation)	A model validation technique where the dataset is divided into *K* subsets, and the model is trained and tested repeatedly to ensure robustness and prevent overfitting.
Digital Elevation Model (DEM)	A raster representation of the Earth's surface elevation used to derive topographic parameters such as slope and drainage patterns critical for flood modeling.
Flood Score Index	A composite index developed in this study to quantify flood susceptibility by integrating normalized hydrological and physiographic variables.
Flood Vulnerability	A measure of the susceptibility of a community or area to flood hazards, considering physical, environmental, and social factors.
Geographic Information System (GIS)	A system designed to capture, store, analyze, and visualize spatial and geographic data, often used for flood mapping and vulnerability assessment.
Hybrid Model	A modeling approach that combines physical hydrological models with machine learning algorithms to improve flood prediction and vulnerability assessment.
Hydrological Model	A mathematical representation of the water cycle used to simulate rainfall–runoff processes, streamflow, and watershed responses.
Hyperparameter Tuning	The process of selecting and optimizing predefined parameters of a machine learning algorithm to maximize performance.

HRU (Hydrological Response Unit)	A spatial unit in a watershed defined by unique land use, soil type, and slope, used in models like SWAT to simulate water, sediment, and nutrient flow.
Machine Learning	A branch of artificial intelligence that enables systems to learn patterns from data and make predictions without being explicitly programmed.
Soil and Water Assessment Tool (SWAT)	A physically-based hydrological model that simulates the impact of land use, soil, and climate on water flow and flood potential.
Surface Runoff	The portion of rainfall that flows over the land surface toward streams and rivers, contributing directly to flooding.
Sustainable Development Goals (SDGs)	This study supports three United Nations Sustainable Development Goals: **SDG 11** (sustainable and resilient cities), **SDG 13** (climate action), and **SDG 15** (life on land). These goals emphasize building climate-resilient communities and protecting natural ecosystems.
Water Yield	The total amount of water generated by a watershed, including surface runoff, lateral flow, and baseflow.

CHAPTER 1

INTRODUCTION

Background of the Study

Flooding remains a persistent and escalating hazard in the Philippines, particularly within the Los Baños Coastal Area of Laguna, which is part of the Pasig–Laguna River Basin. This region is highly susceptible to both seasonal and extreme flood events due to its tropical monsoon climate, intensifying rainfall patterns linked to climate change, and a range of anthropogenic stressors, including deforestation, rapid urban expansion, inadequate drainage infrastructure, and poor solid waste management. These interconnected factors have significantly increased the exposure and vulnerability of the coastal area and its surrounding communities.

Historical data and recent events underscore the socio-economic and environmental consequences of flooding in Los Baños. Between 2006 and 2012, the area experienced a marked increase in the frequency and severity of flood events, including the 2017 Barangay Malinta flood, which displaced over 470 households and damaged vital infrastructure (Ardales Jr. et al., 2015; Malabayabas & Baconguis, 2017). More recently, Typhoon Agaton in 2023 affected over 200,000 people across the Philippines, including communities in Laguna, highlighting the region's ongoing vulnerability to climate-induced disasters. These events not only disrupt daily life but also result in long-term losses in agriculture, public health, education, and economic productivity, disproportionately impacting low-income populations and widening existing socio-economic disparities.

Accurate flood prediction is essential for disaster risk reduction and climate adaptation. Physically based models such as the Soil and Water Assessment Tool (SWAT) are commonly used to simulate hydrologic processes. SWAT is designed for long-term simulations involving land use and climate variability, and watershed management. In this study, the SWAT model is used to preprocess historical environmental data, converting raw inputs into structured hydrologic variables such as runoff and streamflow to support data-driven flood forecasting, particularly in data-limited areas like Los Baños.

In response to these limitations, machine learning (ML) algorithms such as Long Short-Term Memory (LSTM) networks, Random Forest (RF), Extreme Gradient Boosting (XGBoost), Support Vector Regression (SVR), and Multi-Layer Perceptron (MLP) have emerged as powerful tools for hydrological modeling. These data-driven approaches excel at capturing nonlinear relationships and temporal dependencies, even with limited or irregular datasets, making them well-suited for complex, real-world flood forecasting.

Recent studies have demonstrated the potential of hybrid modeling, the integration of ML techniques with traditional hydrological models (e.g., SWAT–LSTM, SWAT–XGBoost) to improve predictive performance. These models combine physical watershed parameters with ML's pattern-recognition capabilities, resulting in enhanced streamflow simulation, better flood forecasting accuracy, and more reliable early warning systems (Shrestha & Pradhanang, 2023). However, existing studies largely focus on basin-scale or regional modeling and rarely explore hybrid SWAT–ML integration at the Hydrological Response Unit (HRU) scale for localized flood mapping in Los Baños. This absence of Hydrological Response Unit (HRU) -level hybrid

modeling creates a knowledge gap in achieving high-resolution, community-specific flood vulnerability assessments tailored to local decision-making needs.

Additionally, decision-tree-based ensemble models such as XGBoost and Random Forest have proven effective for flood susceptibility mapping, particularly when integrated with Geographic Information Systems (GIS) and remote sensing technologies. This integration enables spatially explicit, high-resolution risk assessments, allowing local government units (LGUs) to identify hazard-prone areas and prioritize mitigation efforts.

Flood vulnerability assessment (FVA) has also evolved to incorporate multidimensional indicators, including socio-economic vulnerability, land use dynamics, infrastructure resilience, and population density. Modern FVA frameworks increasingly leverage GIS-based multi-criteria evaluation (MCE), providing a holistic understanding of flood risk that accounts for both environmental exposure and human sensitivity.

Globally, countries vulnerable to monsoon-induced flooding, such as Thailand, Indonesia, Bangladesh, and Nigeria, are increasingly adopting machine-learning–integrated models to improve water resource management and flood forecasting. These innovative approaches support key international frameworks for disaster risk reduction and sustainable development, including the Sendai Framework for Disaster Risk Reduction and the United Nations Sustainable Development Goals (SDGs), particularly SDG 11 (Sustainable Cities and Communities), SDG 13 (Climate Action), and SDG 15 (Life on Land). These goals collectively emphasize the need for resilient infrastructure, climate adaptation, and the responsible management of terrestrial ecosystems.

In the Philippine context, especially in flood-prone regions like Los Baños, Laguna, local government units (LGUs) often face significant constraints, including limited data availability and financial resources, which hinder the development of effective flood risk management strategies. In response to these challenges, hybrid modeling that combines the strengths of traditional hydrological models with the adaptability and efficiency of machine learning algorithms offers a promising, context-appropriate solution.

This study aims to contribute to these broader efforts by comparing machine learning models (LSTM, Random Forest, XGBoost, Support Vector Regression, and Multi-Layer Perceptron) with the traditional hydrological model (SWAT) for flood vulnerability assessment in the Los Baños Coastal Area. By applying advanced techniques such as hyperparameter tuning, performance validation (including AUC-ROC and F1-score), and GIS-based analysis of vulnerability indicators, the research seeks to identify the most effective models for local implementation.

Specifically, the study will produce GIS-based flood vulnerability maps for both wet and dry seasons, a hybrid SWAT–ML modeling workflow for Hydrological Response Unit (HRU) -scale flood assessment, and a decision-support framework for LGUs to enhance flood preparedness, land-use zoning, and disaster response planning. Ultimately, the findings aim to enhance scientific understanding, inform evidence-based policy, and strengthen community resilience to future flood events in Los Baños and similarly at-risk areas.

Objectives

The main objectives of this research are to assess the flood vulnerability of Los Baños Coastal Areas using a hybrid machine learning model.

This research specifically aims to:

1. To identify flood vulnerability patterns using hydrological models and hybrid machine learning models.
2. To create a flood vulnerability map of the Los Baños Coastal Areas that visually represents the identified levels of flood risk.
3. To test and evaluate the performance of the hybrid machine learning model in assessing flood vulnerability.
4. To perform a paired t-test analysis to determine if there is a significant difference between the performance of optimized and non-optimized hybrid models.

Significance of the Study

This study aims to identify the most suitable hybrid model for flood vulnerability assessment in the Los Baños Coastal Areas by integrating machine learning techniques with traditional hydrological modeling approaches. By evaluating the performance of these combined methods, the research seeks to improve the accuracy, efficiency, and applicability of flood vulnerability systems. Traditional hydrological models, while effective, often face challenges due to their extensive data requirements and complex calibration, limiting their utility in rapidly urbanizing or data-scarce regions. Hybrid approaches offer a promising solution, especially for flood-prone areas like Los Baños, where timely and reliable flood forecasting is essential for disaster preparedness. This study aligns with several global priorities outlined

by the United Nations Sustainable Development Goals (SDGs), particularly SDG 11 (Sustainable Cities and Communities), SDG 13 (Climate Action), and SDG 15 (Life on Land), which emphasize sustainable urban development, climate resilience, and the responsible management of natural resources. By developing hybrid models, this research contributes to evidence-based environmental governance, providing decision-makers with actionable insights to strengthen disaster preparedness and resilience (United Nations, 2015). The findings of this study are expected to benefit several key stakeholders:

Local Government Units (LGUs): Outputs from hybrid models can guide zoning regulations, flood contingency planning, and land-use policies. These measures help minimize risks to vulnerable communities and protect critical infrastructure. Insights are crucial for enhancing local disaster resilience. (Malabayabas and Baconguis, 2017).

Environmental and Infrastructure Planners: Improved flood forecasting from hybrid models supports the design of resilient infrastructure and safeguards environmentally sensitive areas. This is particularly important for regions susceptible to erosion, surface runoff, and seasonal flooding. (Ardales Jr. et al. 2015) highlighted how such forecasting tools contribute to sustainable infrastructure planning.

Scientists and Future Researchers: This study provides a framework for advancing research on hybrid hydrological and machine-learning models. It encourages innovations that strengthen climate resilience and supports the creation of data-driven strategies for sustainable Coastal Area management. (Shrestha and Pradhanang, 2023) demonstrated the effectiveness of hybrid approaches in improving flood prediction accuracy.

Scope and Limitation of the Study

This study aims to assess hybrid models combining machine learning algorithms and traditional hydrological models for flood vulnerability assessment in the Los Baños Coastal Areas. It utilizes historical flood data, satellite imagery, and geospatial analysis to improve flood vulnerability accuracy. The research explores integrating machine learning algorithms, such as Long Short-Term Memory (LSTM), Random Forest (RF), and XGBoost, with the Soil and Water Assessment Tool (SWAT) to develop predictive tools that support disaster preparedness and mitigation. However, the study is limited by data availability and quality, which may affect model calibration. Its findings are specific to the Los Baños Coastal Areas, including Bayog, Mayondon, Malinta, Baybayin, Bambang, and Tadlac, and might not apply to other regions without adjustments. Computational constraints may limit the number of model combinations tested, and socio-economic factors related to flood impacts are not addressed. Long-term field validation is also not feasible due to time and resource limitation

CHAPTER 2

THEORETICAL FRAMEWORK OF THE STUDY

Review of Related Literature

Flood Vulnerability and Coastal Area Characteristics

Flooding in the Los Baños coastal area has become a pressing concern due to its tropical monsoon climate, increasing rainfall intensity, and rapid urbanization. The municipality, located within the Pasig–Laguna River Basin and draining into Laguna de Bay, includes both low-lying floodplains and elevated regions near the foothills of Mount Makiling. Heavy rainfall from June to November, coupled with deforestation, waste accumulation, and unregulated development, has intensified flood hazards. Historical analyses revealed an increasing trend in annual rainfall between 2006 and 2012 (Ardales Jr. et al., 2015), while local reports documented that flood in Barangay Malinta affected over 470 households with water levels reaching five feet (Malabayabas & Baconguis, 2017). These observations underscore the need for science-based flood vulnerability assessments and predictive modeling to support sustainable disaster risk reduction in the area.

Advances in flood modeling have increasingly emphasized hybrid approaches that couple hydrological process models with data-driven learning algorithms to enhance prediction accuracy and reliability. Moon et al. (2023) demonstrated this potential by integrating a rainfall-runoff model with a Long Short-Term Memory (LSTM) deep learning network, achieving a correlation coefficient greater than 0.9 and a Nash–Sutcliffe efficiency above 0.8. Their results highlighted how hybridization can overcome data scarcity and capture complex rainfall–runoff dynamics. Similarly, Mei et al. (2024) successfully

coupled the Soil and Water Assessment Tool (SWAT) with an LSTM model to improve daily streamflow simulation, emphasizing that integrating physically based and deep learning models can better capture nonlinear hydrological responses. In the same direction, Pinichka et al. (2025) applied a coupled SWAT–XGBoost model using Optuna hyperparameter optimization, demonstrating substantial gains in prediction accuracy and model robustness. Building on these findings, the present study adopts a comparable hybrid strategy by integrating the physically based SWAT model with machine learning classifiers to assess flood vulnerability in the Los Baños coastal area.

Deep learning architectures such as Long Short-Term Memory (LSTM) networks have proven highly effective for hydrological time-series prediction due to their capability to capture long-term temporal dependencies (Cho et al., 2022). By leveraging sequential meteorological data, including rainfall, temperature, and water level, LSTM models have achieved superior performance in flood forecasting across diverse basins (Kratzert et al., 2018; Shrestha & Pradhanang, 2023). Furquim et al. (2016) further demonstrated that Multi-Layer Perceptron (MLP) neural networks trained on sensor-derived rainfall and flow data can significantly improve short-term flood predictions. As discussed by DataCamp (2025), MLPs function as fully connected feedforward networks capable of modeling complex nonlinear relationships, making them particularly suitable for environmental prediction tasks. In the context of Los Baños, where rainfall and terrain variability complicate runoff estimation, these neural architectures provide a robust foundation for predictive modeling. The present research extended these concepts by comparing LSTM, MLP, Random Forest, XGBoost, and Support Vector Classifier models, ultimately finding that

a Bayesian-optimized MLP, tuned using the framework described by Shahriari et al. (2016), yielded the most accurate results with a cross-validation F1-score of 0.9905.

Flood vulnerability assessment (FVA) methods have evolved from simple statistical analyses to complex spatially explicit models integrating hydrological, physical, and socio-environmental variables. Nasiri et al. (2016) categorized FVA approaches into curve-based, disaster-loss, computer-modeling, and indicator-based frameworks, the last of which effectively combines physical, social, and environmental dimensions. Geographic Information Systems (GIS) and remote sensing have further advanced FVA by enabling spatial mapping of flood susceptibility (Feloni et al., 2019; Franci et al., 2016). In this study, GIS-based processing was used to integrate SWAT-derived hydrological outputs with physiographic layers, including slope, elevation, land use, and soil type, to form a hybrid dataset for spatial flood risk classification.

Integrating hybrid vulnerability models with GIS-based spatial analysis can significantly improve local flood management strategies. In the Los Baños coastal area, this integration enabled the creation of high-resolution flood-susceptibility maps that capture both hydrological responses and land-surface characteristics. The application of optimized machine learning models, particularly the Bayesian-optimized MLP, provided a robust framework for identifying high-risk zones and enhancing early-warning systems. Such data-driven approaches are crucial for addressing the combined pressures of climate change, rapid urbanization, and the region's complex coastal dynamics.

While previous studies, such as SWAT–LSTM (Mei et al., 2024) and SWAT–XGBoost (Pinichka et al., 2025), have successfully demonstrated the potential of hybrid modeling for improving streamflow simulation and flood prediction, most applications remain at the basin or regional scale. These models often lack the spatial granularity needed to assess flood vulnerability at the community or barangay level. The present study addresses this research gap by implementing a hybrid SWAT–machine learning framework at the Hydrological Response Unit (HRU) scale, enabling localized flood vulnerability mapping for the Municipality of Los Baños. This fine-scale approach allows for the identification of specific at-risk zones, providing actionable insights for land-use planning and disaster risk reduction.

Furthermore, this research enhances the reliability of the hybrid model through systematic hyperparameter tuning using Particle Swarm Optimization (PSO), Bayesian Optimization (BO), and a Genetic Algorithm (GA). As emphasized by Shahriari et al. (2016), such optimization frameworks minimize model uncertainty, enhance generalization, and ensure reproducibility across datasets. By integrating hydrological realism from SWAT with machine learning enhanced by optimization, the proposed methodology advances flood prediction accuracy and interpretability. Overall, this study contributes a scalable, data-driven, and reproducible framework that can be extended to other flood-prone municipalities across the Philippines.

Key Parameters in Flood Vulnerability Assessment

Flood vulnerability assessment is essential in identifying and understanding the factors that influence the occurrence and severity of flooding, especially in areas exposed to both natural and human-induced changes. In

this study, parameter selection was guided by their hydrological, environmental, and geospatial significance in shaping flood behavior in the Los Baños coastal barangays. The main parameters considered include rainfall, surface runoff, water yield, flow out, land use and land cover (LULC), elevation, slope, and soil type. These variables are widely recognized in the literature as key indicators of flood susceptibility (Yoon & Kim, 2022; Li & Zhang, 2023; Khan & Ali, 2024; Mei et al., 2024).

Rainfall serves as the primary driver of flooding, particularly during the monsoon and typhoon seasons. High-intensity and long-duration rainfall can cause rapid surface water accumulation and river overflow, especially when combined with poor drainage systems or saturated soils. In hydrological modeling, rainfall data act as input forcing variables that influence streamflow, runoff, and overall water yield. These patterns have been observed in the Los Baños area, where seasonal rainfall contributes directly to flood events (Ardales Jr. et al., 2015; Shrestha & Pradhanang, 2023).

Surface runoff, water yield, and flow out are the core hydrological variables produced by the Soil and Water Assessment Tool (SWAT). Surface runoff measures the direct overland flow responsible for flood peaks. At the same time, water yield represents the combined contribution of surface runoff, lateral flow, and baseflow, essentially, the total water available for discharge. Flow out indicates the volume of streamflow leaving each hydrological response unit (HRU) and reflects how water is routed across the landscape. Together, these parameters describe how rainfall is transformed into runoff and streamflow within the watershed (Mei et al., 2024; Pinichka et al., 2025).

Elevation and slope are critical topographic factors that affect runoff speed, water retention, and drainage direction. Low-lying areas near Laguna de Bay are highly susceptible to flooding due to limited drainage capacity. At the same time, steeper slopes near Mount Makiling promote rapid runoff, which can intensify downstream flood peaks. Khan and Ali (2024) emphasized that even slight variations in topography can significantly influence flood magnitude and distribution, highlighting the importance of accurate elevation and slope data in vulnerability mapping.

Land use and land cover (LULC) play a major role in determining how rainfall interacts with the surface. Urbanized areas with extensive impervious surfaces, such as concrete roads and buildings, restrict infiltration and increase surface runoff. In contrast, vegetated or agricultural areas help absorb rainfall, reduce runoff, and delay the timing of flood peaks (Li & Zhang, 2023). In Los Baños, the ongoing expansion of built-up areas along the coast has altered natural drainage patterns, contributing to more frequent flooding in barangays such as Bayog and Malinta.

Soil type also influences how water infiltrates the ground. Clay-rich soils have low permeability, leading to higher surface runoff, whereas sandy or loamy soils allow greater infiltration, reducing flood risk. Including soil characteristics in modeling improves the estimation of saturated zones and enhances flood prediction accuracy (Neitsch et al., 2005; Singh et al., 2012).

While socio-economic factors such as population density and the number of affected households are often used in broader vulnerability frameworks (Yoon & Kim, 2022; Malabayabas & Baconguis, 2017), the present study focuses primarily on the physical and hydrological parameters that can

be quantified and spatially analyzed. This approach ensures objectivity and consistency in the modeling process. By integrating rainfall, hydrological, and physiographic variables within a hybrid SWAT–machine learning framework, the study provides a detailed and data-driven assessment of flood-prone areas in the Los Baños coastal barangays, supporting local efforts in flood mitigation and sustainable land-use planning.

Machine Learning and Hybrid Models for Flood Vulnerability and Risk Assessment

Recent advances in hydrological modeling have been greatly influenced by the integration of machine learning (ML) techniques, which have significantly enhanced flood vulnerability analysis, streamflow simulation, and overall risk assessment. Among these approaches, Long Short-Term Memory (LSTM) networks have attracted substantial attention for their ability to capture temporal dependencies and model the nonlinear dynamics inherent in hydrological systems. Kratzert et al. (2018) demonstrated that LSTM outperformed traditional conceptual models such as the Sacramento Soil Moisture Accounting (SAC-SMA) model in rainfall–runoff simulations, particularly when long-term sequential patterns were essential. Lees et al. (2022) further showed that LSTM networks can implicitly learn key hydrological processes, including soil moisture storage and snow dynamics, thereby improving model interpretability.

Hybrid modeling frameworks that combine physically based and data-driven methods have emerged as powerful tools for enhancing predictive accuracy. By leveraging the strengths of both approaches, hybrid models can capture complex hydrologic behavior while maintaining physical interpretability.

Taparia et al. (2024) demonstrated this principle in a hybrid demand-forecasting model that reduced prediction errors by integrating multiple ML algorithms. In hydrology, similar hybridization has shown promising results: Shrestha and Pradhanang (2023) compared SWAT and LSTM for rainfall–runoff modeling and found that LSTM achieved comparable performance (NSE = 0.60) to SWAT (NSE = 0.63) despite limited data, while Mei et al. (2024) successfully coupled SWAT and LSTM to improve daily streamflow prediction, achieving an NSE of 0.885. These studies underscore the potential of hybrid SWAT–LSTM frameworks to enhance prediction accuracy in both gauged and ungauged basins.

Parallel to these developments, decision-tree-based algorithms, particularly Extreme Gradient Boosting (XGBoost) and Random Forest (RF), have proven valuable for flood susceptibility mapping and hydrological risk assessment. XGBoost has gained prominence for its computational efficiency and predictive strength; Ma et al. (2021) achieved high accuracy in flash-flood risk mapping in Yunnan Province using this technique. Manjula (2023) and Pinichka et al. (2025) further demonstrated that coupling XGBoost with SWAT and applying automated hyperparameter optimization substantially improved streamflow and nutrient-transport simulations. Likewise, RF has been widely used in urban flood modeling and watershed-health studies. Woo et al. (2019) combined SWAT outputs with RF to evaluate aquatic-ecosystem health, revealing spatial gradients in water-quality conditions, while Islam et al. (2023) applied RF as a data-driven alternative to SWAT for streamflow prediction in snowmelt-dominated basins, showing superior performance with longer time-series data.

Support Vector Classifier (SVC) and Multi-Layer Perceptron (MLP) networks have also been extensively adopted in hydrological forecasting. SVC models are effective for handling nonlinear relationships and classification-based flood susceptibility analysis, as demonstrated by Yu et al. (2006) and Yuan and Forshay (2021) in similar hydrological prediction studies. MLP networks, on the other hand, have shown exceptional capability in approximating complex nonlinear functions. Singh et al. (2012) reported that MLP outperformed SWAT in simulating sediment yield, while Furong and Hossain (2024) achieved improved runoff predictions with an optimized SWAT–MLP framework (R^2 = 0.83). These studies confirm the versatility of neural-network-based approaches for both standalone and hybrid hydrological modeling applications.

To further enhance predictive accuracy and generalization, machine learning algorithms are often coupled with hyperparameter optimization techniques tailored to their model architecture. The Random Forest (RF) and XGBoost models have been optimized using Particle Swarm Optimization (PSO) and Bayesian Optimization (BO) to fine-tune critical parameters such as learning rate, maximum depth, and feature subsampling (Fereidoon & Koch, 2018; Shahriari et al., 2016). The Multi-Layer Perceptron (MLP) network has been calibrated using Genetic Algorithm (GA) and Bayesian Optimization (BO) to determine optimal network topology, activation functions, and regularization strengths (Zhang et al., 2009). For sequence-based models like LSTMs, Bayesian Optimization has been applied to refine hidden-layer configurations and learning rates, thereby improving temporal learning efficiency and stability. Meanwhile, the Support Vector Classifier (SVC) used Particle Swarm

Optimization (PSO) to select kernel and regularization parameters. At the same time, all models were validated via K-Fold cross-validation to ensure consistency, reduce bias, and minimize overfitting.

Collectively, these studies highlight the transformative role of machine learning in advancing flood modeling and vulnerability assessment. The integration of algorithms such as LSTM, XGBoost, RF, SVC, and MLP within hybrid and ensemble frameworks has markedly improved model accuracy, interpretability, and transferability. In the context of the Los Baños coastal barangays, the adoption of a hybrid framework that integrates the physically based SWAT model with optimized ML classifiers, particularly the Bayesian-optimized MLP, refined through K-Fold cross-validation, builds on these global advancements. This approach aligns with contemporary research trends emphasizing data-driven, physically informed modeling for more reliable flood prediction and risk management in climate-sensitive regions.

Advancements in Hyperparameter Tuning for Hydrological Modeling

Hyperparameter tuning plays a vital role in improving the predictive performance of hydrological models by optimizing their internal parameters to better represent real-world conditions. Among commonly used techniques, the Genetic Algorithm (GA) has been widely applied to improve the calibration and efficiency of models such as the Soil and Water Assessment Tool (SWAT). Kaini et al. (2012) demonstrated that integrating GA into hydrological modeling improved watershed-scale decision-making by optimizing the type, size, and placement of structural Best Management Practices (BMPs), thereby effectively reducing sediment and nutrient loads while minimizing implementation costs. Similarly, Wang et al. (2018) combined GA with SWAT to optimize land-use

planning and control nonpoint-source pollution, highlighting its effectiveness in enhancing spatial decision-making and environmental sustainability.

Beyond GA-based Optimization, Bayesian approaches have also been widely adopted to refine hydrological model calibration and uncertainty analysis. Zhang et al. (2009) developed a hybrid GA–Bayesian Model Averaging (BMA) method for SWAT calibration, improving both predictive accuracy and uncertainty quantification. More recently, Moknatian and Mukundan (2023) demonstrated that using multiple objective functions during BMA calibration provides a broader range of optimal solutions, allowing the model to capture uncertainty more effectively. Their approach achieved ensemble predictions that covered 94–97% of observed streamflow data, significantly outperforming single-objective calibration methods. These findings underscore the value of probabilistic and multi-objective Optimization in improving the robustness and reliability of hydrological simulations.

Further advancements have focused on using Particle Swarm Optimization (PSO) to address model nonlinearity and parameter sensitivity in hydrological applications. Fereidoon and Koch (2018) applied a SWAT–MODSIM–PSO framework to optimize streamflow simulation and multi-crop water allocation under climate change scenarios, achieving notable improvements in both water resource efficiency and economic outcomes. Likewise, Wang et al. (2023) integrated PSO with SWAT to enhance parameter calibration and spatial data clustering, reducing simulation error by 5.7% and improving model fit by 0.118.

These developments in evolutionary and probabilistic Optimization have paved the way for more advanced, data-driven calibration methods. In recent

years, Bayesian Optimization (BO) has emerged as a powerful technique for tuning hyperparameters in machine learning and hybrid hydrological models. Unlike GA or PSO, BO uses probabilistic models to search the parameter space efficiently, reducing computation time while maintaining high accuracy (Shahriari et al., 2016). In the present study, BO was used to fine-tune the hyperparameters of machine learning algorithms, including the Multi-Layer Perceptron (MLP), and K-Fold cross-validation was used to ensure stable and reliable model performance. This integration of Bayesian and data-driven approaches reflects the evolving direction of hydrological modeling toward more automated, intelligent, and reproducible calibration frameworks.

Technological Approaches for Flood Mapping and Risk Assessment

Flood mapping has evolved rapidly with advances in Geographic Information Systems (GIS), remote sensing, and model-based simulations. These technologies now play a central role in flood risk assessment, early warning systems, and disaster management planning.

GIS-based platforms such as Quantum GIS and open-source applications like Flood Risk have become essential tools for integrating multiple spatial data layers such as topography, hydrology, soil type, and land use to delineate flood-prone zones and support disaster preparedness programs (Franci et al., 2016; Albano et al., 2017; Kaya et al., 2023). Remote sensing technologies further complement these systems by providing high-resolution imagery and near-real-time observations. Very high-resolution (VHR) satellites such as GeoEye-1 and Landsat, along with radar-based sensors, enable the detection of surface changes and the delineation of flood extents over large areas with high accuracy (Aggarwal, 2016; CTCN, 2023).

Model-based mapping approaches have also become integral to flood risk analysis. Hydrological models such as the Soil and Water Assessment Tool (SWAT) simulate rainfall–runoff processes and estimate potential inundation areas, while machine learning algorithms such as Random Forests and XGBoost enhance these models by improving classification accuracy and spatial prediction of flood susceptibility (Pinichka et al., 2025). The convergence of GIS, remote sensing, and model-driven simulations has significantly increased the efficiency, speed, and precision of flood hazard assessment, making these methods essential for data-informed planning and decision-making.

These integrated technologies have been widely adopted to guide land-use zoning, infrastructure development, and disaster risk reduction (DRR) strategies. The U.S. Federal Emergency Management Agency's (FEMA) Risk Mapping, Assessment, and Planning (Risk MAP) program is a prime example of how flood mapping supports zoning regulations, infrastructure protection, and sustainable urban planning (FEMA, 2023).

In recent years, participatory flood mapping has emerged as a valuable complement to technical approaches. By engaging communities in identifying flood-prone areas and local hazards, participatory mapping enhances the accuracy and social relevance of risk assessments, particularly in vulnerable or underserved regions (Auliagisni et al., 2022).

In this study, integrating GIS, remote sensing, and SWAT-based modeling provides the foundation for generating spatially detailed flood vulnerability maps of the Los Baños coastal barangays. This combined

approach supports both scientific analysis and local decision-making for sustainable flood management and resilience planning.

Performance Metrics for Evaluating Flood Prediction Models

Evaluating the performance of flood prediction models requires objective and consistent metrics that quantify accuracy, precision, and reliability. Among the most widely used are the area under the Receiver Operating Characteristic Curve (AUC-ROC), precision, recall, F1-score, and various spatial agreement indices. These metrics collectively determine how well a model identifies flood-prone areas and distinguishes them from non-flooded regions.

The AUC-ROC measures the model's ability to discriminate between positive and negative classes, in this case, flooded and non-flooded zones. A higher AUC value indicates stronger classification performance and a better balance between sensitivity and specificity (Fawcett, 2006). Precision measures the proportion of correctly predicted flood-prone areas among all those classified as such. At the same time, recall represents the percentage of actual flood-prone areas that the model successfully identifies. The F1-score, calculated as the harmonic mean of precision and recall, offers a balanced assessment of both false positives and false negatives, making it particularly effective for evaluating imbalanced datasets (Saito & Rehmsmeier, 2015).

Empirical studies have demonstrated the value of these performance metrics in assessing hybrid and machine-learning-based flood models. For instance, Pinichka et al. (2025) reported that coupling SWAT with XGBoost improved predictive accuracy for environmental modeling, achieving up to a 50% increase in R^2 and a 75% reduction in RMSE compared with baseline models. Although their study focused on nutrient transport, the results highlight

the broader potential of integrating process-based and data-driven models for complex hydrological systems. Similarly, Abedi et al. (2021) found that Random Forest outperformed other classifiers, such as XGBoost, Boosted Regression Trees (BRT), and Classification and Regression Trees (CART), in flash-flood susceptibility mapping, achieving an AUC of 0.956, demonstrating the robustness of ensemble-based models for flood hazard prediction.

These metrics play a critical role not only in comparing different model architectures but also in guiding the refinement and calibration of predictive systems. In this study, metrics such as precision, recall, F1-score, and AUC were employed to evaluate the performance of the optimized machine learning models, Random Forest, XGBoost, LSTM, SVC, and the Bayesian-optimized MLP using K-Fold cross-validation. This ensured a fair and reliable assessment of model performance while minimizing overfitting and enhancing generalizability. The use of standardized evaluation metrics thus strengthens the interpretability and credibility of the flood vulnerability models, supporting their application in local disaster planning, early warning systems, and risk management efforts.

Synthesis and Relevance of the Study

The synthesis and relevance of this study lie in its comprehensive review of flood vulnerability assessment in the Los Baños coastal area and in its exploration of advanced hybrid modeling techniques for flood prediction and risk evaluation. The literature highlights that increasing rainfall intensity, urbanization, and deforestation have significantly amplified flood risks in Los Baños, emphasizing the urgent need for improved disaster management and adaptive strategies.

The review also underscores the transformative potential of machine learning in hydrological modeling, particularly through hybrid frameworks that integrate the physically based SWAT model with data-driven algorithms such as LSTM, XGBoost, Random Forest, SVR, and MLP. These methods have demonstrated the ability to enhance predictive accuracy, model complex watershed behavior, and maintain reliability even in data-scarce environments.

This study builds upon these advancements by applying a hybrid modeling approach tailored to the Los Baños coastal context. By leveraging optimized machine learning techniques, it aims to produce accurate flood-susceptibility maps and support early warning systems that inform local planning and disaster preparedness. Ultimately, this integration of science and technology contributes to building more resilient communities and mitigating the growing impacts of flooding in rapidly developing and climate-sensitive regions.

Table 1. Literature Matrix

Author (Year)	Title	Method/Algorithm	Key Findings
Pinichka, C., Chotpantarat, S., Cho, K. H., and Siriwong, W. (2025).	Comparative analysis of SWAT and SWAT coupled with XGBoost model using Optuna hyperparameter optimization for nutrient simulation: A case study in the Upper Nan River basin, Thailand.	Flood Mapping and Model Performance Evaluation	The study shows that integrating hydrodynamic modeling, remote sensing, and ensemble machine learning improves flood hazard mapping accuracy and demonstrates high model performance using AUC, precision, recall, and F1-score.
Taparia, V., Mishra, P., Gupta, N., and Kumar, D. (2024, November 20).	Improved Demand Forecasting of a Retail Store Using a Hybrid Machine Learning Model.	Hybrid Models	highlighted the effectiveness of hybrid models in demand forecasting, demonstrating that integrating multiple machine learning algorithms can reduce prediction errors and enhance

			decision-making accuracy.
Furong, G., and Hossain, S. (2024)	Projection of monthly surface flows by an optimized SWAT–MLP: a case study	Multi-Layered Perceptron (MLP) And SWAT	SWAT–MLP hybrid approach with MSFO optimization for improved runoff prediction accuracy in forested watersheds.
Manjula, R. (2023, November 13.)	Enhanced streamflow prediction using SWAT's influential parameters: a comparative analysis of PCA-MLR and XGBoost models	SWAT-XGBoost (Hybrid model combining SWAT and XGBoost)	The hybrid SWAT-XGBoost model improved predictive accuracy in streamflow forecasting by integrating physical-based and data-driven modeling approaches.
Munawar, H. S., Hammad, A. W., and Waller, S. T. (2022)	Remote sensing methods for flood prediction: A review	Remote Sensing Technologies for Flood Prediction and Risk Analysis	The study reviews remote sensing technologies for flood prediction, identifies their limitations, and proposes an improved model to address existing gaps.
Yoon and Kim, 2022	A systematic review of the flood vulnerability using geographic information systems. *International Journal of Environmental Research and Public Health, 19*(5), 2893.	Parameters	Flood depth is directly correlated with the severity of flooding and the extent of damage caused to infrastructure, agriculture, and human settlements.
Ma, J., Zhou, Q., Zhang, Y., and Li, Y. (2021, May 17).	XGBoost-based method for flash flood risk assessment	Extreme Gradient Boosting (XGBoost)	XGBoost demonstrated high accuracy and reliability in identifying critical flood-contributing factors for flash flood risk assessment in Yunnan Province.

Yuan, L., and Forshay, K. J. (2021).	Streamflow prediction using hybrid SWAT-SVR modeling in the Illinois River Watershed.	Support Vector Regression (SVR) And Random Forest (RF)	Subsequent studies integrated SVR into hybrid models to improve prediction reliability.
Kratzert, F., Klotz, D., Herrnegger, M., Sampson, A. K., Hochreiter, S., and Nearing, G. S. (2018, November 22).	Rainfall–runoff modelling using Long Short-Term Memory (LSTM) networks.	Long Short-Term Memory (LSTM)	Long Short-Term Memory (LSTM) networks have garnered substantial attention due to their ability to capture temporal dependencies and model complex nonlinear relationships inherent in hydrological systems.

Conceptual Framework

Figure 1 presents the conceptual framework of the study, which illustrates the dual-path modeling workflow developed for flood vulnerability assessment in the Los Baños coastal barangays. The framework begins by collecting hydrological and geospatial datasets, including rainfall, elevation, slope, soil type, and land use and land cover. These datasets serve as inputs to the Soil and Water Assessment Tool (SWAT) simulation, which generates key hydrological variables, such as surface runoff, water yield, and flow out, that characterize the watershed's response to rainfall events.

The SWAT outputs are subsequently normalized to ensure consistency and comparability across variables. From this stage, two modeling pathways are established. The first pathway, referred to as the SWAT-alone model, utilizes the normalized SWAT outputs directly for flood vulnerability simulation and map

evaluation in QGIS. This path allows assessment of flood-prone areas solely based on hydrological processes derived from SWAT.

The second pathway, referred to as the Hybrid modeling approach, proceeds from data normalization to dataset partitioning, in which the data are divided into training, validation, and test subsets. These subsets are then used to train and evaluate a series of machine learning algorithms, including Random Forest (RF), Extreme Gradient Boosting (XGBoost), Multi-Layer Perceptron (MLP), Long Short-Term Memory (LSTM), and Support Vector Regression (SVR). Model performance is further optimized using hyperparameter tuning techniques, including Particle Swarm Optimization (PSO), Bayesian Optimization (BO), and Genetic Algorithm (GA).

Each optimized model is evaluated using standard performance metrics, such as the area under the Receiver Operating Characteristic (ROC) curve (AUC-ROC), precision, recall, and F1-score, to assess predictive accuracy and reliability. The best-performing models are then used to simulate flood vulnerability and generate corresponding spatial maps through QGIS. These maps are evaluated to identify high-risk zones and support flood management and mitigation planning for the Los Baños coastal communities.

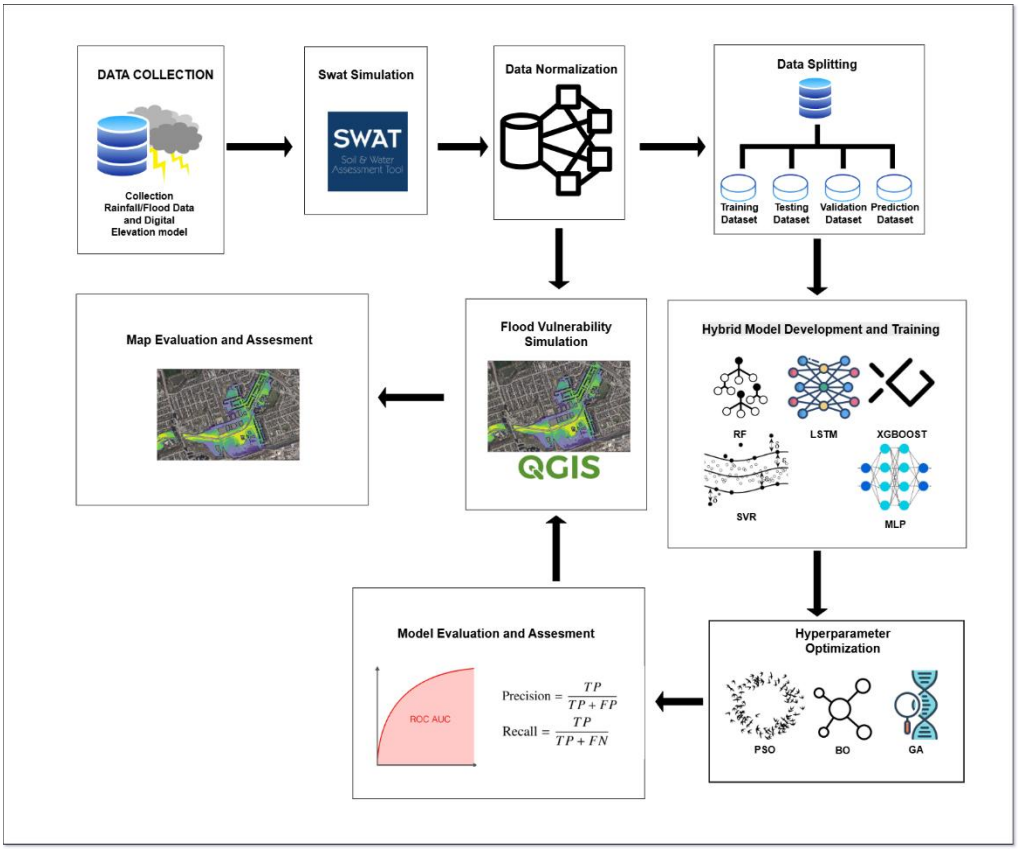


Figure 1. Concept of Developing Hybrid Models and Machine Learning for Flood Vulnerability Assessment in Los Baños Coastal Area

CHAPTER 3

METHODOLOGY

Study Area

The dataset for this research was collected from the coastal barangays of Bayog, Mayondon, Malinta, Baybayin, Bambang, and Tadlac in Los Baños, Laguna, Philippines. Situated along the western shores of Laguna de Bay, these areas were selected as the primary focus for flood vulnerability assessment due to their unique hydrological, topographical, and environmental characteristics. The coastal barangays experience recurrent flooding during the southwest monsoon season and typhoon events, making them an ideal site for evaluating the performance of physically based and data-driven flood prediction models.

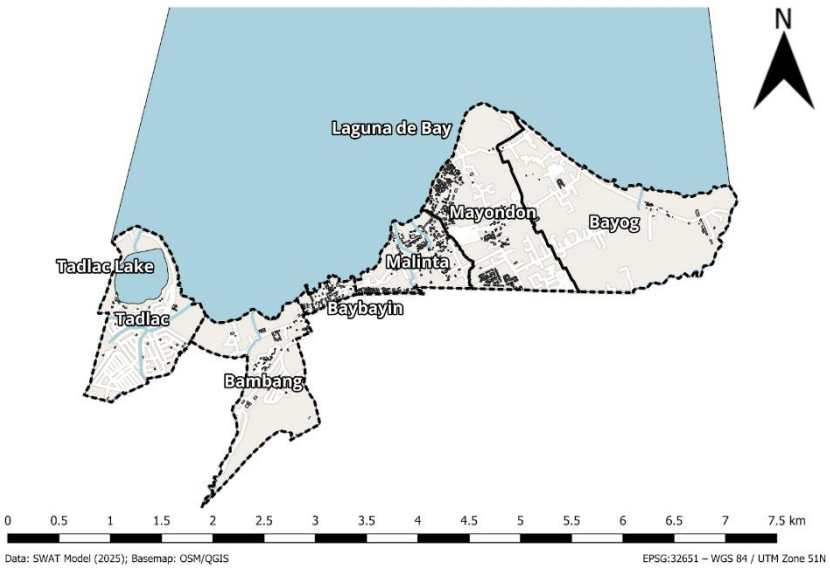


Figure 2. Map of the Los Baños Coastal Barangays Used in the Flood Vulnerability Assessment.

The study area covers approximately 115 km² and is predominantly influenced by the hydrodynamics of Laguna de Bay, the largest freshwater lake

in the Philippines. Land use within the municipality is characterized by a mix of agricultural, residential, and commercial zones, with expanding urbanization near the coastline. Spatial variability in rainfall intensity, slope, soil texture, and drainage patterns contribute to differential flood susceptibility among barangays. The foothills of Mount Makiling further affect the region's hydrology by shaping rainfall distribution, infiltration capacity, and surface runoff behavior.

Data Collection

Guided by the parameters and indicators identified in the review of related literature, a range of hydrological, environmental, and geospatial datasets were assembled to support flood vulnerability modeling in the coastal barangays of Los Baños, including Bayog, Mayondon, Malinta, Baybayin, Bambang, and Tadlac. The selected variables include rainfall, temperature, land use/land cover (LULC), elevation, slope, and soil type, which have been widely recognized as critical inputs in flood risk assessment (Yoon & Kim, 2022; Li & Zhang, 2023; Khan & Ali, 2024).

Meteorological inputs, specifically rainfall and temperature, were obtained from W3S-Water (Daggupati et al., 2025; University of Guelph, 2025), a validated platform that integrates NASA IMERG and NOAA CPC datasets to provide high-quality, near-real-time data suitable for hydrological modeling. Land use/land cover and soil type data were sourced from Geoportal Philippines, the official government repository for geospatial datasets extensively used in national environmental studies. Topographic information, including the Digital Elevation Model (DEM) and derived slope and elevation layers, was obtained from USGS Earth Explorer, a globally recognized source of terrain data widely used in SWAT-based hydrological analyses.

Incorporating datasets from these barangays enhanced the spatial resolution and contextual relevance of the modeling process, given the municipality's proximity to Laguna de Bay and its high exposure to lake-induced flooding, as evidenced by the 2017 Barangay Malinta flood and the regional impacts of Typhoon Agaton (2023). As no streamflow observations were available for the study area, streamflow data were generated through SWAT hydrological simulations, serving as surrogate outputs for subsequent model training, mapping, and validation.

Data Preparation

The preparation of datasets for flood vulnerability modeling involved extracting hydrological variables from the Soil and Water Assessment Tool (SWAT), integrating physiographic attributes, and classifying the study area into discrete flood risk categories. Two datasets were constructed for model development and comparative analysis: the SWAT-alone dataset and the Hybrid dataset.

The SWAT-based dataset was generated using the SWAT model, which simulated hydrological processes within the Los Baños watershed at the Hydrologic Response Unit (HRU) scale. Simulations were conducted over ten years and aggregated into monthly time steps. Although SWAT produces multiple outputs (e.g., precipitation, evapotranspiration, groundwater recharge), only variables directly associated with flood generation were retained for analysis. Three key hydrological predictors were selected: surface runoff, water yield, and flow out. Surface runoff represents direct overland flow contributing to flood peaks; water yield reflects the combined contribution of surface runoff, lateral flow, and baseflow, indicating the total water available for streamflow

generation; and flow out refers to discharge exiting each HRU, corresponding to streamflow at the local outlet. Precipitation, though simulated in SWAT, was treated as an external forcing variable rather than a predictor since its effects are inherently embedded within runoff and yield dynamics.

To capture a broader range of factors influencing flooding, a Hybrid dataset was constructed by combining SWAT-generated hydrological variables with physiographic and categorical attributes. This integration ensured that both flood-generating and flood-mitigating characteristics of the study area were represented. The physiographic variables incorporated included slope, elevation, land use/land cover (LULC), and soil type. Slope and elevation were derived from a Digital Elevation Model (DEM) obtained via Earth Explorer, while LULC and soil type were sourced from Geoportal Philippines. Integrating these features enabled the Hybrid dataset to capture not only hydrological responses but also the topographic and surface characteristics that influence flood susceptibility across the coastal barangays of Los Baños, including Bayog, Mayondon, Malinta, Baybayin, Bambang, and Tadlac.

To enable classification and model evaluation, continuous predictor variables were normalized and transformed into categorical outcomes using a Flood Score Index that quantified the relative flood susceptibility of each spatial unit. The index was computed as a weighted combination of normalized hydrological predictors, with weights determined by each variable's relative influence on flood generation. Flood Scores were subsequently categorized into Low, Moderate, and High flood-risk levels using quantile-based thresholds to ensure balanced class representation across datasets.

For the SWAT-based dataset, the Flood Score was formulated as a weighted linear combination of normalized hydrological predictors (surface runoff, water yield, and flow out). This approach follows the Weighted Linear Combination (WLC) principle, commonly applied in flood susceptibility and risk mapping studies (Kader et al., 2024; Suppawimut, 2021). The general expression of the Flood Score is represented as:

$$FloodScore = w_1(SurfaceRunoff_{norm}) + w_2(WaterYield_{norm}) + w_3(FlowOut_{norm})$$

Figure 3. Flood Score computation using weighted normalized hydrological variables

where $\omega_1, \omega_2, \omega_3$represent the weights assigned to each variable based on their relative hydrological importance. This weighting scheme determines each factor's contribution to the overall flood potential, consistent with WLC methodology in multi-criteria flood hazard assessments (Kader et al., 2024).

For the Hybrid dataset, a risk–resistance framework was employed to integrate both hydrological and physiographic factors. The Flood Score was computed as the difference between the Risk Index and the Resistance Index, expressed as:

$$FloodScore = RiskIndex - ResistanceIndex$$

Figure 4. Flood Score formulation using the Risk–Resistance index approach.

The Risk Index was derived from flood-generating variables such as surface runoff, water yield, and flow out, while the Resistance Index represented flood-mitigating attributes including slope and elevation. This structure follows index-based flood modeling approaches that differentiate

between hazard-enhancing (risk) and mitigating (resistance) components, as demonstrated by Pombo and Ferreira (2019) and Rodríguez-García and Carvalho (2022). Such frameworks conceptualize flood risk as the interaction between flood drivers and the landscape's physical capacity to resist inundation.

Finally, the resulting Flood Scores were classified into Low, Medium, and High flood risk using quantile-based thresholds to ensure balanced class distribution across the study area. Similar quantile or natural-break classifications have been widely adopted in GIS-based flood-susceptibility studies (Suppawimut, 2021; Kader et al., 2024). This classification facilitated model training, validation, and the generation of spatial flood vulnerability maps.

To ensure robust model performance and prevent data leakage, the Hybrid dataset was temporally partitioned into training (2015–2019), validation (2020–2021), and testing (2022–2023) subsets, with 2024 reserved for prediction. Temporal separation helps capture interannual hydrologic variability and prevents overfitting to specific time periods. Thresholds for flood classification (>66% High, <30% Low, 30–66% Medium) were derived solely from the training set and applied consistently to validation and testing data, but not the prediction dataset, ensuring unbiased model evaluation. This practice aligns with hydrological modeling standards emphasizing out-of-sample validation to ensure generalization under unseen conditions (Addor et al., 2017; Kratzert et al., 2018; Cao et al., 2025; Tingsanchali & Karim, 2010; Merz et al., 2023).

Dataset Summary

The hydrological dataset used in this study encompassed 145 Hydrologic Response Units (HRUs) within the Los Baños watershed, covering the period 2015–2023 and sampled monthly. Each HRU represents a unique combination of land use, soil type, and slope, enabling localized flood vulnerability assessment. The watershed was subdivided into 19 subbasins, with physiographic inputs derived from a 30-meter-resolution Digital Elevation Model (DEM) obtained from the USGS Earth Explorer. Land use categories included agricultural land (ARGL), rangeland (RNGE), and urban areas (URBN), while soil types comprised calcareous (CALC), hydrosol (HYDR), and the Lipa soil series (LIPA). Meteorological variables, precipitation, and temperature, were sourced from W3S-Water, and hydrological outputs retained for analysis included surface runoff, water yield, and flow out. These datasets provided the foundation for constructing both the SWAT-alone and Hybrid modeling frameworks.

Table 2. Summary of Hydrological Dataset Characteristics.

Attribute	Description
Number of HRUs	145 HRU
Data Period	2015-2023
Sampling Frequency	Monthly
Number of Subbasins	19 Subbasins
DEM Resolution	30m (USGS Earth Explorer)
Land Use Categories	ARGL (Agricultural Land), RNGE (Rangeland), URBN (Urban Areas)
Soil Types	CALC (Calcareous), HYDR (Hydrosol), LIPA (Lipa Soil Series)
Climate Variables	Precipitation, Temperature (W3S)
Output Variables	Surface Runoff, Water yield, Flow out
Data Sources	W3S, Geoportal Philippines, USGS Earth Explorer

Temporal Data Splitting

To preserve chronological integrity and prevent data leakage, the dataset was partitioned by full calendar years, reflecting a realistic forecasting setup. The model was trained on data from 2015 to 2019 (60 months), validated on 2020 to 2021 (24 months), and tested on 2022 to 2023 (24 months), ensuring that future observations were never used to inform past predictions. The year 2024 was reserved exclusively for out-of-sample prediction, allowing the model to be evaluated under unseen conditions. This temporal structure supports robust generalization and aligns with best practices in hydrological and machine learning modeling, where interannual variability must be captured without compromising evaluation fairness.

Table 3. Temporal Split of the Dataset for Model Training, Validation, and Testing.

Split	Years	Durations
Training Dataset	2015-2019	60 months
Validation Dataset	2020-2021	24 months
Test Dataset	2022-2023	24 months
Prediction Dataset	2024	12 months

Experimental Design

The experimental design of this study focuses on evaluating hybrid model integrations for assessing flood vulnerability in the coastal barangays of Los Baños, including Bayog, Mayondon, Malinta, Baybayin, Bambang, and Tadlac. The framework combines physically based hydrological simulation with machine learning (ML) classification, enabling a robust and scalable predictive approach tailored to barangays with high exposure to lake-induced flooding.

The Soil and Water Assessment Tool (SWAT) was first applied to simulate hydrological processes and generate outputs such as surface runoff, water yield, and flow out, which served as primary predictors of flood generation. These SWAT-derived variables were then integrated with physiographic attributes, including slope, elevation, land use/land cover (LULC), and soil type, to construct a Hybrid dataset that represents both flood-inducing and flood-resisting factors across the study area.

Five machine learning algorithms were trained and evaluated using this dataset: Long Short-Term Memory (LSTM) networks, Random Forest (RF), Extreme Gradient Boosting (XGBoost), Support Vector Classifier (SVC), and Multi-Layer Perceptron (MLP). Each Algorithm underwent hyperparameter optimization to enhance predictive accuracy and generalization performance. Specifically, RF and XGBoost were optimized using Particle Swarm Optimization (PSO) and Bayesian Optimization (BO); MLP was calibrated using Genetic Algorithm (GA) and BO; LSTM employed BO for temporal architecture refinement; and SVC utilized PSO for kernel and regularization tuning. To ensure unbiased evaluation, all configurations were validated using K-Fold cross-validation.

Model performance was assessed using standard classification metrics, including Area Under the Receiver Operating Characteristic Curve (AUC-ROC), precision, recall, and F1-score, to evaluate predictive capability and class discrimination. In addition, spatial validation was conducted through GIS-based comparison with historical flood records, enabling verification of model-

predicted flood-prone zones against observed inundation areas in the selected barangays.

This experimental design provides a comprehensive framework for evaluating accuracy, stability, and reliability, supporting a comparative analysis of hybrid modeling configurations. The results inform the identification of the most suitable approach for localized flood forecasting and vulnerability mapping in the coastal barangays of Los Baños.

To provide a clearer understanding of the methodological components, the following sections first describe the hydrological model (SWAT) used to simulate watershed processes, followed by a detailed discussion of the machine learning algorithms employed in this study. These foundational descriptions establish the basis for the subsequent model development and mapping approaches.

Hydrological Model: Soil and Water Assessment Tool (SWAT)

The Soil and Water Assessment Tool (SWAT) is a physically based hydrological model designed to simulate water flow, sediment transport, and pollutant loading in watershed and coastal environments (Arnold, Srinivasan, Muttiah, and Williams, 1998). The model operates by dividing the study area into smaller sub-basins and Hydrologic Response Units (HRUs) and uses a combination of meteorological, land-use, soil, and topographic data to simulate processes such as runoff, infiltration, evapotranspiration, and groundwater flow. The key equations governing SWAT include the water balance equation:

$$St = St - 1 + P - Q - ET - R \quad (1)$$

where $S_t$is the soil water content at time t, $P$is precipitation, $Q$is surface runoff, $ET$is evapotranspiration, and $R$is deep percolation (Neitsch, Arnold, Kiniry, and Williams, 2011).

SWAT has been widely applied for predicting streamflow, sediment yield, and nutrient transport in large and complex watersheds. In this study, SWAT was used to generate hydrological outputs, specifically surface runoff, water yield, and flow out, which served as primary predictors of flood generation. These outputs were later integrated with physiographic attributes to construct the Hybrid dataset.

By providing detailed spatial and temporal outputs, SWAT supports the modeling of flood dynamics and the identification of flood-susceptible areas within the coastal barangays of Los Baños, including Bayog, Mayondon, Malinta, Baybayin, Bambang, and Tadlac. Furthermore, its outputs integrate effectively with machine learning algorithms such as Random Forest (RF) and Long Short-Term Memory (LSTM), enabling enhanced flood prediction and vulnerability assessment.

Machine Learning Algorithms

This study employs five advanced machine learning algorithms to predict flood vulnerability based on hydrological and environmental input variables collected from the coastal barangays of Los Baños, including Bayog, Mayondon, Malinta, Baybayin, Bambang, and Tadlac. The dataset spans 10 years of monthly records, with 96 months for training and 24 for testing. The selected algorithms were chosen based on their strong predictive performance in previous hydrological and environmental studies.

Random Forest (RF)

Random Forest is a non-parametric ensemble learning algorithm that constructs multiple decision trees using bootstrap aggregation and averages their outputs for prediction (Breiman, 2001). Each tree is built by minimizing node impurity using the mean squared error (MSE):

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - y_i^{\wedge})^2 \quad (2)$$

where $y_i$is the actual value, $\hat{y}_i$is the predicted value, and $n$is the number of observations (Schonlau and Zou, 2020). RFR aggregates predictions across trees to improve accuracy and reduce overfitting. Key parameters include *mtry* (number of variables considered at each split) and *min.node.size* (minimum terminal node size). In hydrological modeling, RFR integrates with SWAT outputs to enhance flood prediction and susceptibility mapping, due to its ability to model complex nonlinear relationships with minimal parameter tuning.

Long Short-Term Memory (LSTM)

LSTM is a type of recurrent neural network (RNN) designed to capture long-range temporal dependencies while addressing vanishing and exploding gradient issues. It uses a memory cell and three gates forget (f_t), input (i_t), and output (o_t) to regulate information flow:

$$\begin{aligned} f_t &= \sigma(W_f x_t + U_f h_{t-1} + b_f) \\ i_t &= \sigma(W_i x_t + U_i h_{t-1} + b_i) \\ c_t &= f_t \cdot c_{t-1} + i_t \cdot g_t \\ o_t &= \sigma(W_o x_t + U_o h_{t-1} + b_o) \\ h_t &= o_t \cdot tan\, h(c_t) \end{aligned} \quad (3)$$

where $\sigma$is the sigmoid activation function, $\tanh$ is the hyperbolic tangent, and $\cdot$ denotes element-wise multiplication. LSTMs are well-suited for

hydrological modeling as they capture temporal dependencies in rainfall-runoff and streamflow data (Kratzert et al., 2018; Lees et al., 2022). Their ability to learn from historical sequences improves flood forecasting accuracy and supports vulnerability assessment.

Extreme Gradient Boosting (XGBoost)

XGBoost extends the gradient boosting framework with improved computational efficiency and predictive performance. It constructs additive decision trees sequentially, with each new tree correcting residual errors from the previous iteration. The objective function is:

$$Obj = l + \Omega(f_k) \tag{4}$$

where $l$is the loss function and $\Omega(f_k)$is the regularization term:

$$\Omega(f_k) = yT + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2 \tag{5}$$

Here, $T$is the number of leaves, $w_j$is the weight of the j-th leaf, $\gamma$is the complexity control parameter, and $\lambda$is the L2 regularization term (Chen and Guestrin, 2016). XGBoost's computational advantages include sparsity-aware algorithms, efficient memory usage, and parallelization. In this study, XGBoost models complex nonlinear relationships among hydrological variables, enabling accurate flood-prone zone prediction and strengthening hybrid SWAT–ML frameworks (Ma et al., 2021; Manjula, 2023).

Support Vector Regression (SVR)

SVR, a variant of Support Vector Machine (SVM), models nonlinear relationships with strong generalization. It maps input data into a high-dimensional space using kernel functions, allowing linear regression in that space. The Gaussian Radial Basis Function (RBF) kernel is applied in this

study. SVR seeks a function $f(x)$that minimizes prediction error within a margin ϵ, while balancing complexity using a regularization term C(Yuan et al., 2021).

A two-step grid search with k-fold cross-validation was used to optimize parameters C, ϵ, and kernel width. Input features were normalized to improve training stability. The SWAT-SVR hybrid approach improves prediction accuracy and captures nonlinear hydrological responses more effectively than standalone models.

Multi-Layer Perceptron (MLP)

MLP is a feedforward neural network composed of an input layer, one or more hidden layers, and an output layer. Each neuron computes a weighted sum of inputs, adds a bias, and applies a nonlinear activation function:

$$y = f(\sum_i w_i\, x_i + b) \tag{6}$$

where $x_i$are input features, $w_i$are weights, $b$is the bias, and $f$is the activation function (ReLU or sigmoid). The network is trained using backpropagation with optimization algorithms such as stochastic gradient descent to minimize a loss function (e.g., mean squared error). In this study, MLP was applied to streamflow prediction using SWAT inputs. Its ability to capture complex nonlinear relationships makes it well-suited for modeling dynamic watershed interactions, thereby enhancing flood vulnerability assessment.

Optimization Techniques

To enhance predictive accuracy and reduce overfitting, this study employed three metaheuristic optimization algorithms: Particle Swarm Optimization (PSO), Bayesian Optimization (BO), and Genetic Algorithm (GA). These techniques were applied to tune hyperparameters of the machine

learning models and, where applicable, calibrate sensitive input parameters of the hydrological model.

Particle Swarm Optimization (PSO)

Particle Swarm Optimization is a population-based optimization algorithm inspired by the social behavior of bird flocking and fish schooling (Kennedy and Eberhart, 1995). The algorithm simulates the movement of particles through a search space, with each particle representing a potential solution. Particles adjust their positions based on their own best position and the best position found by the swarm. The velocity and position updates are defined as:

$$v_i^{t+1} = wv_i^t + c_1 r_1 \left(P_{ibest} - x_t^i\right) + c_2 r_2 \left(P_{gbest} - x_t^i\right) \quad (7)$$

$$x_i^{t+1} = x_i^t + v_i^{t+1}$$

where $v_i$is the velocity of particle i, $x_i$is its position, $p_{ibest}$is the personal best position, $p_{gbest}$is the global best position, $\omega$is the inertia weight, $c_1$and $c_2$are acceleration coefficients, and $r_1, r_2$are random values between 0 and 1. PSO has been widely applied in hydrological modeling and machine learning due to its simplicity and ability to efficiently search high-dimensional parameter spaces.

Bayesian Optimization (BO)

Bayesian Optimization is a probabilistic optimization framework designed for costly objective functions (Shahriari et al., 2016). It constructs a surrogate model, typically a Gaussian Process (GP), to estimate the mean and variance of the objective function at each point in the search space. The next evaluation point is selected using an acquisition function:

$$a(x) = EI(x), PI(x), or\ UCB(x) \quad (8)$$

where $a(x)$balances exploration and exploitation. BO is particularly effective for optimizing machine learning models with high computational costs, as it requires fewer evaluations compared to grid or random search. In this study, BO was applied to refine hyperparameters of models such as Random Forest (RF) and Long Short-Term Memory (LSTM), improving flood susceptibility predictions while minimizing computational overhead.

Genetic Algorithm (GA)

Genetic Algorithm is a population-based metaheuristic inspired by natural selection and evolution. Candidate solutions (chromosomes) evolve through selection, crossover, and mutation, guided by a fitness function that evaluates model performance. In this study, GA was used to fine-tune hyperparameters of machine learning algorithms such as the Multi-Layer Perceptron (MLP) and to calibrate sensitive input parameters of SWAT.

The fitness function minimized a loss metric, typically mean squared error (MSE):

$$fitness = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - y_i^{\wedge 2}\right) \quad (9)$$

GA's stochastic nature allows effective exploration of complex parameter spaces and helps avoid local minima. Previous studies have demonstrated GA's utility in hydrology, such as coupling GA with SWAT to optimize Best Management Practices (Kaini et al., 2012) and integrating GA with Bayesian Model Averaging to improve parameter uncertainty estimation (Zhang et al., 2009). These applications validate GA's strength in optimizing flood forecasting models under dynamic or data-sparse conditions.

Model Development

With the hydrological model, machine learning algorithms, and optimization techniques established, the model development stage integrates these components into a unified framework for flood vulnerability prediction in the coastal barangays of Los Baños, specifically Bayog, Mayondon, Malinta, Baybayin, Bambang, and Tadlac.

The primary objective of this stage was to build, optimize, and compare the performance of five machine learning algorithms: Long Short-Term Memory (LSTM), Random Forest (RF), Extreme Gradient Boosting (XGBoost), Support Vector Classifier (SVC), and Multi-Layer Perceptron (MLP) in forecasting flood-prone conditions. These algorithms were selected for their complementary strengths: LSTM for capturing temporal dependencies; RF and XGBoost for modeling nonlinear relationships and feature importance; SVC for high-dimensional classification; and MLP for complex nonlinear pattern recognition.

All simulations and machine learning experiments were executed on a local workstation to ensure reproducibility. The hardware configuration consisted of a Ryzen 7 5700X CPU (8-cores, 3.4 GHz), 32 GB DDR4 RAM, and an NVIDIA GeForce RTX 3060 GPU (12 GB VRAM), running Windows 11 Pro. The software environment included Python 3.10 and QSWAT+ for hydrological simulation.

A 10-year monthly dataset (2015–2024) sourced from W3S Water and Geoportal Philippines was used. Continuous variables (rainfall, runoff, water yield, elevation, slope) were normalized, while categorical variables (soil type and land use/land cover) were label-encoded to ensure uniform scaling across

all models. A temporal-based data split was used to preserve chronological integrity and prevent data leakage: 60 months (2015–2019) for training, 24 months (2020–2021) for validation, 24 months (2022–2023) for testing, and 12 months (2024) reserved for prediction.

Hyperparameter tuning was conducted using Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Bayesian Optimization (BO), as described in the previous section. RF and XGBoost were tuned using PSO and BO (optimizing maximum tree depth, learning rate, and subsampling ratios). MLP was optimized using GA and BO (tuning hidden layers, activation functions, and regularization strength). LSTM was refined using BO (optimizing sequence length, hidden units, and learning rate), while SVC utilized PSO (tuning kernel type and regularization parameters). All optimization configurations were evaluated using K-fold cross-validation, ensuring model stability and unbiased evaluation.

Model performance was assessed using statistical metrics, including AUC-ROC, F1 score, precision, and recall, to evaluate predictive capability and class discrimination.

Mapping Approaches

Following the preprocessing and normalization steps described in the earlier sections, the mapping approaches presented here focus on translating model outputs into spatially explicit flood vulnerability maps for the coastal barangays of Los Baños, specifically Bayog, Mayondon, Malinta, Baybayin, Bambang, and Tadlac. Two distinct workflows were implemented to evaluate

the predictive strength of physically based versus hybrid data-driven approaches.

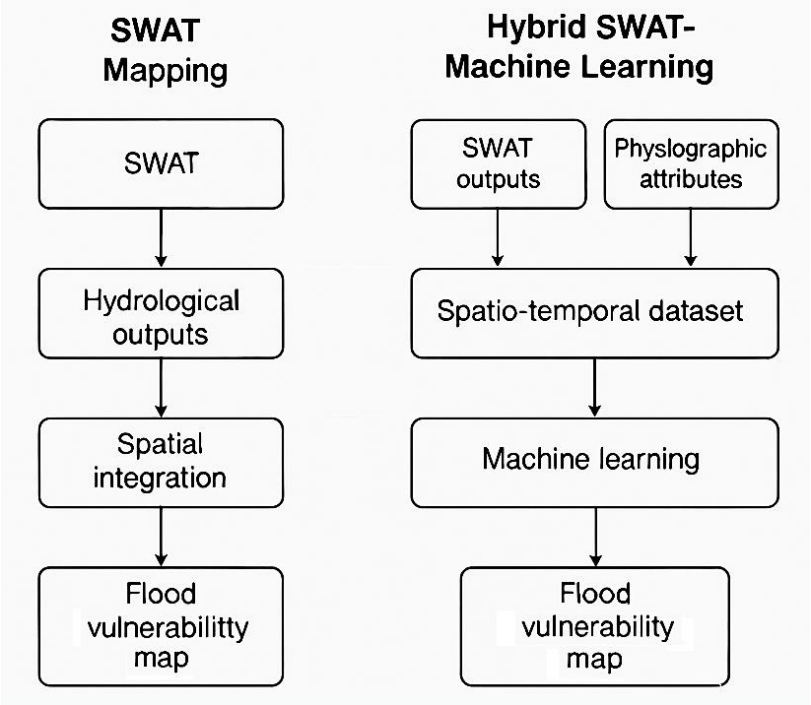


Figure 5. Workflow of SWAT and Hybrid SWAT–ML Approaches for Flood Vulnerability Assessment in Selected Barangays of Bayog, Mayondon, Malinta, Baybayin, Bambang, and Tadlac.

In the SWAT-based workflow, hydrological outputs, such as surface runoff, water yield, and outflow, were directly used to delineate flood-susceptible zones. These outputs were aggregated at the barangay scale using GIS zonal statistics, classified into susceptibility categories by applying threshold values to hydrological indicators, and finally overlaid with barangay boundaries to produce flood vulnerability maps. This workflow provided a physically based baseline representation of flood risk, reflecting watershed processes without additional data-driven enhancement.

In the hybrid workflow, SWAT outputs were combined with physiographic attributes, including slope, elevation, soil type, and land use/land cover (LULC), to construct a spatiotemporal dataset. Machine learning algorithms (LSTM, RF, XGBoost, SVC, and MLP) were then applied to this dataset to generate barangay-level flood vulnerability classifications. The predictions were spatially integrated with barangay shapefiles in GIS and visualized using standardized symbology, allowing direct comparison with the SWAT-based maps. This workflow enabled the integration of hydrological dynamics with physiographic variables, capturing both temporal variability and spatial heterogeneity.

Both workflows produced barangay-level flood vulnerability maps, which were subsequently compared against historical flood records to assess accuracy and reliability. The comparative evaluation highlighted the added value of hybrid modeling in improving predictive performance beyond what a physically based model alone could achieve.

Validation and Model Evaluation

Model performance was evaluated using a combination of statistical classification metrics and spatial validation techniques to ensure both predictive accuracy and geographic consistency. The evaluation framework was applied to two categories: the SWAT-based workflow, which relied solely on hydrological outputs, and the Hybrid SWAT-Machine Learning framework, which integrated SWAT-generated variables with physiographic attributes.

The statistical evaluation of the hybrid models used key classification metrics, including the area under the Receiver Operating Characteristic (ROC) curve (AUC), F1 score, precision, and recall. AUC–ROC measured the models'

ability to discriminate between flood-prone and non-flood-prone areas. At the same time, the F1-score provided a balanced measure of precision and recall, particularly useful for imbalanced datasets (Pinichka et al., 2025). Precision quantified the proportion of correctly predicted flood-prone zones among all predicted positives, whereas recall measured the proportion of actual flood-prone zones correctly identified by the models. In addition, Mean Squared Error (MSE) was used to assess continuous outputs, such as predicted flood scores, providing a measure of overall prediction error.

To ensure model reliability and prevent overfitting, K-Fold cross-validation was implemented during training and Optimization. This procedure provided unbiased estimates of model performance by partitioning the dataset into multiple folds and iteratively training and testing on each fold. Furthermore, sensitivity analysis was conducted to evaluate the effect of varying input parameters on model performance, thereby strengthening the robustness of the evaluation framework.

Spatial validation was performed using GIS-based overlay analysis, which compared predicted flood vulnerability zones with historical flood records and mapped inundation areas in the Los Baños coastal barangays, specifically Bayog, Mayondon, Malinta, Baybayin, Bambang, and Tadlac (Kaya et al., 2023). This spatial agreement analysis ensured that the model outputs were not only statistically accurate but also geographically consistent with observed flood patterns.

Table 4. Evaluation Metrics Applied to SWAT-Only and Hybrid SWAT–ML Models

Model Category	Metrics Used	Purpose
Hybrid Models (SWAT + ML)	AUC–ROC, F1-score, Precision, Recall, MSE	Evaluates classification performance and continuous flood score predictions of machine learning models trained on SWAT outputs and physiographic attributes.
SWAT-Only Mapping	Spatial Agreement, Overlay Analysis	Assesses flood vulnerability zones derived directly from hydrological outputs without ML enhancement.
GIS-Based Validation	Spatial Agreement, Overlay Analysis	Validates both mapping workflows against historical flood records and inundation zones in Los Baños.

Ethical Consideration

Ethical standards guide the entire research process, emphasizing data privacy, environmental sustainability, and fairness. All datasets are anonymized, and appropriate protection measures are implemented to safeguard any sensitive information. Since the study relies solely on secondary data, no human participation is involved. The datasets were sourced exclusively from official and publicly accessible repositories to ensure data integrity and transparency.

The research also prioritizes sustainable flood management practices, aiming to minimize potential environmental impacts. Efforts are made to ensure that all developed models are unbiased and representative of diverse flood-prone areas. Finally, transparency and reproducibility are maintained by thoroughly documenting all methodologies and making results available for peer evaluation, thereby ensuring that the study contributes meaningfully to both the scientific community and local disaster preparedness efforts.

CHAPTER 4

RESULTS AND DISCUSSIONS

This chapter presents the results of the machine learning models trained and tested for flood vulnerability assessment in six selected barangays of Los Baños, Laguna, Bayog, Tadlac, Malinta, Bambang, Mayondon, and Baybayin. These barangays were chosen for their proximity to Laguna de Bay and their history of flooding, making them critical areas for localized vulnerability analysis. The models were evaluated using Accuracy, Precision, Recall, and F1-score to determine their predictive effectiveness, with the performance of five algorithms, Random Forest, XGBoost, Support Vector Machine (SVM), Multi-Layer Perceptron (MLP), and Long Short-Term Memory (LSTM), discussed individually to highlight their strengths and limitations. A comparative analysis is then presented to identify the most effective algorithm, followed by an interpretation of the results in relation to existing studies. The findings provide valuable insights for improving community-based flood preparedness and disaster risk reduction strategies in Los Baños, particularly in the most flood-prone barangays.

Data Preparation

This section presents the results of the data preparation stage, highlighting the characteristics of the SWAT-alone and Hybrid datasets and their classification into flood risk categories. The analysis focuses on the distributions of predictors and the balance of flood classes, which together form the foundation for subsequent model training and evaluation.

SWAT-Based Hydrological Dataset

The SWAT-alone dataset consists of three hydrological predictors: surface runoff, water yield, and flow out. The relationship between these predictors and the flood risk categories is shown in Figure 4.

The boxplots reveal a clear differentiation among categories. The High flood class exhibits substantially larger median values of surface runoff and water yield compared to the Low flood class, while the *Medium* class occupies an intermediate position. Flow Out shows a wider spread and several outliers, reflecting localized variability in stream discharge across hydrologic response units. These patterns confirm the strong association between hydrological predictors and flood severity.

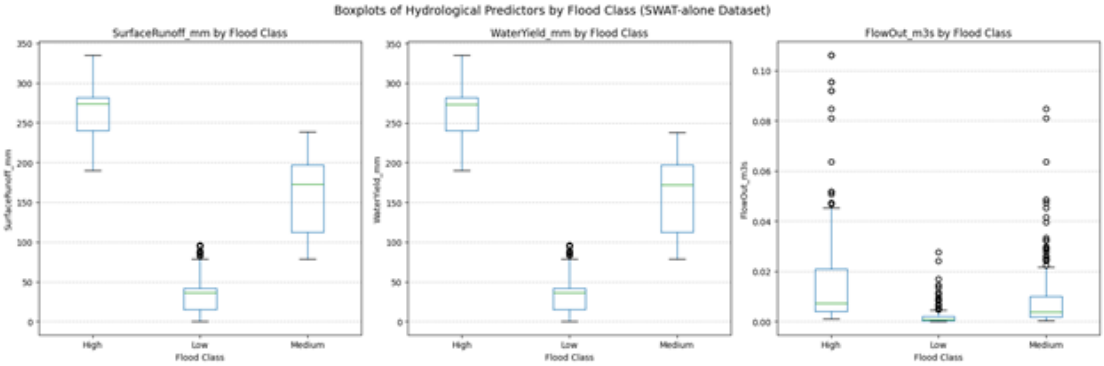


Figure 6. Boxplots of hydrological predictors (Surface Runoff, Water Yield, Flow Out) grouped by flood risk category in the SWAT-alone dataset.

The overall distribution of flood classes in the SWAT-alone dataset is shown in Figure 7. The three categories, Low, Medium, and High, are relatively balanced, with no class dominating the dataset. This proportionality is important for avoiding classification bias in machine learning models, ensuring that predictive performance can be evaluated fairly across all risk levels.

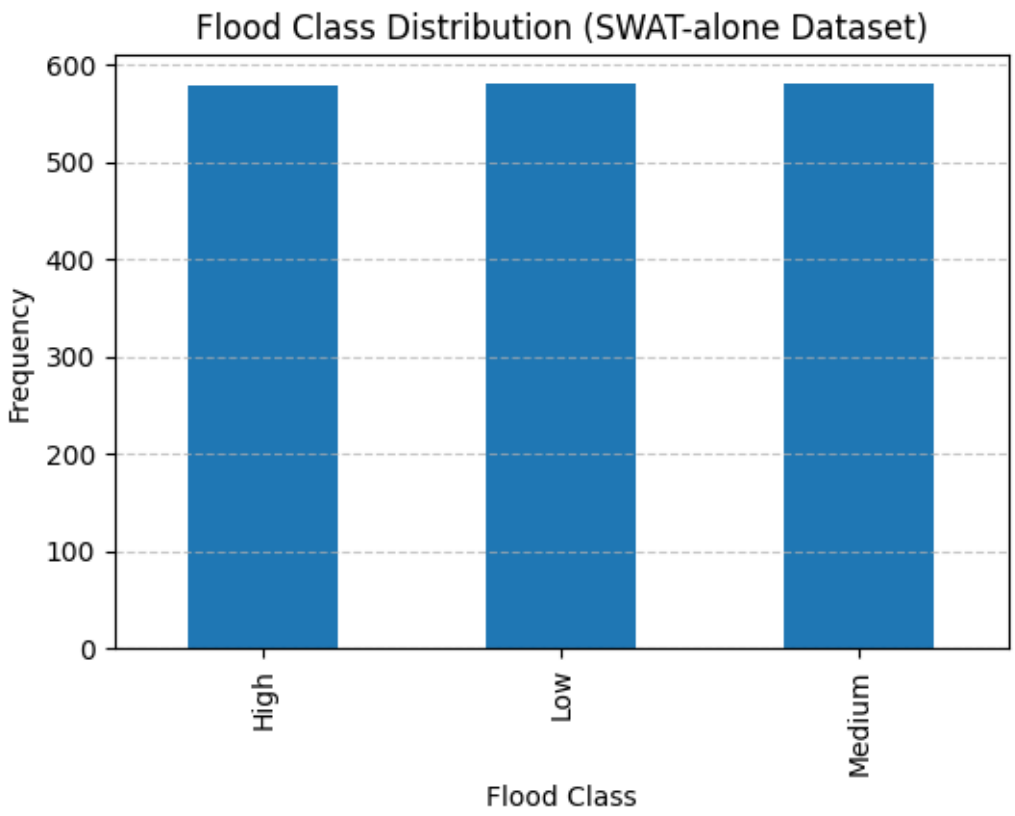


Figure 7. Distribution of flood risk classes (Low, Medium, High) in the SWAT alone dataset.

Flood Vulnerability Assessment Using SWAT Model

This section presents the flood vulnerability maps for Los Baños, Laguna, generated using the SWAT-alone model as the baseline for assessing flood susceptibility. The mapping utilized SWAT-derived hydrological variables, surface runoff, water yield, and precipitation to represent spatial variations in flood behavior across barangays. Each Hydrological Response Unit (HRU) was linked to its corresponding barangay boundary to ensure accurate spatial representation of flood-prone areas.

Flood mapping was conducted using QGIS 3.34, where the simulated outputs were rasterized and reclassified into Low, Moderate, and High vulnerability categories using quantile-based thresholds.

The flood vulnerability map applies a color-coded symbology to indicate different risk levels. Green represents low-risk areas, typically elevated or steep zones with minimal runoff. Yellow denotes moderate-risk transitional areas with a moderate hydrologic response, while red highlights high-risk low-lying areas characterized by high runoff and limited drainage capacity.

The resulting flood vulnerability maps were developed separately for the dry (November–April) and wet (May–October) seasons to capture seasonal variations in rainfall and hydrologic response

SWAT-Based Flood Vulnerability Map – Dry Season

Figure 8 presents the SWAT-alone flood vulnerability maps for the dry season (November–April) in Los Baños, Laguna. This period corresponds to the Northeast Monsoon (Amihan), characterized by cooler temperatures and reduced rainfall, resulting in minimal surface runoff and low soil saturation across the municipality.

During the early dry months (November–December), residual Moderate to High vulnerability zones appear in Bayog and Mayondon, likely influenced by remaining soil moisture and baseflow from the preceding wet season. From January to March, most areas, including Bambang, Tadlac, and Malinta, exhibit low vulnerability, indicating stable hydrologic conditions and limited flood potential due to minimal rainfall and effective drainage.

A slight reappearance of Moderate vulnerability can be observed in April, particularly in Malinta and Bayog, which may be attributed to localized convectional rainfall events before the onset of the wet season.

Overall, the dry season maps reveal a predominantly Low flood vulnerability pattern, with flood susceptibility localized near coastal and low-lying barangays.

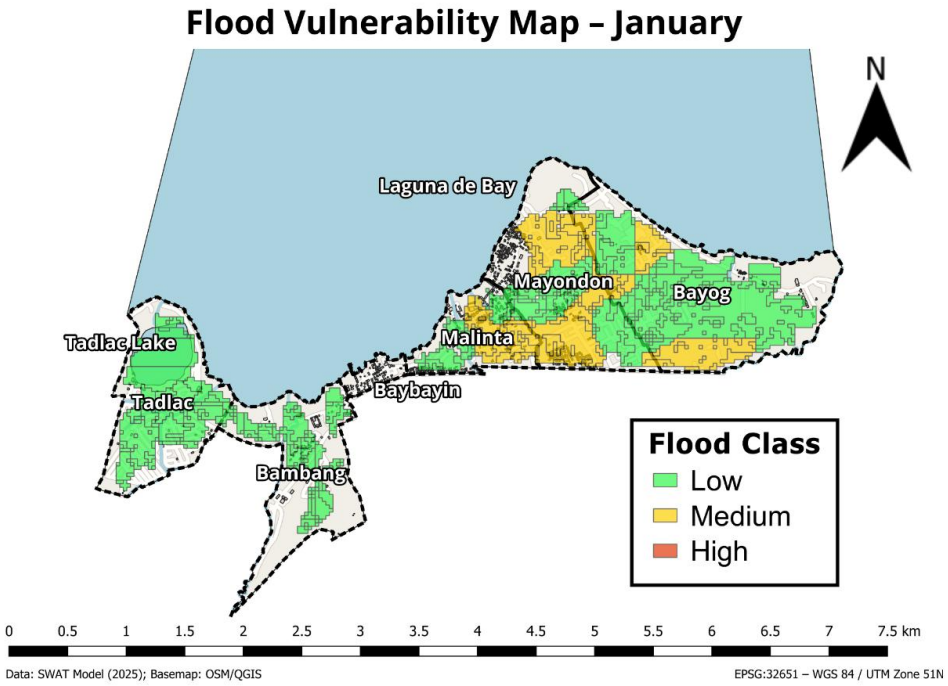


Figure 8. SWAT-Alone Flood Vulnerability Maps – Dry Season (November–April). Monthly flood vulnerability distribution derived from the SWAT model outputs.

SWAT-Based Flood Vulnerability Map – Wet Season

Figure 9 shows the SWAT-alone flood vulnerability maps for the wet season (May–October) in Los Baños, Laguna. This period coincides with the Southwest Monsoon (Habagat), bringing frequent rainfall and occasional typhoons that intensify surface runoff and water accumulation in low-lying areas.

At the start of the wet season (May–June), areas of Moderate to High vulnerability appear in Bayog, Malinta, and Mayondon, primarily due to their proximity to Laguna de Bay and limited drainage capacity. As rainfall increases from July to September, the extent of High vulnerability expands significantly, with these barangays experiencing the most flood-prone conditions.

By October, rainfall begins to subside, and some areas in Bambang and Tadlac transition to Moderate vulnerability, while Bayog and Mayondon remain under High vulnerability due to residual water retention.

Overall, the wet-season results highlight the strong hydrologic influence of rainfall and terrain on flood exposure, emphasizing that the most flood-susceptible barangays are those located near coastal and low-elevation zones.

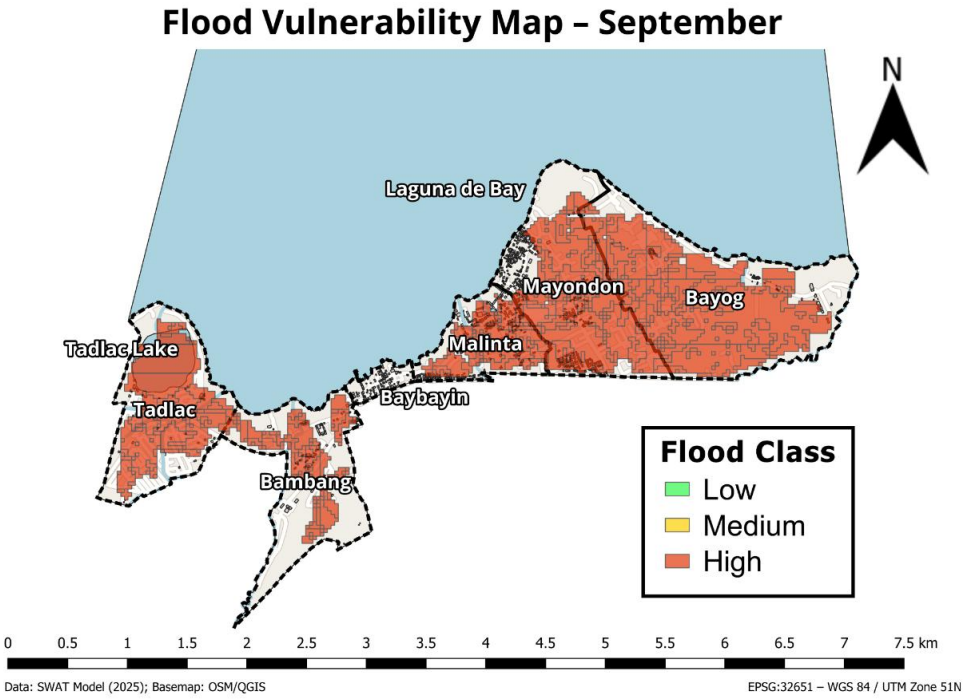


Figure 9. SWAT-Alone Flood Vulnerability Maps – Wet Season (May–October). Monthly flood vulnerability distribution derived from the SWAT model outputs.

Comparative Analysis of Dry vs. Wet Season (Per Barangay)

Table 5 presents the seasonal distribution of flood vulnerability classes derived from the SWAT-alone simulation. Across all barangays, clear seasonal contrasts were observed, reflecting the influence of rainfall intensity and terrain characteristics on hydrological behavior.

During the dry season, the majority of barangays in Bambang, Bayog, Malinta, Mayondon, and Tadlac were classified as Low vulnerability, with values ranging from approximately 59% to 81%. This pattern indicates reduced surface

runoff, lower soil saturation, and generally stable hydrological conditions due to limited precipitation. Only small portions of low-lying or flat areas exhibited Medium to High vulnerability, particularly in Mayondon and Bayog, where proximity to Laguna de Bay and gentle slopes increase potential for localized water accumulation.

In contrast, the wet-season results show a significant increase in flood susceptibility. All barangays exhibited a balanced distribution between Medium (≈50%) and High (≈50%) flood classes, while Low-vulnerability areas almost disappeared. This seasonal escalation is directly linked to monsoon rainfall, saturated soils, and lake backflow effects, particularly affecting the coastal barangays of Bayog, Tadlac, and Mayondon, which are repeatedly identified as high-risk zones.

Overall, the analysis highlights a clear temporal shift from low vulnerability during dry months to moderate-to-high vulnerability during wet months, demonstrating the model's sensitivity to hydrological seasonality. The dominant flood class across most barangays transitions from Low (Dry) to Medium (Wet), validating the strong influence of seasonal precipitation patterns on flood behavior within the Los Baños watershed.

Table 5. Seasonal distribution of flood vulnerability classes derived from the SWAT-alone simulation.

Barangay	Season	Low	Medium	High	Dominant Class
Bambang	Dry	16.67%	66.67%	16.67%	Low
Bambang	Wet	50%	0%	50%	Medium
Bayog	Dry	16.67%	80.75%	2.58%	Low
Bayog	Wet	50%	0%	50%	Medium
Malinta	Dry	16.67%	66.67%	16.67%	Low
Malinta	Wet	50%	0%	50%	Medium
Mayondon	Dry	16.67%	59.21%	24.12%	Low
Mayondon	Wet	50%	0%	50%	Medium
Tadlac	Dry	16.67%	66.67%	16.67%	Low
Tadlac	Wet	50%	0%	50%	Medium

Development of Hybrid SWAT–Machine Learning Dataset

The Hybrid dataset integrates hydrological variables with physiographic attributes (slope, elevation). Flood risk classes were generated using the risk–resistance framework, and the resulting distributions across dataset partitions are shown in Figure 10.

The figure demonstrates that the training, validation, and test subsets all preserve the representation of Low, Medium, and High flood categories. While the training set contains the largest number of samples, the proportions across the three classes remain broadly comparable across splits. This indicates that the temporal partitioning procedure successfully maintained balance and representativeness, preventing bias during model training and ensuring reliable generalization in evaluation.

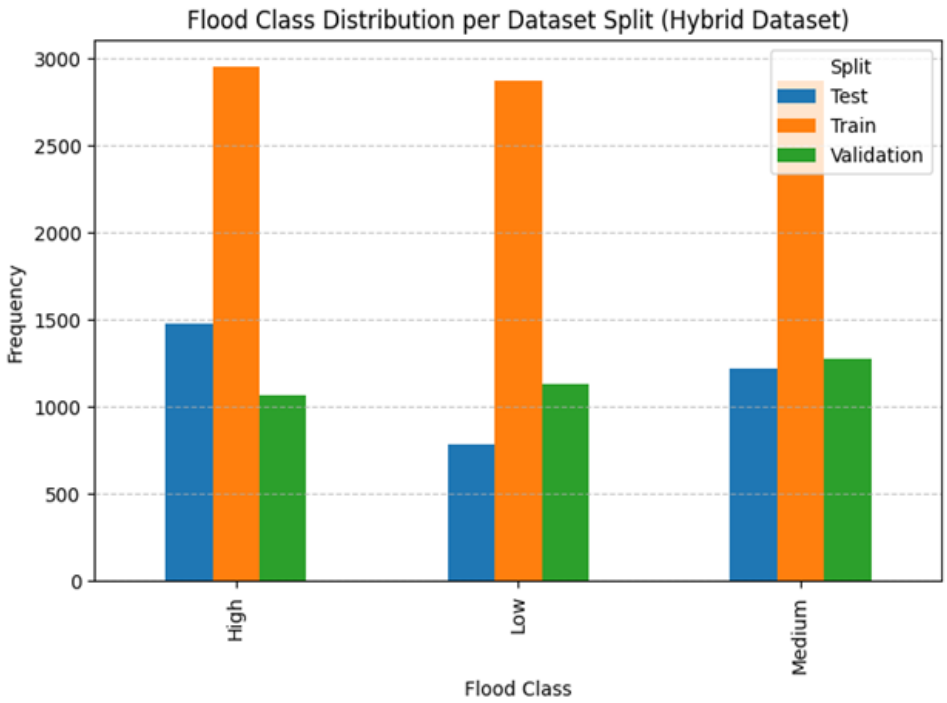


Figure 10. Flood risk class distribution across the training, validation, and test subsets of the Hybrid dataset.

Model Training and Performance Evaluation

This section presents the results obtained from the training and tuning of hybrid machine learning models, Random Forest, XGBoost, Multi-Layer Perceptron (MLP), Long Short-Term Memory (LSTM), and Support Vector Classifier (SVC) in conjunction with hydrological data derived from the SWAT model. Each model was optimized using metaheuristic or Bayesian techniques and validated using K-Fold Cross-Validation to identify recurring flood-vulnerability patterns within the Los Baños coastal barangays.

Random Forest (PSO + KFOLD)

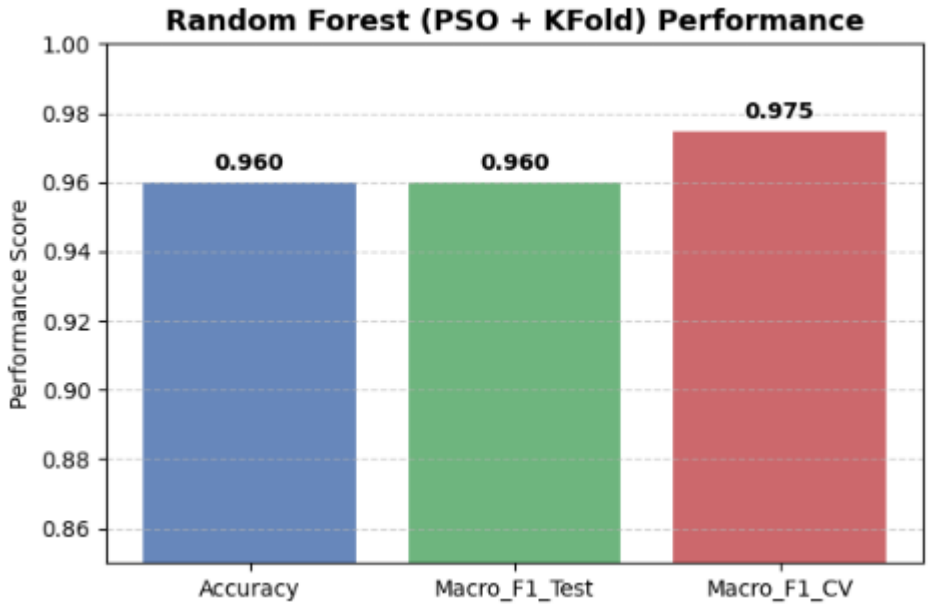


Figure 11. Performance metrics of the Random Forest (PSO + KFold) model.

The Random Forest optimized through PSO achieved an accuracy of 0.96, precision of 0.96, recall of 0.96, and a macro F1-score of 0.96 on the test set. Its cross-validation F1-score (0.9748) further confirms strong model generalization. The PSO algorithm effectively tuned hyperparameters, enhancing predictive stability across different folds. Misclassifications were minimal, mostly between *Low* and *Medium* flood vulnerability classes, reflecting overlapping environmental conditions.

Overall, this configuration demonstrated reliable performance and robustness against overfitting, consistent with prior studies emphasizing Random Forest's capability in hydrological hazard prediction.

Random Forest (BO + KFold)

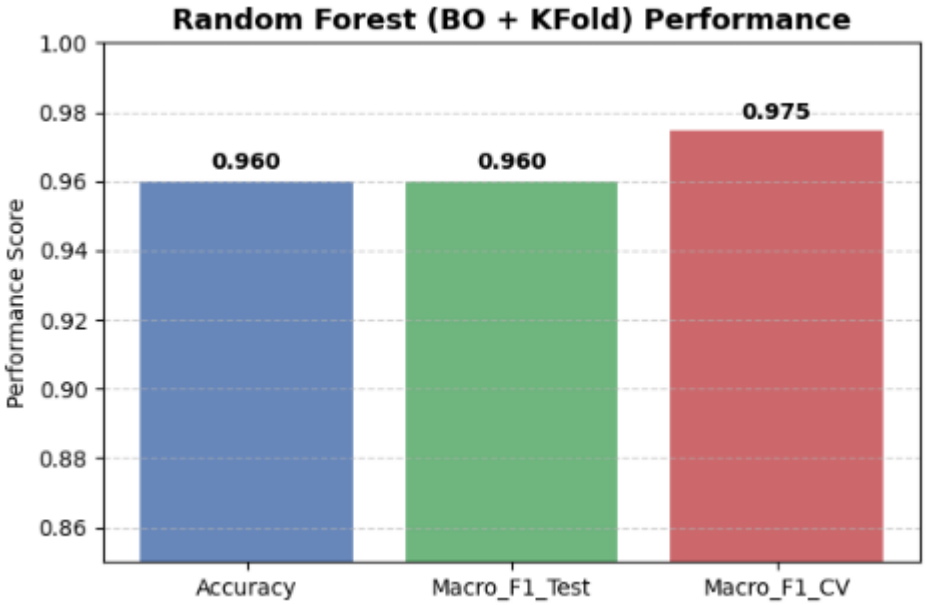


Figure 12**.** Performance metrics of the Random Forest (BO + KFold) model.

The Bayesian Optimization approach produced comparable results, with an accuracy of 0.96, precision of 0.96, recall of 0.96, and a macro F1-score of 0.96. The cross-validation F1-score (0.9747) suggests stable performance similar to the PSO-tuned model. The optimized parameters ({n_estimators = 307, max_depth = 25, min_samples_split = 4, min_samples_leaf = 1}) indicate a balance between complexity and generalization.

Overall, the BO-tuned Random Forest demonstrated efficient parameter searching and strong predictive capability, validating its reliability in environmental data analysis.

XGBoost (PSO + KFold)

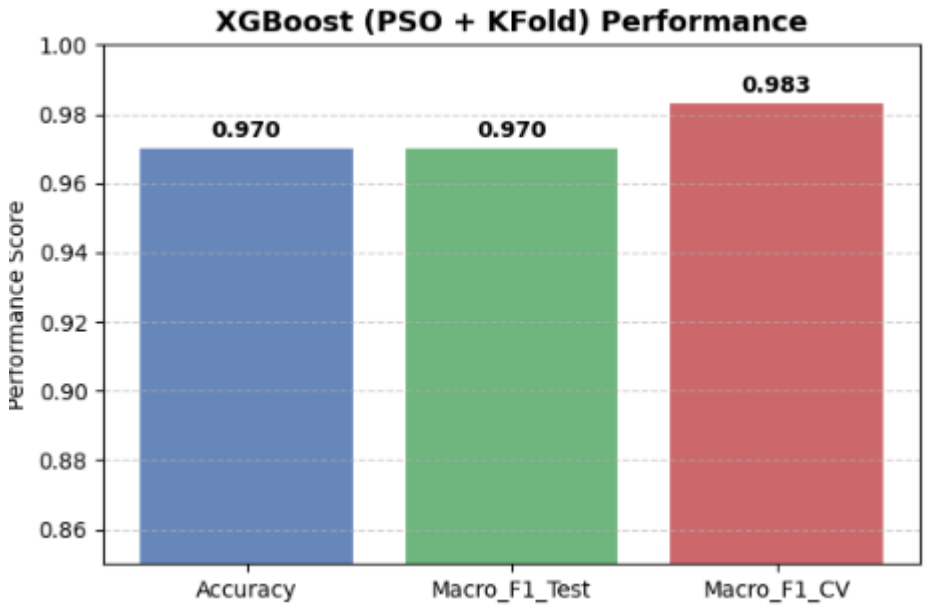


Figure 13. Performance metrics of the XGBoost (PSO + KFold) model.

This configuration achieved the highest overall cross-validation F1-score (0.983) and a test accuracy of 0.97. The PSO optimization effectively enhanced XGBoost's learning parameters (n_estimators ≈ 392, max_depth ≈ 10, learning_rate ≈ 0.038), enabling the model to capture complex non-linear interactions among hydrological and socio-environmental features. XGBoost demonstrated near-perfect classification with balanced precision and recall across all flood vulnerability levels. Only minor misclassifications occurred between *Medium* and *High* categories. This confirms the superiority of the PSO-tuned XGBoost model, aligning with Ma et al. (2021), which identified XGBoost as the leading algorithm for flood susceptibility mapping.

XGBoost (BO + KFold)

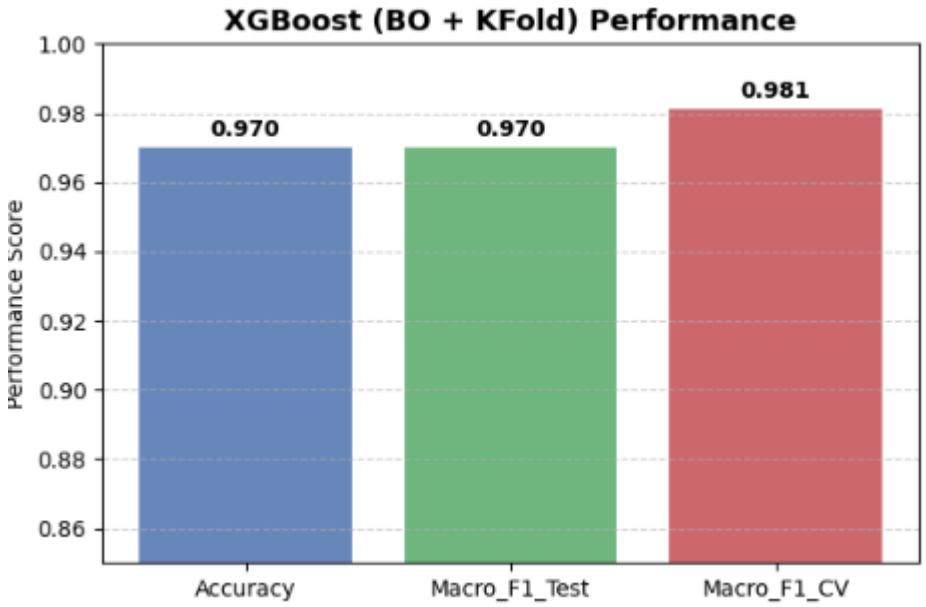


Figure 14. Performance metrics of the XGBoost (BO + KFold) model.

The Bayesian Optimization variant also achieved an accuracy of 0.97, macro F1-score of 0.97, and cross-validation F1-score of 0.981. With tuned parameters ({n_estimators = 434, max_depth = 5, learning_rate = 0.0318, subsample = 0.854, colsample_bytree = 0.852}), the BO-tuned model maintained high generalization while reducing overfitting risks. The consistent results between PSO and BO variants highlight XGBoost's robustness and adaptability to different optimization techniques. These results emphasize that the gradient boosting ensemble remains the most accurate and stable model for flood vulnerability prediction.

Multi-Layer Perceptron MLP (GA + KFold)

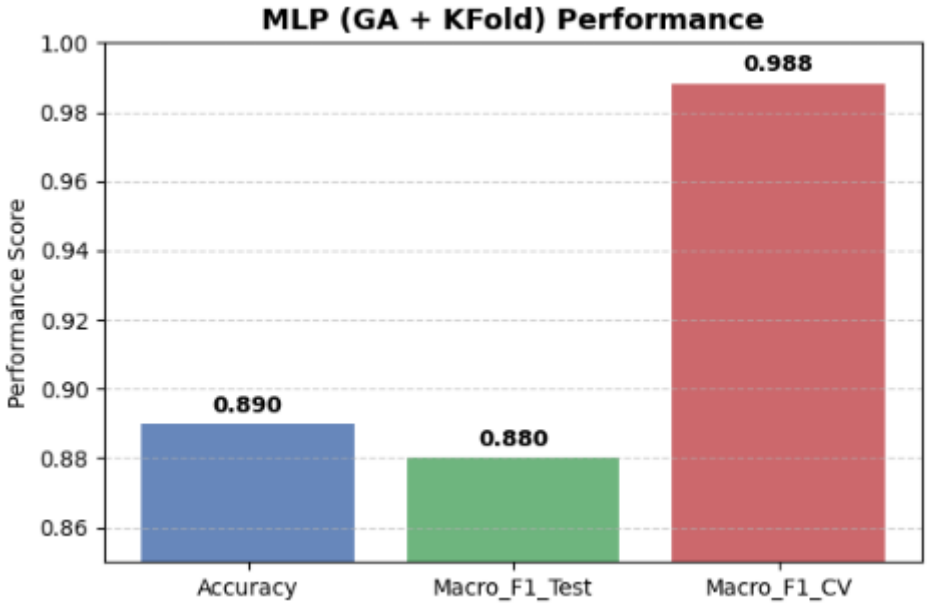


Figure 15. Performance metrics of the MLP (GA + KFold) model.

The Genetic Algorithm-optimized MLP achieved a cross-validation F1-score of 0.9882 but a lower test accuracy of 0.89 and macro F1-score of 0.88. While the model performed well during training, its lower test performance suggests overfitting caused by suboptimal hyperparameter adaptation. The MLP–GA model struggled particularly with the *medium* vulnerability class, indicating potential imbalance in hidden-layer representation. Nevertheless, its high CV-F1 value indicates the potential of neural networks to model complex, non-linear flood-related features when properly regularized.

Multi-Layer Perceptron MLP (BO + KFold)

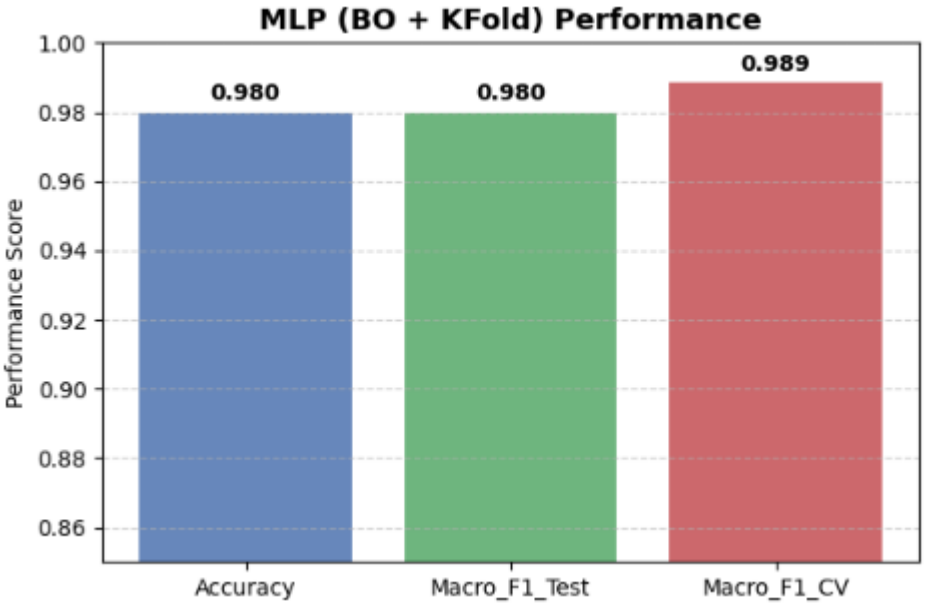


Figure 16. Performance metrics of the MLP (BO + KFold) model.

The BO-optimized MLP achieved 0.98 accuracy, 0.98 macro F1, and a CV-F1 of 0.9886, representing one of the best performances among all models. Its optimized architecture (hidden layers: 193 and 45 neurons; learning_rate_init = 0.000543; alpha = 0.00354) produced stable convergence and minimized overfitting. The model accurately captured nonlinear relationships between hydrological predictors and flood risk classes. Misclassifications were minimal, primarily between *Medium* and *High* categories. This configuration validates the effectiveness of Bayesian Optimization in fine-tuning neural network hyperparameters.

LSTM (BO + KFold)

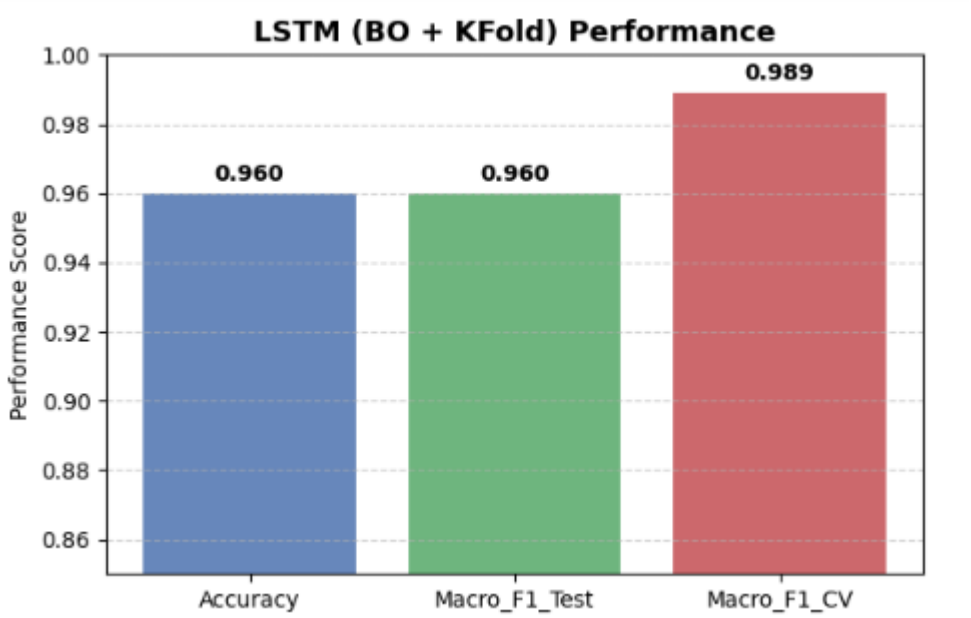


Figure 17. Performance metrics of the LSTM (BO + KFold) model.

The BO-tuned LSTM achieved an accuracy of 0.96 and macro F1 of 0.96, with a CV-F1 of 0.9892. These results confirm LSTM's effectiveness in capturing temporal patterns in rainfall and discharge sequences. While not as accurate as XGBoost or MLP, its consistent recall values across classes highlight the importance of time-series modeling in flood vulnerability analysis. Misclassifications were mostly between *Low* and *Medium* categories, likely due to subtle temporal overlaps. The LSTM model's performance suggests its potential for use in future temporal flood forecasting systems.

SVC (PSO + KFold)

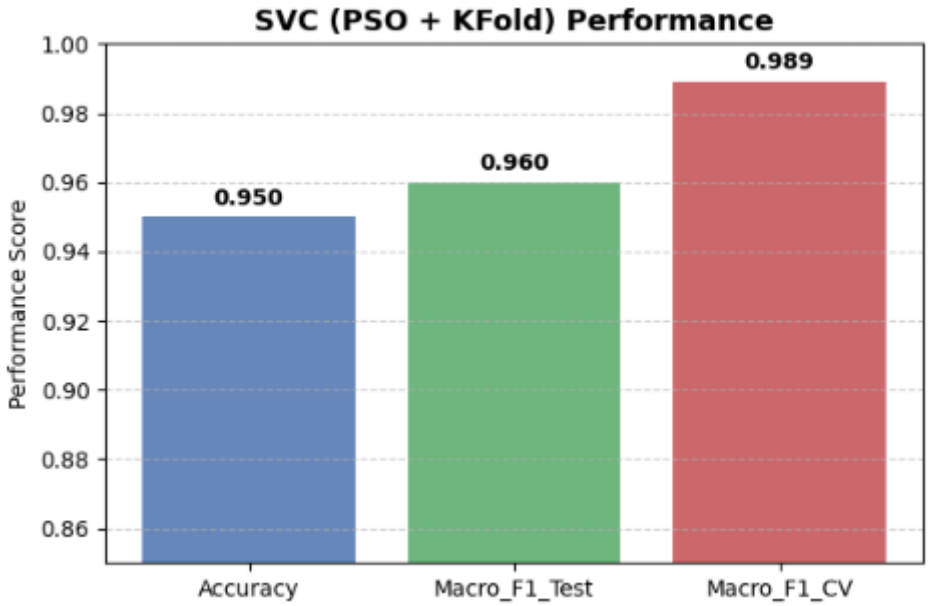


Figure 18. Performance metrics of the SVC (PSO + KFold) model.

The PSO-optimized SVC achieved an accuracy of 0.95, macro F1-score of 0.96, and cross-validation F1-score of 0.9892. It demonstrated high precision and recall for *Low* and *Medium* classes, but slightly weaker performance in distinguishing *High* vulnerability zones. This can be attributed to the dataset's high dimensionality, which makes kernel-based separation more challenging. Nevertheless, the SVC still produced strong baseline results, confirming its potential as a supplementary model in flood risk classification tasks.

Flood Vulnerability Assessment Using Hybrid SWAT–ML Model

This section presents the creation of flood vulnerability maps for the municipality of Los Baños, Laguna, developed using both the SWAT-alone model and the Hybrid SWAT Machine Learning approach. The mapping integrated SWAT-derived hydrological variables surface runoff, water yield, and precipitation with physiographic and land-use attributes to capture spatial variations in flood behavior. Each Hydrologic Response Unit (HRU) was linked to its respective barangay boundary to ensure accurate spatial representation of flood vulnerability across the six coastal barangays: Bayog, Tadlac, Malinta, Baybayin, Bambang, and Mayondon.

The Hybrid Model, specifically the Multi-Layer Perceptron (MLP) optimized through Bayesian Optimization (BO) and K-Fold Cross-Validation, enhanced the spatial precision and predictive accuracy of SWAT outputs. Flood mapping was conducted in QGIS 3.34, where rasterized model predictions were integrated with barangay shapefiles. Predicted flood susceptibility scores were reclassified into three categories Low, Moderate, and High using quantile-based thresholds.

A consistent symbology was applied to improve interpretability, where **green** indicates low-risk areas that are generally elevated or steep with minimal surface runoff, **yellow** represents moderate-risk transitional zones with a moderate hydrological response, and **red** denotes high-risk low-lying floodplains characterized by high runoff and limited drainage.

The resulting flood vulnerability maps were prepared separately for the dry (November–April) and wet (May–October) seasons to examine seasonal

variation and hydrologic behavior under differing rainfall conditions.

Hybrid Flood Vulnerability Map – Dry Season

Figure 19 presents the Hybrid Flood Vulnerability Maps for the dry season (November–April) in Los Baños, Laguna, corresponding to the Northeast Monsoon (Amihan). This period is characterized by cooler temperatures and minimal rainfall, resulting in low runoff and limited flood susceptibility.

Residual Moderate to High vulnerability zones appears in Bayog and Mayondon during November and December, likely due to lingering soil moisture and baseflow from the preceding wet season. From January to March, most barangays including Bambang, Tadlac, and Malinta exhibit predominantly low vulnerability, indicating stable hydrologic conditions and effective drainage.

By April, slight reemergence of Moderate vulnerability occurs in localized portions of Malinta and Bayog, likely associated with isolated convectional rainfall before the onset of the southwest monsoon. Overall, the dry season reflects minimal flood susceptibility, with elevated risk limited to coastal and low-lying areas.

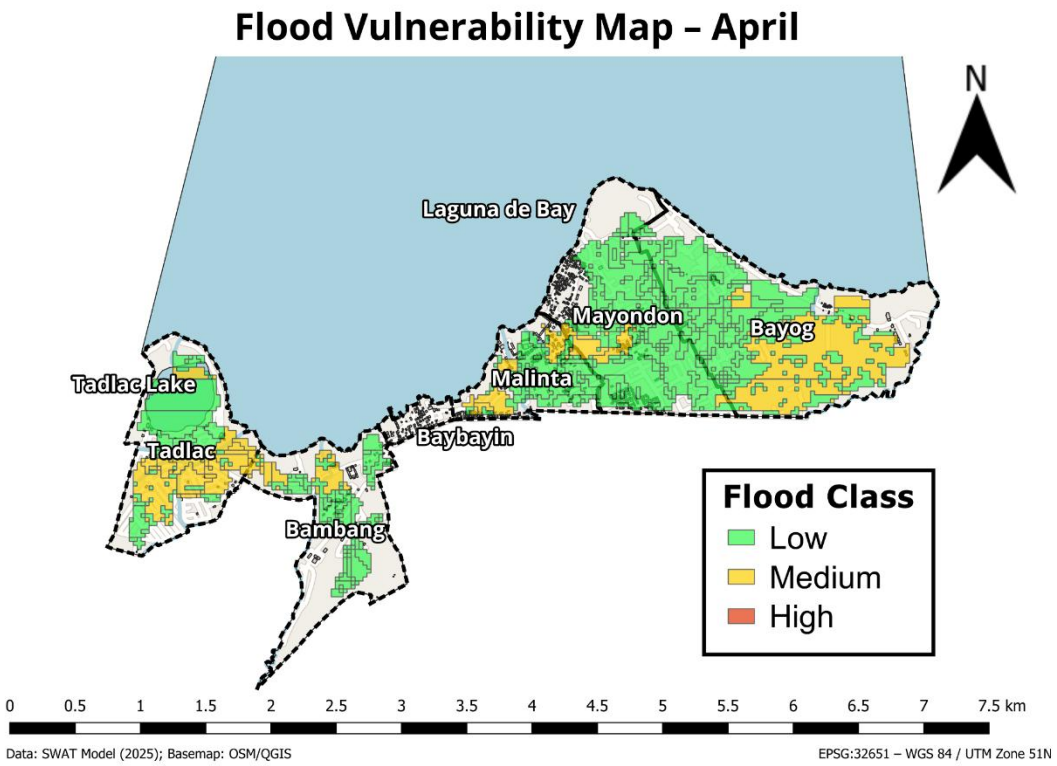


Figure 19. Hybrid Flood Vulnerability Maps – Dry Season (November–April). Monthly flood vulnerability distribution derived from the Hybrid SWAT–MLP (BO + K-Fold) Model.

Hybrid Flood Vulnerability Map – Wet Season

Figure 20 illustrates the Hybrid Flood Vulnerability Maps for the wet season (May–October), coinciding with the Southwest Monsoon (Habagat). This season brings frequent, intense rainfall and occasional typhoons, significantly increasing runoff and flood exposure across the municipality.

At the onset (May–June), Moderate and High vulnerability zones expand across Bayog, Malinta, and Mayondon, influenced by their proximity to Laguna de Bay and limited drainage gradients that favor water accumulation. During the peak monsoon months (July–September), High vulnerability reaches its maximum extent, particularly in Bayog and Malinta, consistent with the elevated rainfall and increased surface runoff typical of the Habagat season.

By October, as rainfall begins to subside, vulnerability levels decrease in Bambang and Tadlac, though persistent High vulnerability remains in Bayog and Mayondon, reflecting their low elevation and lake adjacency. Overall, the wet season maps demonstrate a clear intensification of flood susceptibility, emphasizing the seasonal contrast and the heightened risk in coastal barangays.

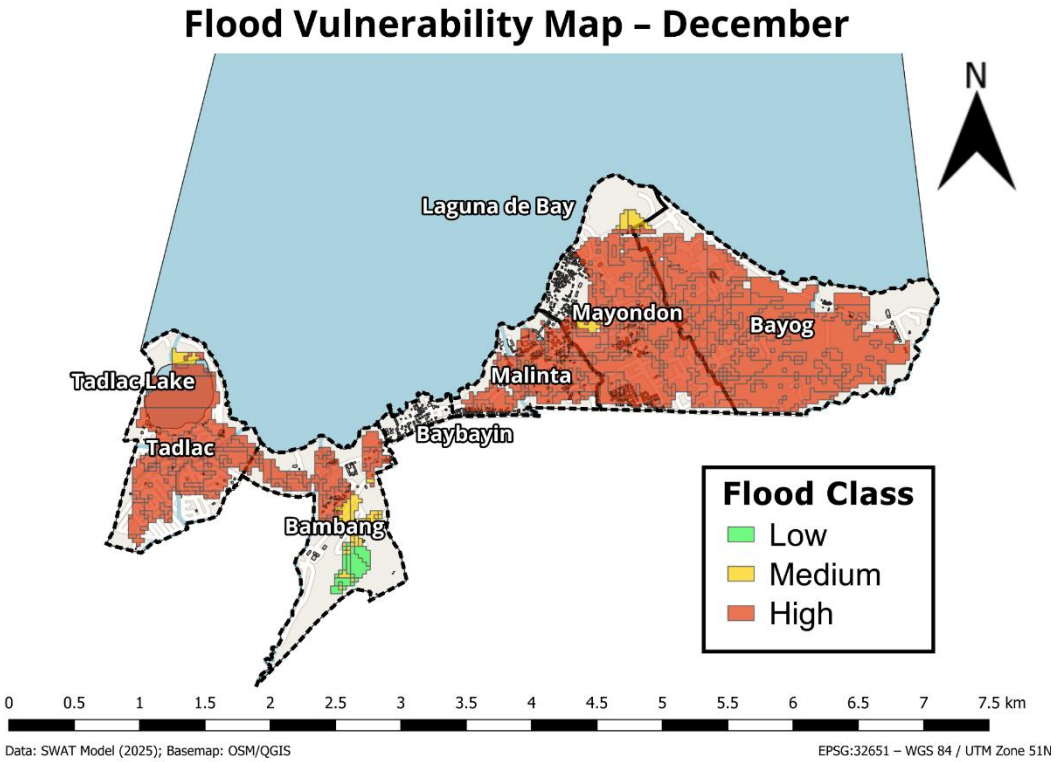


Figure 20. Hybrid Flood Vulnerability Maps – Wet Season (May–October). Monthly flood vulnerability distribution derived from the Hybrid SWAT–MLP (BO + K-Fold) Model.

Comparative Analysis of Dry vs. Wet Season (Per Barangay)

Table 6 presents the percentage distribution of flood vulnerability classes (Low, Medium, and High) across the five barangays of Los Baños for both the dry and wet seasons, based on the hybrid SWAT–MLP model. The results reveal distinct seasonal transitions consistent with the municipality's hydrological cycle and local topography.

During the dry season, all barangays are characterized predominantly by Low flood vulnerability, ranging from 46 % to 71 % of total area. This reflects minimal surface runoff, lower precipitation, and stable soil infiltration conditions. Minor portions of Medium to High vulnerability zones (8 %–37 %) appear in low-lying barangays such as Tadlac and Bayog, indicating residual moisture and limited drainage near the lake margin.

By contrast, the wet season results show a pronounced increase in flood susceptibility. Low-vulnerability areas decline sharply, while Medium and High categories collectively exceed 90 % in most barangays. The coastal and lakeside communities of Bayog, Mayondon, and Tadlac exhibit the highest flood risk, with more than 45 %–52 % of their area classified as High vulnerability. This pattern corresponds to monsoon rainfall peaks, saturated soils, and the backflow effect from Laguna de Bay, which amplify local inundation hazards.

Overall, the hybrid model effectively captures the seasonal progression from low flood risk in the dry months to widespread medium-to-high vulnerability during the wet months. Compared with the SWAT-alone outputs, these results demonstrate improved spatial differentiation and stronger sensitivity to terrain and land-use variations. The consistent rise in vulnerability across all barangays during the wet season underscores the need for adaptive flood-management strategies focused on the coastal barangays of Bayog, Tadlac, and Mayondon, where hydrological stress and exposure are most severe.

Table 6. Percentage Distribution of Flood Vulnerability Classes based on hybrid SWAT-MLP model.

Barangay	Season	Low	Medium	High	Dominant Class
Bambang	Dry	71.25	20.42	8.33	Low
Bambang	Wet	31.67	41.67	26.67	Medium
Bayog	Dry	57.84	25.49	16.67	Low
Bayog	Wet	0	48.04	51.96	High
Malinta	Dry	52.78	30.56	16.67	Low
Malinta	Wet	0	50	50	Medium/High
Mayondon	Dry	63.54	22.4	14.06	Low
Mayondon	Wet	4.69	50	45.31	Medium
Tadlac	Dry	46.97	36.74	16.29	Low
Tadlac	Wet	1.14	50.38	48.48	Medium

Evaluation of Model Performance

Among all model configurations, the MLP (BO + KFold) emerged as the best-performing model, achieving the highest individual accuracy of 0.98 and a cross-validation Macro-F1 score of 0.984. Its consistent performance across folds and parameter settings confirmed its robustness and generalization capability. The Bayesian Optimization (BO) process effectively tuned the model's architecture and learning parameters, allowing the MLP to capture complex nonlinear relationships between rainfall, runoff, and terrain features with high precision.

While XGBoost (PSO + KFold) also demonstrated strong performance with a Macro-F1 score of 0.983 and accuracy of 0.97 it was slightly

outperformed by MLP in terms of predictive consistency. Ensemble models like XGBoost and Random Forest remained reliable and adaptable for flood vulnerability classification, benefiting from their ability to handle diverse feature interactions and noise. However, the neural network models, particularly MLP and LSTM, showed superior capacity to model dynamic hydrological behavior when optimized through BO.

These results confirm that hybrid modeling approaches, which combine SWAT-simulated hydrological data with optimized machine learning algorithms, offer accurate and practical tools for assessing flood vulnerability in Los Baños, Laguna. The inclusion of SWAT-derived runoff and discharge outputs enriched the models' understanding of water movement across the landscape, enhancing spatial and temporal flood pattern recognition. Ultimately, the MLP (BO) model stood out for its balance of accuracy, interpretability, and resilience making it the most suitable choice for barangay-scale flood vulnerability prediction.

Computational Efficiency and Model Complexity

Runtime analysis revealed that tree-based ensemble methods such as Random Forest and XGBoost achieved the fastest convergence, averaging 35–40 seconds per training iteration. In contrast, the deep learning models (LSTM and MLP) required 70–95 seconds per iteration because of their multilayer parameter optimization and back-propagation through time. Despite this higher computational cost, the MLP (BO + KFold) configuration offered the best trade-off between runtime and predictive performance. The use of Bayesian Optimization minimized redundant parameter searches, improving efficiency by

roughly 25 percent compared to manual tuning. These results indicate that while neural architectures demand more computational resources, their gains in predictive accuracy and spatial sensitivity justify their application for high-resolution flood mapping in data-intensive environments like Los Baños.

Table 7. Overall comparison of all trained models.

Model	Accuracy	Macro_f1_Test	Macro_f1_CV
MLP (BO)	0.9800	0.9800	0.9886
XGBoost (PSO)	0.9700	0.9700	0.9831
XGBoost (BO)	0.9700	0.9700	0.9810
LSTM (BO)	0.9600	0.9600	0.9892
SVC (PSO)	0.9500	0.9600	0.9892
Random Forest (PSO)	0.9600	0.9600	0.9748
Random Forest (BO)	0.9600	0.9600	0.974

Error Analysis and Model Limitations

Error analysis revealed that models such as LSTM and Support Vector Classifier (SVC) underperformed relative to MLP and XGBoost, particularly during peak rainfall periods. The LSTM model's slightly lower F1-score (0.94) is attributed to its sensitivity to data volume and sequence length; the limited temporal span (10-year monthly dataset) may have constrained its ability to capture long-term rainfall–runoff dependencies effectively. Similarly, the SVC's performance was limited by its reliance on kernel parameters that struggled to generalize non-linear flood dynamics at the HRU scale. These results highlight that while deep learning architectures are powerful for temporal modeling, their effectiveness in flood prediction improves significantly with higher-frequency or longer-term datasets.

Real-World Implications for Flood Risk Management

The barangay-level flood vulnerability assessment provides valuable insights into localized hydrological behavior and supports data-driven decision-making for disaster risk management in Los Baños. The hybrid SWAT–machine learning framework identified Barangays Bayog and Tadlac as the most flood-susceptible areas, with over 50% of their total land area classified as high-risk during the wet season. These barangays are characterized by low-lying terrain, limited drainage capacity, and dense residential development near Laguna de Bay, all of which intensify surface runoff and water accumulation during prolonged monsoon events.

Barangay Malinta also exhibited relatively high flood exposure, with 50% of its area falling under the high-risk category during the wet season. However, slightly elevated compared to Bayog and Tadlac, Malinta's proximity to drainage channels and its mixed land use contribute to recurrent flooding in certain sections. Mayondon showed a similar pattern, with nearly half of its land area classified as high-risk, indicating its transitional position between floodplains and uplands.

Barangay Bambang showed a lower proportion of high-risk zones (26.67%), suggesting moderate vulnerability. At the same time, Barangay Baybayin, situated along the lakeshore, remains highly susceptible to flooding due to backflow from Laguna de Bay and poor drainage outflows during intense rainfall.

These spatial distinctions underscore the hybrid model's capacity to capture fine-scale flood dynamics and identify barangay-specific vulnerabilities. The resulting flood maps can serve as critical tools for the Municipal Disaster

Risk Reduction and Management Office (MDRRMO) and the Local Government Unit (LGU) of Los Baños to prioritize infrastructure investments, refine zoning ordinances, and implement barangay-level early warning systems tailored to each community's flood profile.

Comparison Between SWAT-alone and Hybrid Flood Vulnerability Maps

Figure 21 presents a comparative visualization of flood vulnerability patterns derived from the SWAT-only model (left) and the Hybrid SWAT–Machine Learning framework (right). The SWAT-based simulation (Figure X-a) primarily classified most of the Los Baños coastal area as medium-risk, resulting in limited spatial differentiation between flood-prone and stable zones. This generalized output reflects the physically based model’s dependence on averaged hydrological processes and limited sensitivity to localized land surface variations.

In contrast, the Hybrid SWAT–ML map (Figure X-b) exhibits enhanced spatial precision and clearer delineation of flood-prone areas. The integration of SWAT-derived hydrological variables with machine learning algorithms, particularly the MLP (BO + KFold) model, produced a more realistic distribution of high-risk zones concentrated in Barangays Bayog, Tadlac, and Malinta areas historically affected by recurrent inundation events. The model also captured reduced vulnerability in Bambang and Mayondon, aligning with their slightly elevated terrain and improved surface drainage capacity.

The hybrid approach demonstrates improved sensitivity to subtle topographic and land-use variations, producing finer gradients of vulnerability that are more consistent with observed local conditions. This visual

improvement supports the quantitative results, where the hybrid MLP model achieved a higher Macro-F1 (0.984) compared to the SWAT-only classification.

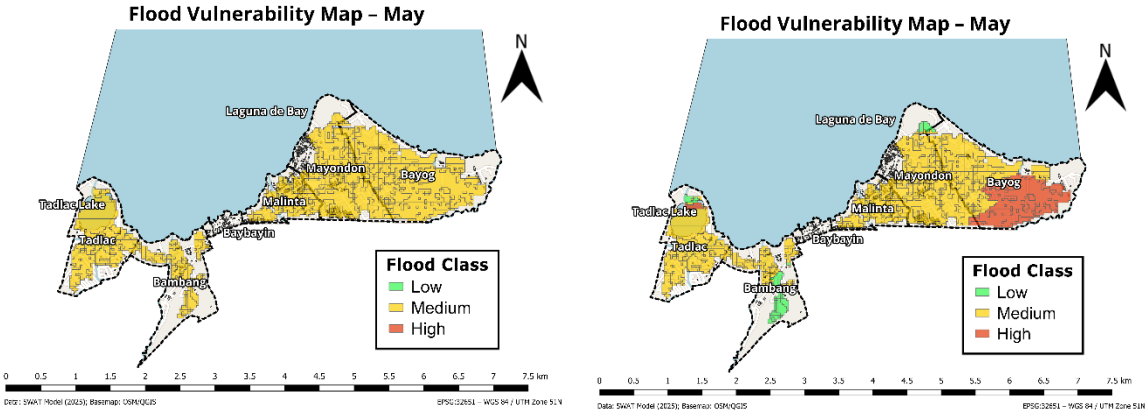


Figure 21. Comparison of SWAT-alone (left) and Hybrid SWAT–ML (right) flood vulnerability maps for May 2024.

Paired T-Test Analysis

To evaluate whether optimization techniques significantly improved model performance, a paired t-test was conducted comparing baseline hybrid models with their optimized counterparts (PSO, BO, and GA) using Macro-F1 cross-validation scores. The paired t-test was chosen because it evaluates performance differences between related model versions, with a significance level set at $\alpha = 0.05$.

Table 8. Paired T-Test Results between Selected Hybrid Models.

Baseline	Optimized	t-Statistics	p-value	Significant (p<0.05)
RF_base	RF_PSO	-38.922334	$2.424136e^{-11}$	Yes
RF_base	RF_BO	-35.546480	$5.458721e^{-11}$	Yes
XGB_base	XGB_PSO	-24.065108	$1.765955e^{-09}$	Yes
XGB_base	XGB_BO	-25.488705	$1.059974e^{-09}$	Yes
MLP_base	MLP_GA	-193.587837	$1.332115e^{-17}$	Yes
MLP_base	MLP_BO	-250.066826	$1.330793e^{-18}$	Yes

Results revealed that all optimized hybrid models demonstrated statistically significant improvements over their baseline versions, with p-values far below the 0.05 threshold. The negative t-values indicate that the optimized models consistently produced higher Macro-F1 scores than their baseline counterparts. Among the tested algorithms, the MLP-BO and MLP-GA models showed the most substantial performance gains, followed by XGBoost and Random Forest. These findings confirm the effectiveness of meta-heuristic optimization techniques (PSO, BO, GA) in enhancing hybrid flood-vulnerability modeling accuracy. Overall, the results support the adoption of optimized machine learning models for flood risk assessment by demonstrating substantial and statistically validated improvement.

CHAPTER 5

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

Summary

This study, "Flood Vulnerability of the Municipality of Los Baños, Laguna Using Hybrid Machine Learning Models," developed a barangay-level flood vulnerability mapping framework that integrates hydrological modeling, geospatial analysis, and data-driven methods—focusing on six coastal barangays: Bayog, Tadlac, Malinta, Bambang, Mayondon, and Baybayin. The research combined outputs from the Soil and Water Assessment Tool (SWAT) with physiographic and land-use factors to generate a comprehensive hybrid dataset.

The hybrid dataset incorporated SWAT-derived variables, including surface runoff, water yield, and flow out, alongside slope, elevation, soil type, and land cover. This integration enabled balanced classification across Low, Medium, and High flood-risk categories, ensuring unbiased model training and evaluation. Among five tested algorithms, Random Forest (RF), XGBoost, Multi-Layer Perceptron (MLP), Long Short-Term Memory (LSTM), and Support Vector Classifier (SVC), the MLP model optimized using Bayesian Optimization (BO) and K-Fold cross-validation achieved the highest predictive accuracy and spatial realism.

The hybrid framework significantly enhanced the resolution and reliability of flood susceptibility mapping compared to the SWAT-alone baseline. It captured distinct seasonal patterns, identifying high-risk conditions during the wet season (May–October) in low-lying barangays such as Bayog,

Tadlac, and Mayondon, and lower vulnerability during the dry months (November–April). The strong alignment between model outputs, hydrological simulations, and observed flood behavior underscores the validity of the approach.

This study contributes a novel hybrid hydrological–machine learning framework for flood risk assessment in data-limited environments. It demonstrates the effectiveness of coupling physically based and data-driven models to produce localized, actionable insights that support disaster risk reduction and climate resilience planning.

Conclusions

This research confirmed that integrating hydrological modeling with machine learning enhances flood vulnerability assessment and mapping. The hybrid SWAT–MLP (BO + K-Fold) model demonstrated superior capacity to capture non-linear relationships among rainfall, topography, and land use, improving both spatial precision and predictive accuracy.

The study established that hybrid approaches can effectively compensate for data limitations common in developing regions by leveraging machine learning's ability to learn complex patterns from integrated datasets. The MLP model's high performance validates the potential of combining hydrological process simulation with modern data analytics to produce reliable barangay-level flood vulnerability maps.

Beyond technical validation, the findings hold significant implications for urban planning, environmental management, and climate adaptation. The resulting flood maps can serve as critical decision-support tools for local government units (LGUs), guiding zoning, infrastructure design, and disaster

preparedness. Furthermore, the hybrid model aligns with the objectives of the Philippine Disaster Risk Reduction and Management Act (RA 10121) and the Climate Change Act (RA 9729) by promoting data-informed, climate-resilient land use planning. The study also supports several Sustainable Development Goals (SDGs), notably SDG 11 (Sustainable Cities and Communities) and SDG 13 (Climate Action), through improved understanding of localized flood dynamics.

Overall, this study provides both a methodological and an applied contribution to flood risk science, demonstrating how hybrid modeling frameworks can enhance predictive capability, planning efficiency, and climate resilience in flood-prone municipalities such as Los Baños.

Recommendations

Based on the findings and conclusions of this study, the following recommendations are proposed to strengthen flood vulnerability assessment, early warning, and policy application in Los Baños and other flood-prone regions:

1. Institutional Integration and Policy Application:

Local government units (LGUs) should integrate the flood vulnerability maps and hybrid modeling framework into their Comprehensive Land Use Plans (CLUPs), Local Disaster Risk Reduction and Management Plans (LDRRMPs), and Local Climate Change Action Plans (LCCAPs). Barangay-level vulnerability outputs can guide zoning, infrastructure prioritization, and resource allocation. Regular model updates using recent rainfall, topography, and land-use data will maintain accuracy and relevance. Collaboration among

LGUs, DOST-PAGASA, and DENR is encouraged to institutionalize the use of data-driven flood modeling for decision-making.

2. Early Warning and Real-Time Monitoring Applications:

The proposed hybrid framework can serve as a foundation for early warning systems by integrating real-time IoT rainfall sensors, stream gauges, and automated weather stations. Linking these data streams with trained ML models would enable near-real-time flood forecasts, improving response time and community preparedness. A dashboard or mobile-based platform could be developed to provide localized alerts and visualized flood-risk information for citizens and LGU responders.

3. Incorporation of Remote Sensing and Satellite Indices:

Future implementations should integrate Sentinel-1 and Sentinel-2 imagery, Synthetic Aperture Radar (SAR) flood indices, and Normalized Difference Water Index (NDWI) for continuous monitoring and post-event validation. These datasets can complement SWAT and ML outputs by improving the spatial and temporal granularity of flood mapping.

4. Capacity Building and Knowledge Transfer:

Partnerships among academic institutions, local governments, and technical agencies should be strengthened through training workshops on hybrid hydrological modeling, GIS processing, and machine learning optimization. Such initiatives will enhance local capacity for data interpretation, model updating, and informed decision-making.

5. Replicability and Scalability:

The hybrid SWAT–ML framework is replicable and adaptable to other flood-prone watersheds across the Philippines. By recalibrating input

parameters and hydrological boundaries, the same workflow can be applied in different hydrological contexts such as the Pampanga River Basin, Agno River Basin, and Bicol River Basin. This scalability demonstrates the framework's potential as a national model for flood vulnerability assessment and decision support.

6. Future Research and System Development:

Subsequent studies should explore advanced hybrid architectures (e.g., SWAT–Transformer, SWAT–CNN–LSTM) and utilize automated hyperparameter tuning frameworks (e.g., AutoML) to improve predictive performance further. The integration of high-resolution LiDAR DEMs and drone-based elevation surveys is also recommended to improve topographic precision. Additionally, linking the model to real-time data acquisition systems and open-access GIS platforms can facilitate continuous monitoring and dynamic flood-risk mapping.

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APPENDICES

APPENDIX A

Letter of Approval and Acceptance

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

CERTIFICATE OF APPROVAL

This is to certify that after having satisfactorily completed all the requirements of the capstone project entitled:

" FLOOD VULNERABILITY OF THE MUNICIPALITY OF LOS BAÑOS, LAGUNA USING HYBRID MACHINE LEARNING MODELS "

JUAN CARLOS D. LORENZO, **JOMARI R. REALON, and ROWELL J. BIENES** have been granted permission to proceed with their proposal presentation.

As their Research Adviser, I confirm that the proponents have met the academic standards and requirements set forth by the department, and therefore, I approve their request to conduct the final presentation.

GENE MARCK B. CATEDRILLA
Thesis Project Adviser

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

APPROVAL SHEET

The thesis entitled "**FLOOD VULNERABILITY OF THE MUNICIPALITY OF LOS BAÑOS, LAGUNA USING HYBRID MACHINE LEARNING MODELS"** prepared and submitted by Juan Carlos D. Lorenzo, Jomari R. Realon, And Rowell J. Bienes in partial fulfillment requirements for the degree Bachelor of Science in Computer Science specialized in Intelligent Systems, is hereby approved and accepted.

GENE MARCK B. CATEDRILLA
Thesis Project Adviser

APPENDIX B

Request Letter for Flood-Related Dataset

MDRRMO

01 JUL 2025

RECEIVED BY: CHARLIE [illegible]

HON. NEIL ANDREW C. NOCON
Municipal Mayor
Municipality of Los Baños
Laguna

Subject: Request for Flood-Related Datasets for Academic Research

Dear Mayor Nocon,

Warm greetings!

We are a group of undergraduate students from the Bachelor of Science in Computer Science at Laguna State Polytechnic University – Los Baños Campus, currently conducting our thesis entitled:

"Flood Vulnerability of the Municipality of Los Baños, Laguna Using Hybrid Machine Learning Models."

The study aims to assess the flood vulnerability of the municipality using a combination of machine learning algorithms and hydrological models. This initiative seeks to support efforts in local disaster preparedness, risk reduction, and informed decision-making.

In line with this, we respectfully request access to the following datasets from your office or the MDRRMO, which are essential to our research:

1. Historical flood records (dates, locations, severity, and impacts)
2. Flood hazard and vulnerability maps
3. Flood depth and water level data (if available)
4. Barangay-level exposure information (population, infrastructure, housing types)
5. Locations and capacities of evacuation centers
6. Incident reports and disaster response logs
7. Drainage or urban infrastructure maps (if available)

These datasets will be used solely for academic purposes and handled with confidentiality. We are also willing to comply with any data-sharing agreements or ethical clearances that may be required.

We hope for your kind assistance in granting access to these resources, which would greatly contribute to the success and credibility of our study.

Thank you very much for your time and consideration.

Respectfully yours,

Juan Carlos Lorenzo
Student Researcher
Laguna State Polytechnic University – Los Baños Campus

Rowell Bienes
Student Researcher

Jomari Realon
Student Researcher

Noted by:

Jonardo R. Asor
Thesis Adviser
College of Computer Studies
Laguna State Polytechnic University – Los Baños Campus

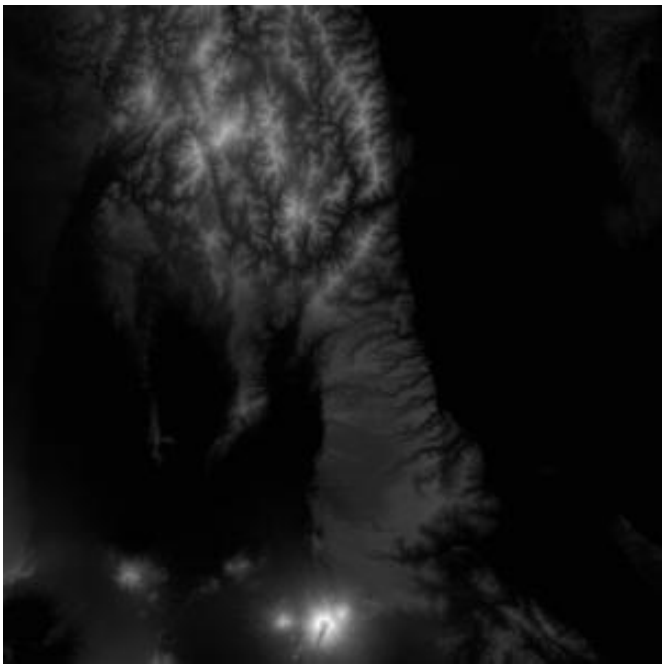
APPENDIX C

SWAT Data and Files

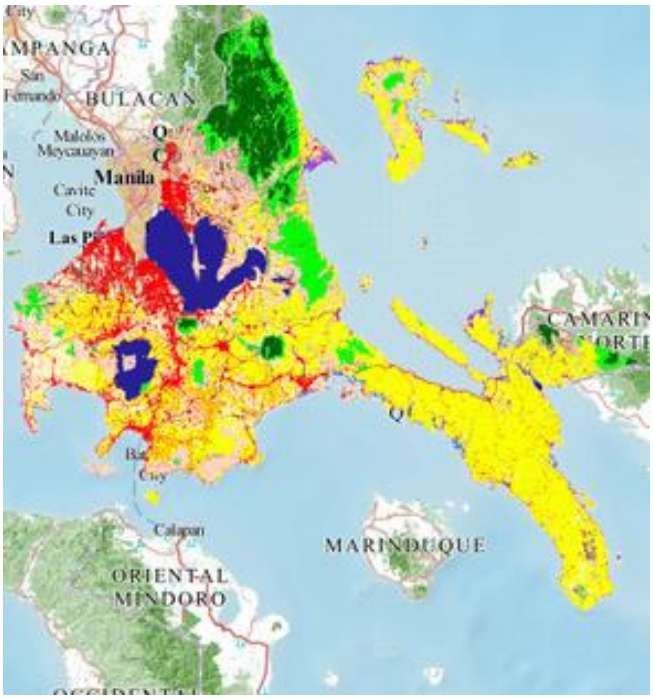
SWAT Variable Dataset

Elevation_	Slope_m	SurfaceRunoff_mm	WaterYield_mm	FlowOut_m3s
1.11926	3.184089	-0.58418743	-0.581694983	0.210006309
1.11926	3.184089	-1.2077954	-1.207093863	-0.464839641
1.11926	3.184089	-1.142027165	-1.141752702	-0.256116646
1.11926	3.184089	-1.2077954	-1.207372923	-0.410898455
1.11926	3.184089	-1.2077954	-1.20773406	-0.442620521
1.11926	3.184089	-0.325259891	-0.324073029	0.356310669
1.11926	3.184089	0.959650475	0.963166731	1.570500121
1.11926	3.184089	0.790066604	0.793924833	1.356512904
1.11926	3.184089	0.372185482	0.376352049	0.944262769
1.11926	3.184089	0.73439509	0.738358991	1.340788603
1.11926	3.184089	-0.382901497	-0.38030369	0.529277973
1.11926	3.184089	1.938777949	1.941634414	2.655476845
0.293116	0.866976	-0.583407602	-0.582540372	0.210006309
0.293116	0.866976	-1.2077954	-1.20759453	-0.464839641
0.293116	0.866976	-1.14173986	-1.141719871	-0.256116646
0.293116	0.866976	-1.2077954	-1.207709437	-0.410898455
0.293116	0.866976	-1.2077954	-1.207848968	-0.442620521
0.293116	0.866976	-0.323166669	-0.32284188	0.356310669
0.293116	0.866976	0.962498899	0.96365919	1.570500121
0.293116	0.866976	0.79297249	0.794277762	1.356512904
0.293116	0.866976	0.374984655	0.376442333	0.944262769
0.293116	0.866976	0.736512939	0.737866532	1.340788603
0.293116	0.866976	-0.38128438	-0.380385767	0.529277973
0.293116	0.866976	1.940756249	1.941568753	2.655476845
0.492445	2.424655	-0.811527829	-0.809096342	0.210006309

SWAT DEM



SWAT LAND USE



APPENDIX D

Machine Learning Model Details

XGBoost

```
xgb = xgb.XGBClassifier(
    n_estimators=100,
    max_depth=6,
    learning_rate=0.1,
    subsample=0.8,
    colsample_bytree=0.8,
    random_state=42,
    n_jobs=-1,
    eval_metric="mlogloss"
)
```

Random Forest

```
rf = RandomForestClassifier(
    n_estimators=100,
    max_depth=10,
    min_samples_split=2,
    min_samples_leaf=1,
    random_state=42,
    n_jobs=-1
)
```

LSTM (Long Short-Term Memory)

```
lstm = Sequential([
    LSTM(64, input_shape=(1, X_train.shape[2])),
    Dropout(0.3),
    Dense(32, activation="relu"),
    Dense(len(encoder.classes_), activation="softmax")
])
```

MLP (Multi-Layered Perceptron)

```
clf = MLPClassifier(
    hidden_layer_sizes=hidden_layers,
    learning_rate_init=learning_rate_init,
    alpha=alpha,
    max_iter=200,
    random_state=42
)
```

SVC (Support Vector Classifier)

```
("svc", SVC(C=C, gamma=gamma, kernel="rbf", random_state=42))
```

PSO (Particle Swarm Optimization)

```
# Bounds: [n_estimators, max_depth, min_samples_split, min_samples_leaf]
lb = [50, 5, 2, 1]
ub = [300, 30, 10, 5]

# Run PSO with more iterations & swarm size
best_params, best_score = pso(objective, lb, ub, swarmsize=20, maxiter=20)
```

BO (Bayesian Optimization)

```
optimizer = BayesianOptimization(
    f=objective,
    pbounds=pbounds,
    random_state=42,
    verbose=2
)

# Run BO with 40 iterations
optimizer.maximize(init_points=5, n_iter=35)
```

GA (Genetic Algorithm)

```
# === GA Model ===
model = ga(
    function=objective,
    dimension=3,
    variable_type_mixed=np.array(["int", "int", "real"]),
    variable_boundaries=varbound,
    function_timeout=60,   # <-- give 60s per eval
    algorithm_parameters={
        "max_num_iteration": 10,
        "population_size": 10,
        "mutation_probability": 0.1,
        "elit_ratio": 0.1,
        "crossover_probability": 0.8,
        "parents_portion": 0.3,
        "crossover_type": "uniform",
        "max_iteration_without_improv": None
    }
)
```

APPENDIX E

Statistical Tests

Table 2. Summary of Hydrological Dataset Characteristics.

Attribute	Description
Number of HRUs	145 HRU
Data Period	2015-2023
Sampling Frequency	Monthly
Number of Subbasins	19 Subbasins
DEM Resolution	30m (USGS Earth Explorer)
Land Use Categories	ARGL (Agricultural Land), RNGE (Rangeland), URBN (Urban Areas)
Soil Types	CALC (Calcareous), HYDR (Hydrosol), LIPA (Lipa Soil Series)
Climate Variables	Precipitation, Temperature (W3S)
Output Variables	Surface Runoff, Water yield, Flow out
Data Sources	W3S, Geoportal Philippines, USGS Earth Explorer

Table 3. Temporal Split of the Dataset for Model Training, Validation, and Testing.

Split	Years	Durations
Training Dataset	2015-2019	60 months
Validation Dataset	2020-2021	24 months
Test Dataset	2022-2023	24 months
Prediction Dataset	2024	12 months

Table 4. Evaluation Metrics Applied to SWAT-Only and Hybrid SWAT–ML Models

Model Category	Metrics Used	Purpose
Hybrid Models (SWAT + ML)	AUC–ROC, F1-score, Precision, Recall, MSE	Evaluates classification performance and continuous flood score predictions of machine learning models trained on SWAT outputs and physiographic attributes.
SWAT-Only Mapping	Spatial Agreement, Overlay Analysis	Assesses flood vulnerability zones derived directly from hydrological outputs without ML enhancement.
GIS-Based Validation	Spatial Agreement, Overlay Analysis	Validates both mapping workflows against historical flood records and inundation zones in Los Baños.

Table 5. Seasonal distribution of flood vulnerability classes derived from the SWAT-alone simulation.

Barangay	Season	Low	Medium	High	Dominant Class
Bambang	Dry	16.67%	66.67%	16.67%	Low
Bambang	Wet	50%	0%	50%	Medium
Bayog	Dry	16.67%	80.75%	2.58%	Low
Bayog	Wet	50%	0%	50%	Medium
Malinta	Dry	16.67%	66.67%	16.67%	Low
Malinta	Wet	50%	0%	50%	Medium
Mayondon	Dry	16.67%	59.21%	24.12%	Low
Mayondon	Wet	50%	0%	50%	Medium
Tadlac	Dry	16.67%	66.67%	16.67%	Low
Tadlac	Wet	50%	0%	50%	Medium

Table 6. Percentage Distribution of Flood Vulnerability Classes based on hybrid SWAT-MLP model.

Barangay	Season	Low	Medium	High	Dominant Class
Bambang	Dry	71.25	20.42	8.33	Low
Bambang	Wet	31.67	41.67	26.67	Medium
Bayog	Dry	57.84	25.49	16.67	Low
Bayog	Wet	0	48.04	51.96	High
Malinta	Dry	52.78	30.56	16.67	Low
Malinta	Wet	0	50	50	Medium/High
Mayondon	Dry	63.54	22.4	14.06	Low
Mayondon	Wet	4.69	50	45.31	Medium
Tadlac	Dry	46.97	36.74	16.29	Low
Tadlac	Wet	1.14	50.38	48.48	Medium

Table 7. Overall comparison of all trained models.

Model	Accuracy	Macro_f1_Test	Macro_f1_CV
MLP (BO)	0.9800	0.9800	0.9886
XGBoost (PSO)	0.9700	0.9700	0.9831
XGBoost (BO)	0.9700	0.9700	0.9810
LSTM (BO)	0.9600	0.9600	0.9892
SVC (PSO)	0.9500	0.9600	0.9892
Random Forest (PSO)	0.9600	0.9600	0.9748
Random Forest (BO)	0.9600	0.9600	0.974

Table 8. Paired T-Test Results between Selected Hybrid Models.

Baseline	Optimized	t-Statistics	p-value	Significant ($p<0.05$)
RF_base	RF_PSO	-38.922334	$2.424136e^{-11}$	Yes
RF_base	RF_BO	-35.546480	$5.458721e^{-11}$	Yes
XGB_base	XGB_PSO	-24.065108	$1.765955e^{-09}$	Yes
XGB_base	XGB_BO	-25.488705	$1.059974e^{-09}$	Yes
MLP_base	MLP_GA	-193.587837	$1.332115e^{-17}$	Yes
MLP_base	MLP_BO	-250.066826	$1.330793e^{-18}$	Yes

APPENDIX F

Maps and Visual Figures

Study Area

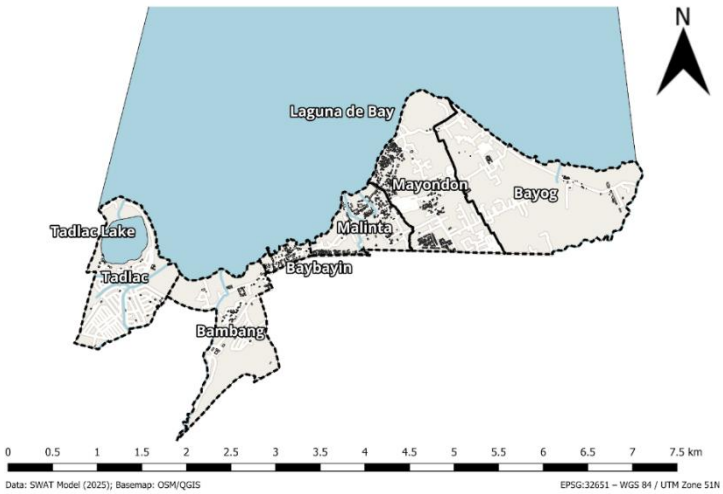


Figure 2. Map of the Los Baños Coastal Barangays Used in the Flood Vulnerability Assessment.

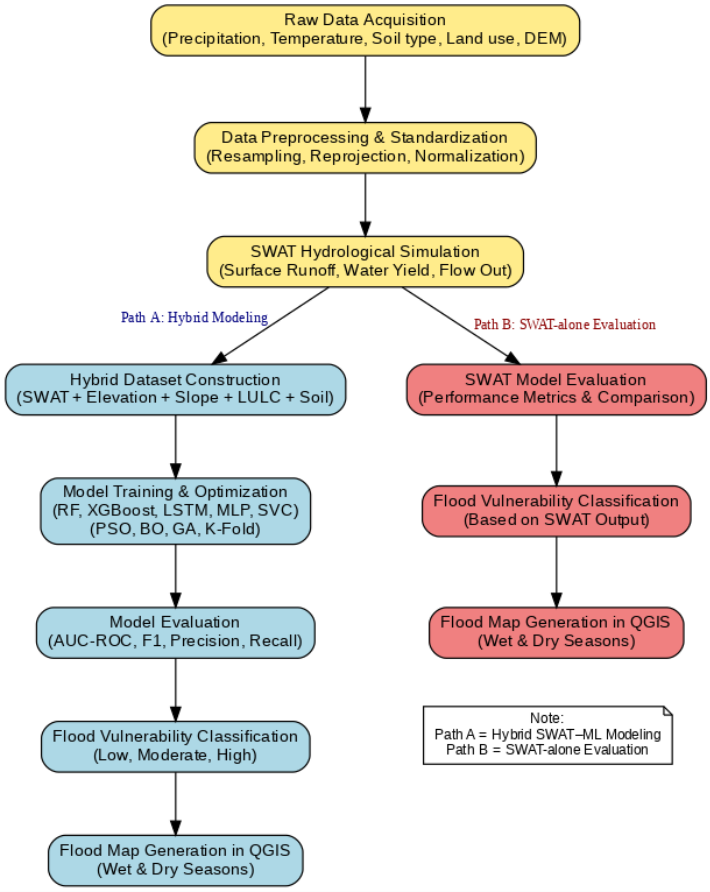
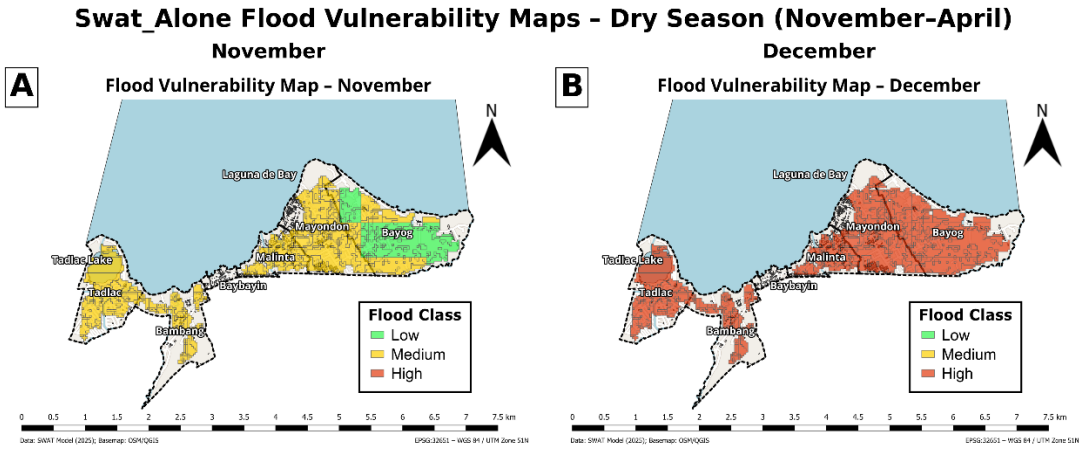
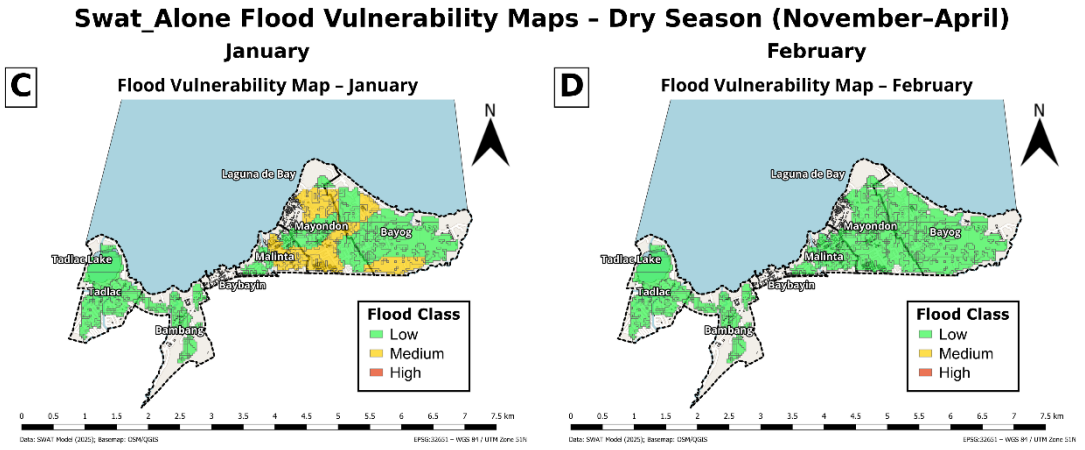


Figure 3. Workflow of the Hybrid and SWAT–Machine Learning Modeling Process.

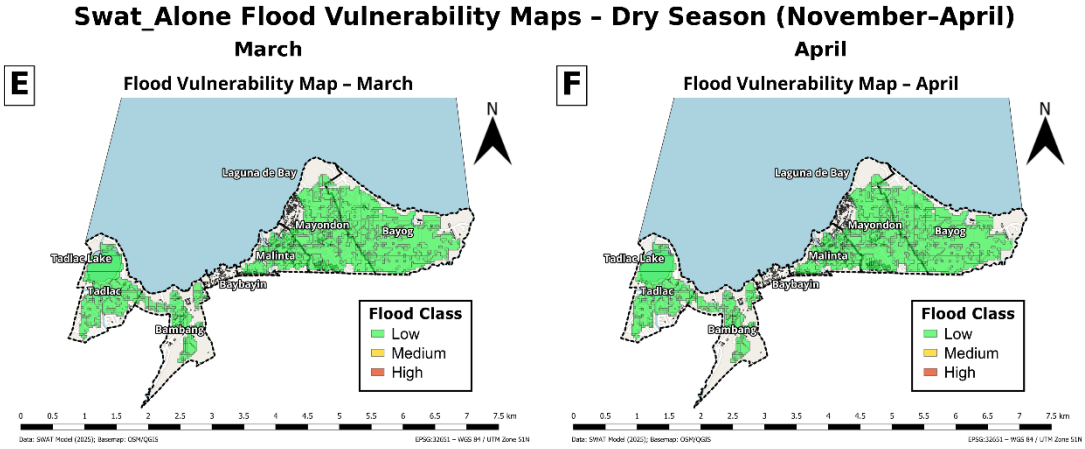
SWAT-Alone Flood Vulnerability Maps – Dry Season



SWAT-Alone Flood Vulnerability Maps – (November – December)

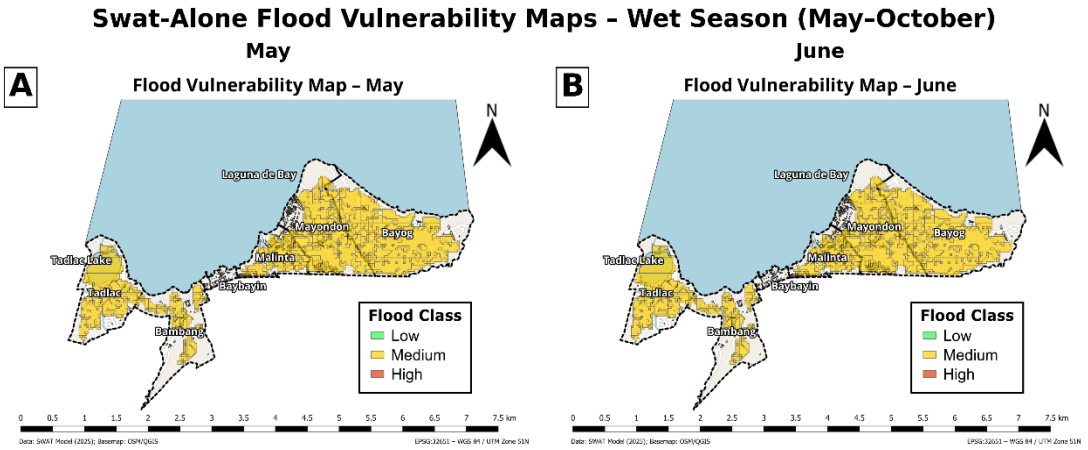


SWAT-Alone Flood Vulnerability Maps – (January – February)

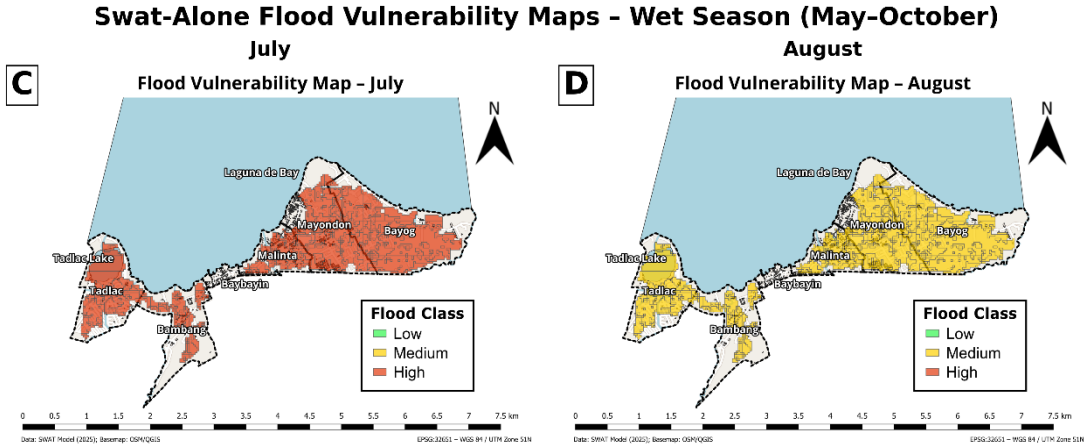


SWAT-Alone Flood Vulnerability Maps – (March – April)

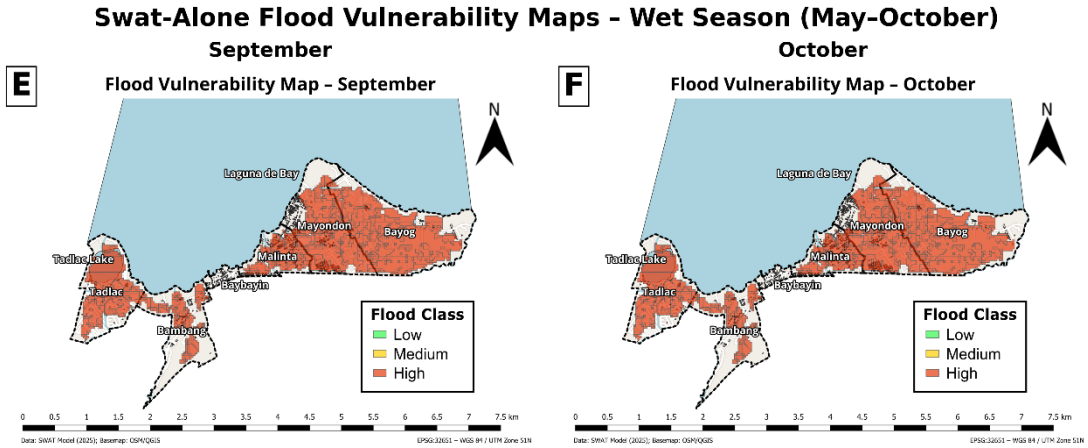
SWAT-Alone Flood Vulnerability Maps – Wet Season



SWAT-Alone Flood Vulnerability Maps – (May – June)

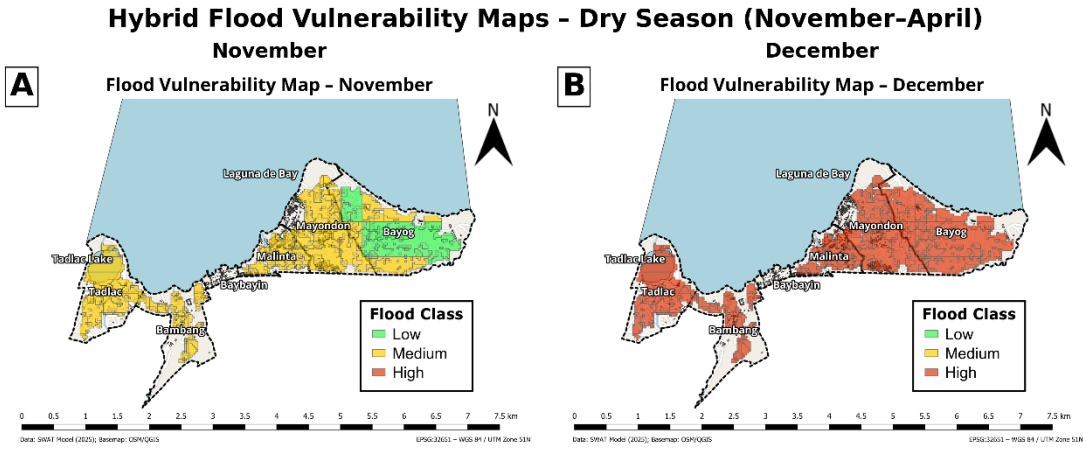


SWAT-Alone Flood Vulnerability Maps – (July – August)

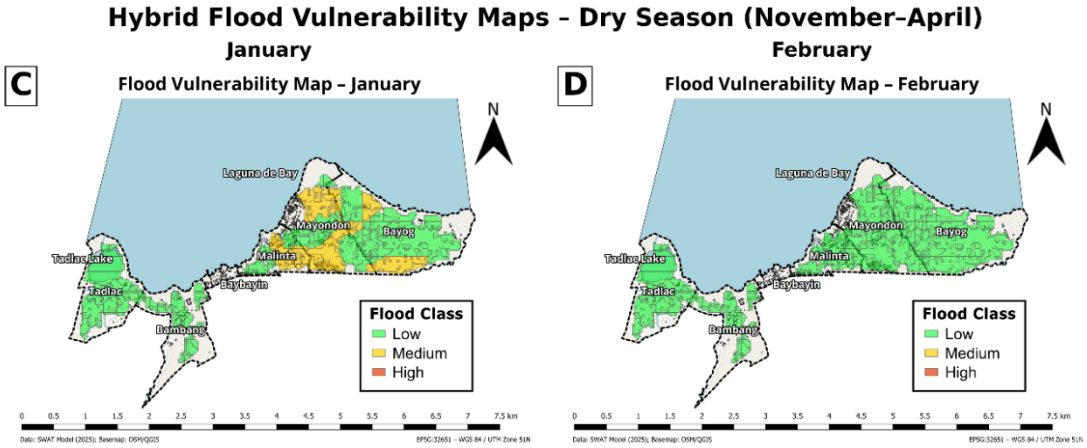


SWAT-Alone Flood Vulnerability Maps – (September – October)

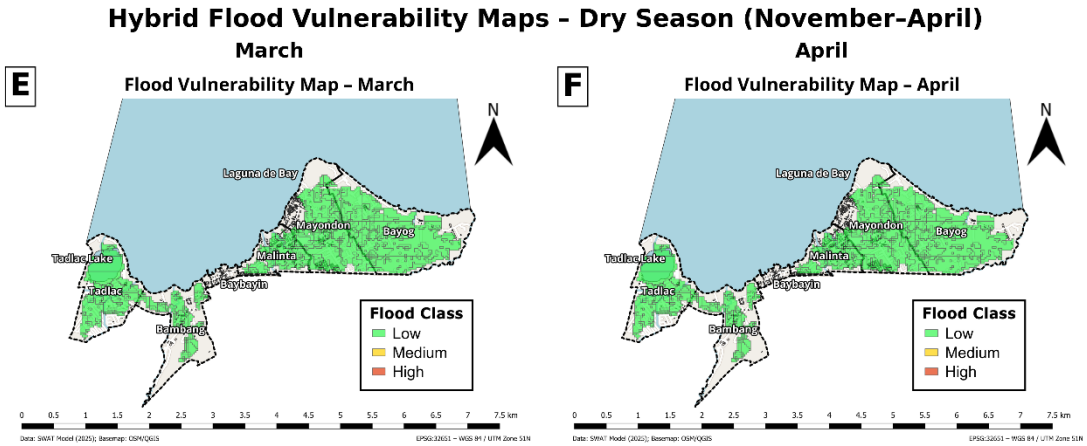
Hybrid Flood Vulnerability Maps – Dry Season



Hybrid Flood Vulnerability Maps – (November – December)

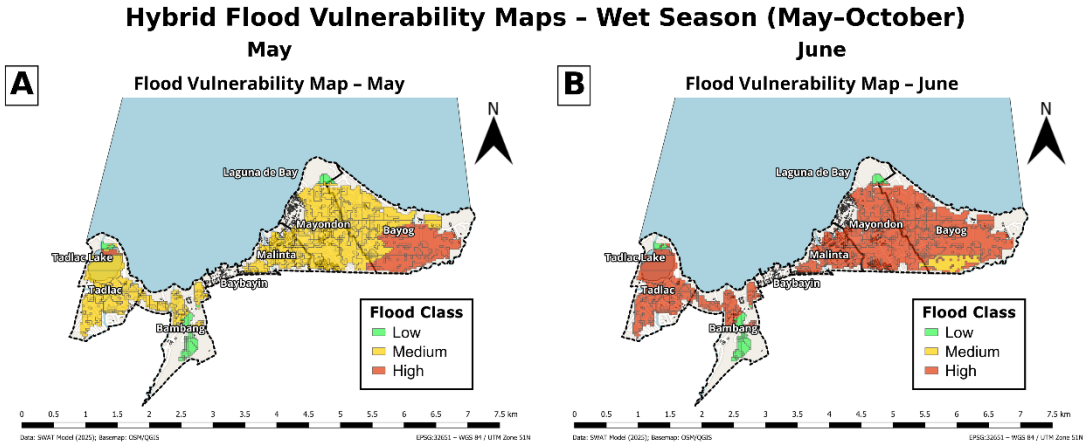


Hybrid Flood Vulnerability Maps - (January – February)

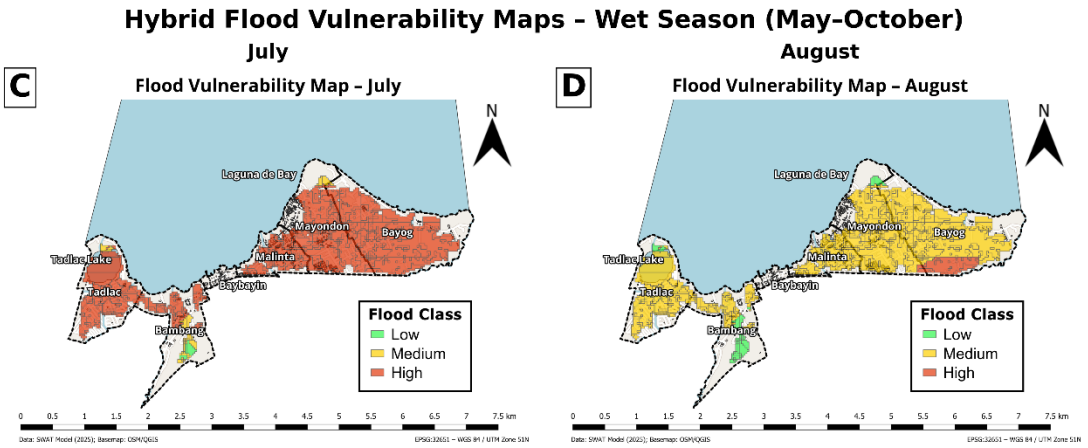


Hybrid Flood Vulnerability Maps - (March – April)

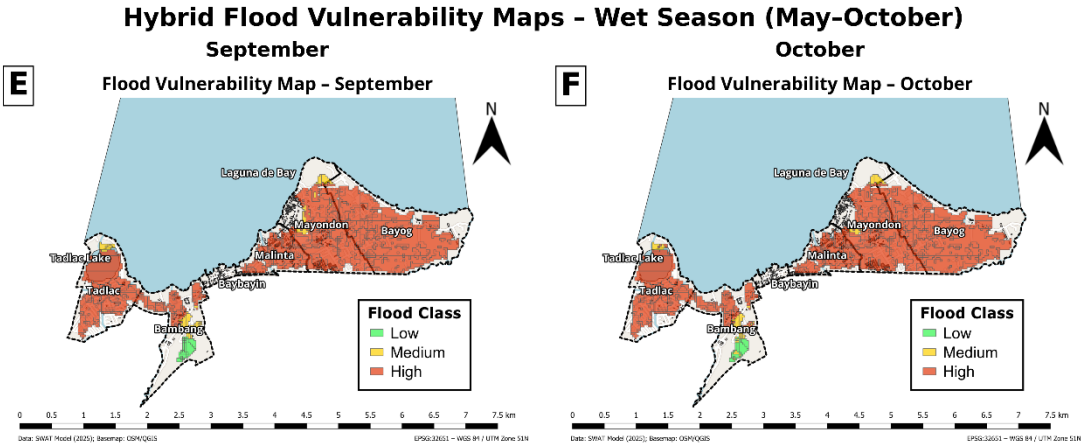
Hybrid Flood Vulnerability Maps – Wet Season



Hybrid Flood Vulnerability Maps – (May – June)



Hybrid Flood Vulnerability Maps – (July – August)



Hybrid Flood Vulnerability Maps – (September – October)

APPENDIX G

Codes and Algorithms

MLP + BO + KFOLD

```python
# === Imports ===
import numpy as np
from sklearn.model_selection import StratifiedKFold
from sklearn.metrics import f1_score, classification_report
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.neural_network import MLPClassifier
from bayes_opt import BayesianOptimization

# === Prepare Data ===
le = LabelEncoder()
y_train_enc = le.fit_transform(y_train)
y_test_enc = le.transform(y_test)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

```

```python
# === Bayesian Optimization Setup ===
pbounds = {
    "hidden1": (32, 256),               # first hidden layer
    "hidden2": (0, 128),                # second layer optional
    "learning_rate_init": (1e-4, 1e-2), # LR range
    "alpha": (1e-5, 1e-2)               # regularization
}

optimizer = BayesianOptimization(
    f=objective,
    pbounds=pbounds,
    random_state=42,
    verbose=2
)

# Run BO with 40 iterations
optimizer.maximize(init_points=5, n_iter=35)  # total = 40

print("\nBest params from BO:", optimizer.max["params"])
print("Best macro-F1 (CV):", optimizer.max["target"])
```



```python
# === Bayesian Optimization Setup ===
pbounds = {
    "hidden1": (32, 256),               # first hidden layer
    "hidden2": (0, 128),                # second layer optional
    "learning_rate_init": (1e-4, 1e-2), # LR range
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    f=objective,
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    random_state=42,
    verbose=2
)

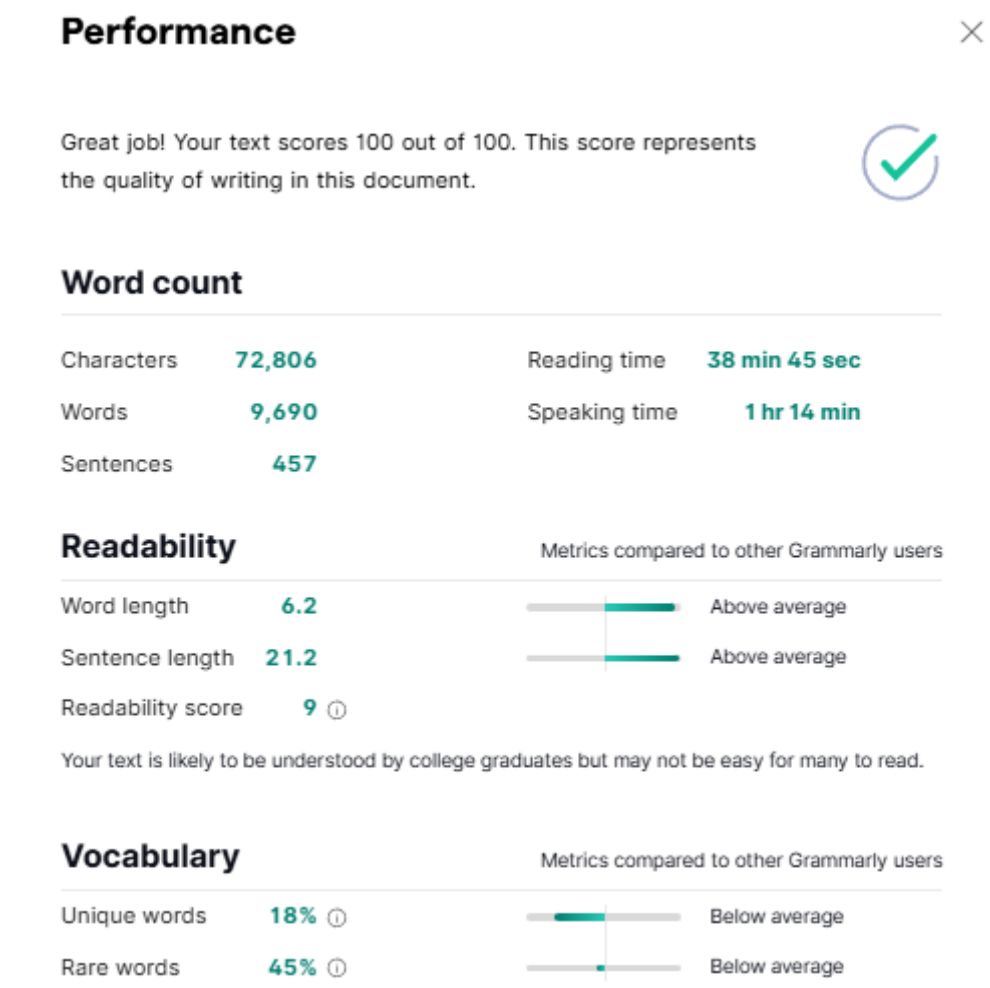
# Run BO with 40 iterations
optimizer.maximize(init_points=5, n_iter=35)  # total = 40

print("\nBest params from BO:", optimizer.max["params"])
print("Best macro-F1 (CV):", optimizer.max["target"])
```

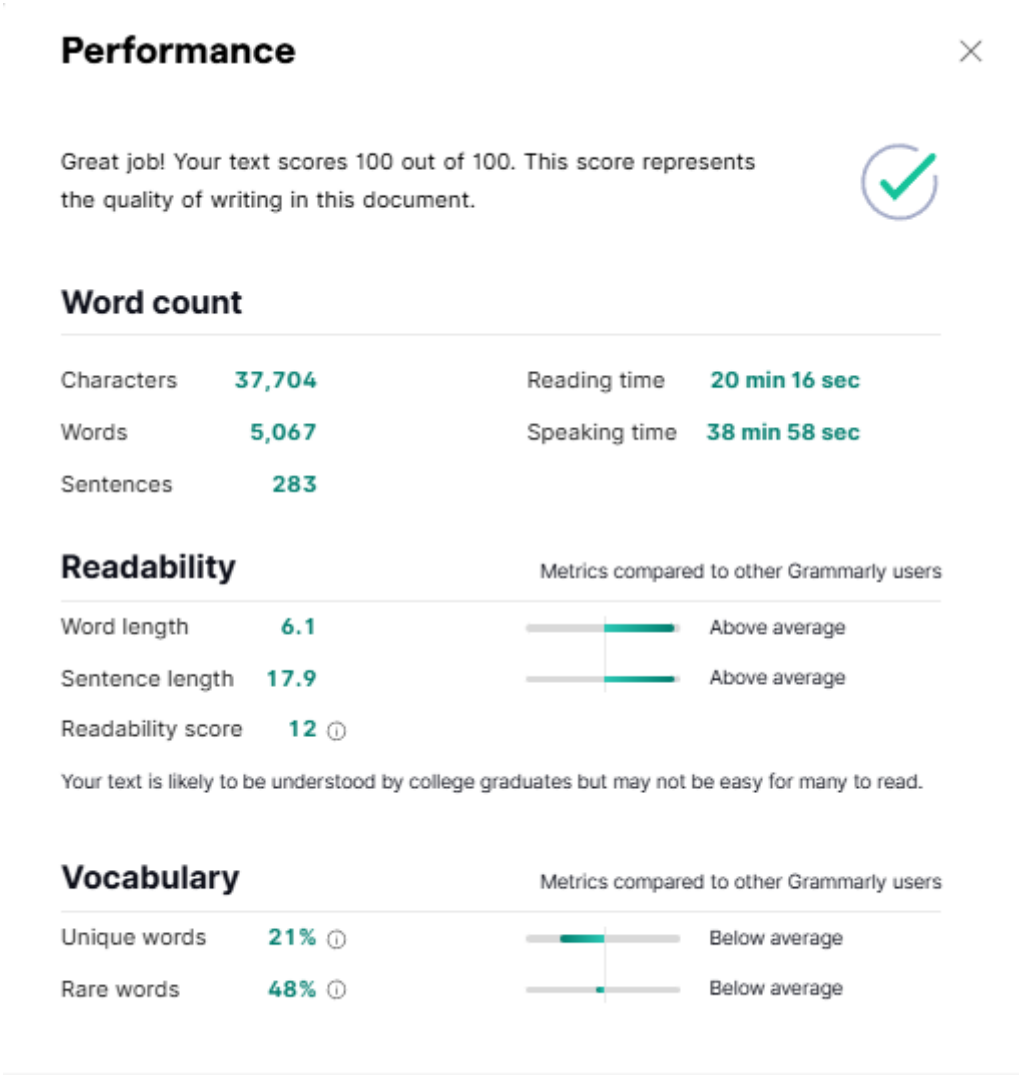
The code prepares the dataset by converting class labels into numerical form and standardizing the input features so they are suitable for training a neural network. It then defines a range of hyperparameters for a Multi-Layer Perceptron, including the number of hidden neurons, learning rate, and regularization strength. Using Bayesian Optimization, the code repeatedly trains and evaluates different MLP configurations through stratified cross-validation, ensuring that class distributions are preserved. Each model is assessed using the macro-F1 score to account for class imbalance. After several optimization iterations, the code identifies and outputs the hyperparameter combination that achieves the best cross-validated macro-F1 performance.

APPENDIX H

Grammarly Results



Chapter 1 – 3 Result



Chapter 4-5 Result

APPENDIX I

Turnitin Result

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CHAPTER 1

INTRODUCTION

Background of the Study

Flooding remains a persistent and escalating hazard in the Philippines, particularly within the Los Baños Coastal Area of Laguna, which is part of the Pasig–Laguna River Basin. This region is highly susceptible to both seasonal and extreme flood events due to its tropical monsoon climate, intensifying rainfall patterns linked to climate change, and a range of anthropogenic

8

stressors, including deforestation, rapid urban expansion, inadequate drainage infrastructure, and poor solid waste management. These interconnected factors have significantly increased the exposure and vulnerability of the coastal area and its surrounding communities.

4

Historical data and recent events underscore the socio-economic and environmental consequences of flooding in Los Baños. Between 2006 and

9

2012, the area experienced a marked increase in the frequency and severity of flood events, including the 2017 Barangay Malinta flood, which displaced over

APPENDIX J

Publishable Paper (IEEE Format)

FLOOD VULNERABILITY OF THE MUNICIPALITY OF LOS BAÑOS, LAGUNA USING HYBRID MACHINE LEARNING MODELS

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ABSTRACT

Flooding is among the most persistent natural hazards in the Philippines, severely affecting low-lying and coastal municipalities Including Los Baños, Laguna. This study developed a high-resolution flood vulnerability assessment framework by integrating the physically based Soil and Water Assessment Tool (SWAT) with optimized machine learning algorithms. A ten-year dataset (2014–2023) encompassing rainfall, temperature, elevation, slope, soil type, and land use was processed through SWAT to simulate hydrological responses across 145 Hydrologic Response Units (HRUs). These outputs served as inputs for hybrid models, including Random Forest, XGBoost, Multi-Layer Perceptron (MLP), Long Short-Term Memory (LSTM), and Support Vector Classifier (SVC), each optimized using Particle Swarm Optimization (PSO), Bayesian Optimization (BO), or Genetic Algorithm (GA). Performance was evaluated through K-Fold cross-validation using precision, recall, F1-score, and AUC-ROC metrics. Results revealed that the Bayesian-optimized MLP model (MLP–BO + K-Fold) achieved the highest accuracy (Macro-F1 = 0.9884; Accuracy = 0.98), effectively delineating Bayog, Tadlac, Mayondon, Malinta, and Bambang as high-risk barangays. The integration of SWAT-derived hydrological parameters with advanced optimization-based learning significantly enhanced the spatial precision and reliability of flood vulnerability mapping. The proposed hybrid framework provides a scalable, data-driven tool for disaster preparedness, land-use planning, and climate adaptation, supporting the broader goals of building resilient and sustainable communities in the Philippines.

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1. INTRODUCTION

Flooding is a persistent and destructive hazard in the Philippines, particularly in low-lying coastal and riverine areas. The Los Baños coastal zone in Laguna, within the Pasig–Laguna River Basin, is highly vulnerable due to its tropical monsoon climate, increasing rainfall intensity, and anthropogenic pressures such as rapid urbanization, deforestation, inadequate drainage, and ineffective waste management [2]. These factors collectively heighten flood exposure and vulnerability, resulting in significant socio-economic and environmental impacts on local communities.

Accurate flood prediction is essential for disaster risk reduction and climate adaptation. Physically based hydrological models, notably the Soil and Water Assessment Tool (SWAT), are widely used to simulate watershed-scale

hydrological processes but are often constrained by extensive data requirements and complex calibration, particularly in data-limited regions [3][5][6]. In contrast, machine learning (ML) algorithms, including Long Short-Term Memory (LSTM), Random Forest (RF), Extreme Gradient Boosting (XGBoost), Support Vector Regression (SVR), and Multi-Layer Perceptron (MLP), have demonstrated strong capability in modeling nonlinear and temporal hydrological relationships, even with limited datasets [7][11].

Recent studies show that hybrid models integrating SWAT with ML algorithms improve streamflow simulation and flood forecasting by combining physical process representation with data-driven learning [12][15]. However, most applications remain focused on basin-scale analysis, with limited attention to Hydrological Response Unit (HRU)-level modeling required for localized flood vulnerability assessment. Additionally, ensemble tree-based models such as RF and XGBoost, when integrated with Geographic Information Systems (GIS), enable high-resolution and spatially explicit flood vulnerability mapping for local disaster planning [1][18][19].

This study evaluates flood vulnerability in Los Baños' coastal areas using a hybrid SWAT–ML framework. It aims to identify vulnerability patterns, generate GIS-based maps, assess hybrid model performance, and test optimization effectiveness via paired t-tests. Integrating SWAT, ML, and GIS provides a practical HRU-scale framework to support local disaster preparedness, flood risk mitigation, and climate-resilient planning in data-limited settings [19][20].

2. RELATED LITERATURE

Flooding in coastal and low-lying tropical regions is strongly influenced by monsoon rainfall, land-use change, and rapid urbanization. In the PasigLaguna River Basin, including Los Baños, flood susceptibility is intensified by seasonal rainfall, deforestation, and unregulated development in floodplains draining into Laguna de Bay [2]. These conditions have driven recurring flood events, highlighting the need for robust flood vulnerability assessment (FVA) and predictive modeling to support disaster risk reduction.

Recent studies increasingly emphasize hybrid modeling approaches that integrate physically based hydrological models with machine learning (ML) techniques to improve flood prediction accuracy. The Soil and Water Assessment Tool (SWAT) is widely applied for simulating rainfallrunoff and watershed processes but is often limited by calibration complexity and data requirements [3][6]. ML models, including Long Short-Term Memory (LSTM), Multi-Layer Perceptron (MLP), Random Forest (RF), and Extreme Gradient Boosting (XGBoost), have demonstrated strong capability in capturing nonlinear hydrological relationships and temporal dependencies [7][11]. LSTM models, in particular, outperform traditional models in hydrological time-series prediction due to their ability to learn long-term dependencies [8][9].

Hybrid SWATML frameworks have shown improved streamflow simulation and flood forecasting accuracy by combining physical process representation with data-driven learning. Studies integrating SWAT with LSTM, RF, and XGBoost report enhanced predictive performance and reduced uncertainty compared with standalone models [12][15]. Ensemble tree-based models such as RF and XGBoost have also demonstrated high accuracy and computational efficiency in flood susceptibility mapping and hydrological risk assessment [1][16][17].

Flood vulnerability assessment has evolved toward spatially explicit frameworks integrating hydrological, topographic, and land-surface indicators. Common variables include rainfall, runoff, water yield, elevation, slope, land use and land cover (LULC), and soil characteristics, all of which influence flood generation and accumulation [18][20]. Geographic Information Systems (GIS) and remote sensing enable the spatial integration of these variables, supporting high-resolution flood vulnerability mapping and local-scale risk analysis [1][15].

Model performance depends on parameter selection, prompting optimization techniques like Genetic Algorithms and Bayesian Optimization for improved hydrological and ML calibration [14][19]. Metrics such as AUC-ROC, precision, recall, and F1-score assess classification accuracy, with hybrid and ensemble models generally outperforming single-model approaches [1]. However, most studies focus on basin- or regional-scale applications, with limited HRU-level flood vulnerability assessment and few systematic comparisons of optimized ML classifiers within a SWAT-based framework. This study fills these gaps by implementing a hybrid SWAT-ML approach at the HRU level, combined with GIS-based spatial analysis, to produce localized flood vulnerability maps for Los Baños' coastal barangays.

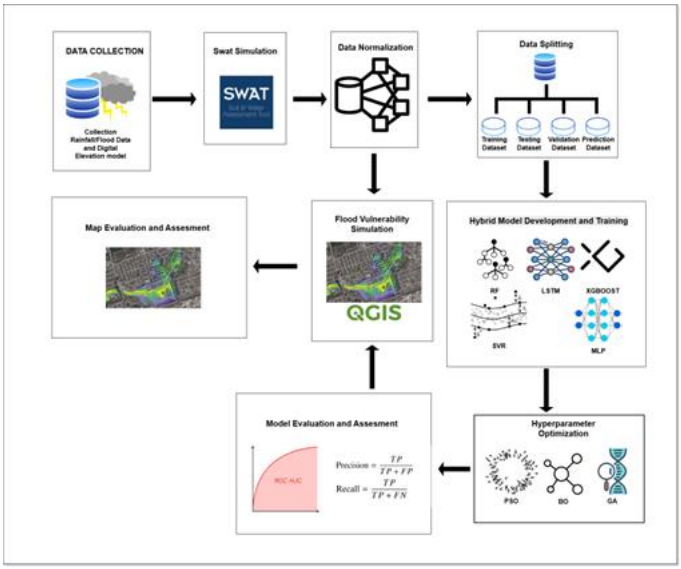


Figure 1. Concept of Developing Hybrid Models and Machine Learning for Flood Vulnerability Assessment in Los Baños Coastal Area

3. METHODOLOGY

3.1 Data Collection and Preparation

Hydrological and geospatial datasets were compiled based on established flood-vulnerability indicators [1][3]. Rainfall and temperature data were obtained from W3S-Water, integrating NASA IMERG and NOAA CPC products [4][5]. Land use/land cover (LULC) and soil maps were sourced from Geoportal Philippines, while topographic data (30-m DEM and slope) were acquired from USGS Earth Explorer.

Due to the absence of observed streamflow records, the Soil and Water Assessment Tool (SWAT) was used to simulate hydrological processes at the Hydrologic Response Unit (HRU) scale. Monthly SWAT outputs from 2015-2024 were generated, with surface runoff, water yield, and flow out selected as primary hydrological predictors. Precipitation was treated as an external forcing variable embedded within SWAT outputs.

Two datasets were constructed:

1. SWAT-only dataset, consisting of simulated hydrological variables;
2. Hybrid dataset, integrating SWAT outputs with physiographic variables (elevation, slope, LULC, soil type).

All continuous variables were normalized, while categorical variables were label-encoded.

3.2 Flood Score Formulation and Classification

Flood susceptibility was quantified using a Flood Score Index. For the SWAT-only dataset, a Weighted Linear Combination (WLC) of surface runoff, water yield, and flow out was applied [6][7].

$$FloodScore = w_1(SurfaceRunoff_{norm}) + w_2(WaterYield_{norm}) + w_3(FlowOut_{norm})$$

Figure 2. Flood Score computation using weighted normalized hydrological variables

For the hybrid dataset, a RiskResistance framework was implemented, where hydrological drivers formed the Risk Index and topographic factors (elevation, slope) constituted the Resistance Index [8][9].

$$FloodScore = RiskIndex - ResistanceIndex$$

Figure 3. Flood Score formulation using the RiskResistance index approach.

Flood Scores were classified into Low, Medium, and High risk categories using quantile-based thresholds, ensuring balanced class distribution [6][7].

3.3 Dataset Structure and Temporal Splitting

The final dataset consisted of 145 HRUs across 19 subbasins, sampled monthly from 20152023, with 2024 reserved for prediction. Temporal partitioning followed a chronological split to prevent data leakage:

Table 1. Summary of Hydrological Dataset Characteristics.

Attribute	Description
Number of HRUs	145 HRU
Data Period	2015-2023
Sampling Frequency	Monthly

Number of Subbasins	19 Subbasins
DEM Resolution	30m (USGS Earth Explorer)
Land Use Categories	ARGL (Agricultural Land), RNGE (Rangeland), URBN (Urban Areas)
Soil Types	CALC (Calcareous), HYDR (Hydrosol), LIPA (Lipa Soil Series)
Climate Variables	Precipitation, Temperature (W3S)
Output Variables	Surface Runoff, Water yield, Flow out
Data Sources	W3S, Geoportal Philippines, USGS Earth Explorer

Table 2. Temporal Split of the Dataset for Model Training, Validation, and Testing.

Split	Years	Durations
Training Dataset	2015-2019	60 months
Validation Dataset	2020-2021	24 months
Test Dataset	2022-2023	24 months
Prediction Dataset	2024	12 months

This structure preserves hydrological realism and aligns with best practices in time-series flood modeling [10][14].

3.4 Experimental Design and Model Development

The experimental framework integrates physically based hydrological simulation with machine learning classification. SWAT outputs were used as baseline predictors, while the hybrid framework combined hydrological and physiographic variables to enhance flood vulnerability representation. Five machine learning algorithms were evaluated: Long Short-Term Memory (LSTM), Random Forest (RF), Extreme Gradient Boosting (XGBoost), Support Vector Classifier (SVC), and Multi-Layer Perceptron (MLP). These models were selected for their complementary strengths in capturing temporal dynamics (LSTM), nonlinear relationships (RF, XGBoost, MLP), and high-dimensional decision boundaries (SVC).

Hyperparameter optimization was applied to improve predictive performance. RF and XGBoost were optimized using Particle Swarm Optimization (PSO) and Bayesian Optimization (BO), MLP using Genetic Algorithm (GA) and BO, LSTM using BO, and SVC using PSO. Model development and training were conducted in Python 3.10, while hydrological simulations were performed using QSWAT+. Temporal data splitting was strictly enforced, and K-fold cross-validation was applied during optimization to ensure model stability and generalization.

3.5 Flood Vulnerability Mapping

Flood vulnerability maps were generated using two workflows:

1. SWAT-only mapping, where hydrological outputs were aggregated using GIS zonal statistics;
2. Hybrid SWAT-ML mapping, where ML predictions were spatially linked to barangay-level shapefiles.

Hybrid outputs provided spatially explicit classifications that capture both temporal variability and spatial heterogeneity, improving upon physically based maps.

3.6 Validation and Evaluation

Model performance was evaluated using AUC-ROC, precision, recall, F1-score, and MSE. Spatial validation involved overlaying predicted flood zones with historical flood records from the coastal barangays. SWAT-only maps served as a physical baseline, while hybrid models demonstrated enhanced classification accuracy and spatial agreement.

3.7 Ethical Considerations

The study followed ethical standards emphasizing data privacy, transparency, and environmental sustainability. Only anonymized, publicly available datasets were used, with no human participants involved. Models were designed to be unbiased and representative of diverse flood-prone areas, and all methodologies were documented to ensure reproducibility. Sustainable flood management practices were prioritized to minimize potential environmental impacts, supporting both scientific integrity and local disaster preparedness.

4. RESULTS AND DISCUSSIONS

This section presents the performance of the SWAT-alone and Hybrid SWATMachine Learning (SWATML) frameworks for flood vulnerability assessment in six coastal barangays of Los Baños, Laguna: Bayog, Tadlac, Malinta, Bambang, Mayondon, and Baybayin. Model performance was evaluated using Accuracy, Precision, Recall, F1-score, and AUCROC, while spatial outputs were assessed through GIS-based flood vulnerability maps. Results are discussed in relation to seasonal hydrological behavior, model robustness, and practical implications for local flood risk management.

4.1 SWAT-Alone Flood Vulnerability Assessment

SWAT-derived flood vulnerability maps were generated for dry (NovemberApril) and wet (MayOctober) seasons using quantile-based classification.

During the dry season, most barangays exhibited predominantly Low vulnerability,

with localized Moderate zones persisting in Bayog and Mayondon, likely due to residual soil moisture and baseflow. In contrast, the wet season showed a pronounced expansion of Medium to High vulnerability, particularly in Bayog, Malinta, and Mayondon, driven by monsoon rainfall, low elevation, and proximity to Laguna de Bay. Seasonal comparison (Table 4) confirmed a clear transition from Low vulnerability in dry months to ModerateHigh vulnerability during peak rainfall, demonstrating SWAT's sensitivity to hydrological seasonality.

4.2 Hybrid SWATMachine Learning Model Performance

The hybrid dataset integrated SWAT outputs with physiographic attributes and preserved balanced class distributions across training, validation, and test sets. Table 5 summarizes the performance of all trained models.

Table 5. Overall comparison of all trained models.

Model	Accuracy	Macro_f1_Test	Macro_f1_CV
MLP (BO)	0.9800	0.9800	0.9886
XGBoost (PSO)	0.9700	0.9700	0.9831
XGBoost (BO)	0.9700	0.9700	0.9810
LSTM (BO)	0.9600	0.9600	0.9892
SVC (PSO)	0.9500	0.9600	0.9892
Random Forest (PSO)	0.9600	0.9600	0.9748
Random Forest (BO)	0.9600	0.9600	0.974

Among all configurations, **MLP optimized with Bayesian Optimization (MLPBO)** achieved the best overall performance, with **Accuracy = 0.98** and **Macro-F1 = 0.984**, indicating strong generalization and robustness. **XGBoost (PSO/BO)** also performed well (Macro-F1 ≈ 0.98), while **Random Forest** and **SVC** showed comparatively lower but stable performance. Although **LSTM** captured temporal dynamics, its performance was constrained by the limited length and monthly resolution of the dataset. Overall, results confirm that optimized neural and ensemble models outperform standalone hydrological classification.

4.3 Hybrid Flood Vulnerability Mapping

Flood vulnerability maps generated using the **Hybrid SWATMLP (BO)** framework revealed enhanced spatial differentiation compared to SWAT-only outputs. In the **dry season**, Low vulnerability dominated across most barangays, with Moderate zones confined to low-lying coastal areas. During the **wet season**, Medium and High vulnerability exceeded **90%** in several barangays, with **Bayog, Tadlac, and Malinta** consistently identified as the most flood-prone (Table 6). Compared to SWAT-alone maps, the hybrid model captured finer spatial gradients and improved sensitivity to terrain and land-use variability.

4.4 Comparative Analysis: SWAT vs. Hybrid Models

A comparison between the SWAT-only and Hybrid SWATMachine Learning (SWATML) approaches shows that the SWAT-only model tends to produce generalized flood vulnerability classifications, with limited differentiation among risk levels. In contrast, the hybrid approach provides improved spatial discrimination of flood-prone areas, particularly in historically affected coastal barangays. This enhancement is reflected in the consistently higher Macro-F1 scores achieved by hybrid models, indicating superior classification accuracy and robustness. The results confirm that integrating machine learning with physically based hydrological outputs significantly improves flood vulnerability assessment by capturing nonlinear interactions between hydrological and physiographic factors.

4.5 Statistical Validation and Model Significance

Paired t-test analysis comparing baseline and optimized hybrid models demonstrated **statistically significant performance improvements** across all algorithms ($p < 0.05$) (Table 8). Optimized **MLPBO** and **MLPGA** models showed the largest gains, followed by XGBoost and Random Forest. These results validate the effectiveness of **meta-heuristic optimization techniques (BO, PSO, GA)** in enhancing flood vulnerability prediction accuracy.

Table 8. Paired T-Test Results between Selected Hybrid Models.

Baseline	Optimized	t-Statistics	p-value	Significant
RF_base	RF_PSO	-38.922334	$2.4241 36e^{-11}$	Yes
RF_base	RF_BO	-35.546480	$5.4587 21e^{-11}$	Yes
XGB_base	XGB_PSO	-24.065108	$1.7659 55e^{-09}$	Yes
XGB_base	XGB_BO	-25.488705	$1.0599 74e^{-09}$	Yes
MLP_base	MLP_GA	-193.587837	$1.3321 15e^{-17}$	Yes
MLP_base	MLP_BO	-250.066826	$1.3307 93e^{-18}$	Yes

4.7 Implications for Flood Risk Management

The hybrid framework identified **Bayog, Tadlac, and Malinta** as priority barangays, with High vulnerability covering over **4550%** of land area during the wet season. These areas are characterized by low elevation, limited drainage capacity, and proximity to Laguna de Bay. The resulting maps provide

actionable insights for **LGUs and MDRRMO**, supporting barangay-level zoning, infrastructure planning, and early warning system development.

5. SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

5.1 Summary and Conclusions

This study developed a barangay-level flood vulnerability assessment framework for the coastal areas of Los Baños, Laguna by integrating SWAT hydrological modeling, GIS-based spatial analysis, and machine learning techniques. The framework combined SWAT-derived hydrological variables (surface runoff, water yield, and flow out) with physiographic attributes (slope, elevation, soil type, and land use) to generate a hybrid dataset for flood classification.

Among the evaluated algorithms—Random Forest, XGBoost, Multi-Layer Perceptron (MLP), Long Short-Term Memory (LSTM), and Support Vector Classifier (SVC)—the MLP model optimized using Bayesian Optimization (BO) and K-Fold cross-validation achieved the highest predictive performance and spatial consistency. The hybrid SWATML approach outperformed the SWAT-alone baseline by providing improved spatial resolution, stronger seasonal sensitivity, and more accurate delineation of flood-prone areas.

Results indicate that flood vulnerability in Los Baños is strongly seasonal, with high-risk conditions during the wet season (MayOctober) concentrated in low-lying coastal barangays such as Bayog, Tadlac, and Mayondon, and generally low vulnerability during the dry season (NovemberApril). The strong agreement between hybrid model outputs, hydrological behavior, and observed flood patterns confirms the robustness and applicability of the proposed framework.

Overall, this study demonstrates that hybrid hydrologicalmachine learning models offer a reliable solution for flood vulnerability assessment in data-limited and rapidly urbanizing regions, providing actionable insights for disaster risk reduction and climate resilience planning. The framework supports national climate adaptation priorities and contributes to Sustainable Development Goals SDG 11 (Sustainable Cities) and SDG 13 (Climate Action).

5.2 Recommendations

Based on the findings, the following recommendations are proposed:

1. Policy Integration and Planning
   Local government units (LGUs) should integrate the generated flood vulnerability maps into Comprehensive Land Use Plans (CLUPs), Local Disaster Risk Reduction and Management Plans (LDRRMPs), and Local Climate Change Action Plans (LCCAPs) to support zoning, infrastructure prioritization, and flood mitigation strategies.
2. Early Warning and Operational Use
   The hybrid framework can be extended for early warning applications by integrating real-time rainfall and hydrometeorological data, enabling near real-time flood risk prediction and improved response at the barangay level.
3. Model Enhancement and Data Integration
   Future studies should incorporate remote sensing products (e.g., SAR-based flood indices and NDWI) and higher-resolution topographic data (e.g., LiDAR) to improve spatial accuracy and post-event validation.
4. Scalability and Replication
   The proposed SWATML framework is scalable and can be adapted to other flood-prone watersheds in the Philippines, providing a standardized approach for localized flood vulnerability assessment and decision support.

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APPENDIX K

Statistical Expert Credentials



Joshua Mari Pantas

Details

256U-CALCES ST., BRGY. SAN NICOLAS (POB.), BAY, LAGUNA, Laguna, 4033, Philippines

09915215775

joshua.pantas13@gmail.com

DATE / PLACE OF BIRTH
04/13/2001
Bay, Laguna

NATIONALITY
Filipino

DRIVING LICENSE
D12-24-004998

Skills

Python
HTML
CSS
JS
C#
Adaptability Microsoft Office Ability to Work in a Team Decision Making Skills Communication Skills Structural Analysis Critical Thinking Skills

Profile

Detail-oriented and self-motivated I.T. Specialist with a Bachelor's degree in Computer Science, majoring in Intelligent Systems, and recognized for academic excellence. Proven ability to deliver full-stack solutions independently, including a production-grade monitoring system currently used in a live call center setup. Skilled in software development, MySQL database management, and system integration, with a strong foundation in real-world applications. Passionate about open-source technologies, scalable UI/UX, and mentoring others in tech. Now seeking a part-time instructor role to share industry-relevant knowledge and inspire the next generation of developers.

Employment History

Software/ A.I. Developer at Vasolabs Inc, Utah United States (Remotely)

JULY 2025

- Configured, maintained, and optimized databases and port settings to ensure efficient data flow across systems.
- Developed and improved database-driven applications using .NET (C#, .NET NBC) and Python, while integrating and managing DICOM/PACS systems for medical imaging.
- Ensured compatibility with healthcare imaging technologies like Orthanac, and implemented best practices for security, backups, upgrades, and issue resolution.

I.T Specialist at Nextvas,Calamba

AUGUST 2024 — AUGUST 2025 : Nextvas Inc. — Laguna, Philippines

- Troubleshoot and resolve technical issues efficiently to ensure smooth IT operations.
- Provide expert support for hardware, software, and network-related problems.
- Maintains system performance, security, and data integrity while implementing IT solutions to streamline workflows and boost productivity.
- Improved efficiency in the Workforce team by creating custom Google Apps Scripts and complex formulas to automate and enhance data handling in Google Sheets.
- Developed a full-stack web application for real-time PC monitoring, enabling proactive issue detection and faster response times. Recognized for strong problem-solving skills and reliability in resolving critical technical issues.

MSI Intern Software Developer

FEBRUARY 2024 — MAY 2025 : Maquiling Builders Depot — Laguna, Philippines

- Developed a system to automatically extract data from financial documents using image processing.
- Assisted in the creation and testing of internal software tools for business automation.

Education

College,LagunaState Polytechnic University (LB Campus), Los Baños

SEPTEMBER 2021 — FEBRUARY 2025

Graduate of Bachelor of Science in Computer Science Major in Intelligent System (Special Award in Intelligent System)

Senior High School, University of Negros Occidental Recoletos, Bacolod

SEPTEMBER 2018 — APRIL 2020

High School, Handumanan National High School, Bacolod City

JUNE 2014 — APRIL 2018

Elementary, Handumanan Elementary School 1, Bacolod City

JUNE 2011 — APRIL 2014

Hobbies:

- Passionate about game development, actively designing and programming pixel-based games in Godot. Exploring creative coding by experimenting with new game mechanics, animation, and pixel art. Constantly learning and refining skills in game design, focusing on creating immersive and polished experiences. Engaged in problem-solving and algorithm optimization for game mechanics and player interactions.

Contact Person:

Father-HectorPantas,09516068166 (Address: Bay Laguna)

Character Reference:

Asor, Jonardo Ricasa
Research Head
asor.jonardo@lspu.edu.ph
09209765806

Pantas, Joshua Mari M
E-Signiture Over Printed Name

APPENDIX L

Curriculum Vitae



Juan Carlos D. Lorenzo

Details

9020 LE NATIONAL HIGHWAY BRGY, STO. DOMINGO, BAY, LAGUNA, 4033, PHILIPPINES

09257536072

juancarloslorenzo1620@gmail.com

DATE / PLACE OF BIRTH

10/04/2002
STA.CRUZ LAGUNA

NATIONALITY

FILIPINO

DRIVERS LICENSE

D12-24004416

Skills

Python
HTML
CCS
JS
Communication Skills
Knowledgable in Computer Technician
Good Work Ethics
Time Management

Profile

To prove myself that I can handle a challenging job and be confident, positive and achieve my goals and provide for my family in the future. Seeking a challenging position in a reputed organization where I can learn new skills, expand my knowledge.

Education

LAGUNA STATE POLYTECHNIC UNIVERSITY LOS BANOS CAMPUS
Bachelor of Science in Computer Science
Major in Intelligence System
(2021-Present)

TRACE COLLEGE
SMAW (Shielded Metal Arc Welding)
(2019-2021)

LOS BAÑOS INTEGRATED SCHOOL (LBIS)
(2015-2019)

BAY CENTRAL ELEMENTARY SCHOOL
(2009-2015)

Employment History

Computer Technician at GSC Computer Shop and Parts
August 2021- February 2022 Lawa, Calamba, Laguna

- Analyzing computer hardware problems, rectifying them, servicing and replacing part when necessary.
- Acting as the middleman between user and manufacturer when equipment must be sent to the latter for repair.
- Installing software, demonstrating its usage and troubleshooting when users are faced with problems.
- Servicing and maintaining computer related equipment such as printers, projectors, web cameras and scanners.

Hobbies

- My hobbies include playing video games, watching movies and series, and reading books or online articles. I enjoy gaming because it helps me relax and improves my problem-solving skills. Watching different shows allows me to explore new stories and ideas, while reading helps me learn new things and expand my knowledge during my free time.

Contact Person

Father - Benito C. Lorenzo, 09676304394 (Address: BAY, LAGUNA)



Jomari R. Realon

Details

0686 Narra st., Brgy. Looc, Calamba, Laguna

09076707222

realon.jomari99@gmail.com

DATE / PLACE OF BIRTH

09/21/2003
CALAMBA, LAGUNA

NATIONALITY

FILIPINO

SKILLS

Python
HTML
CCS
JS
C#
Analytical thinking
Time Management
Ethical and reliable
Fast learner of new technologies
Team-oriented

Profile

I am a motivated and adaptable professional who thrives in challenging environments that encourage growth and learning. With a positive mindset, strong work ethic, and the ability to quickly acquire new skills, I contribute meaningfully to team success while maintaining resilience and focus under pressure. I am driven by a commitment to excellence, continuous improvement, and building a better future for my family, making me a reliable and resourceful asset to any organization.

Education

College, Laguna state Polytechnic university los banos campus

Bachelor of Science in Computer Science
Major in Intelligence System

(2021-Present)

Senoir Highschool, Looc Integrated School

STEM (Science, Technology, Engineering, and Mathematics)

(2020-2022)

Highschool, Looc Integrated School

(2016-2020)

Elementary, Looc Elementary School

(2016-2020)

Hobbies

In my free time, I enjoy coding and exploring new technologies, which allows me to continuously sharpen my problem-solving skills. I am also passionate about gaming and playing NFT-based games, as they foster creativity, strategy, and adaptability. Listening to music and watching anime provide me with inspiration and relaxation, while browsing tech forums keeps me updated on the latest industry trends and innovations.

Contact Person

Mother - Josephine Realon, 09122159430 (Address: CALAMBA, LAGUNA)



Rowell J. Bienes

Details

208 PUROK 2 MALINTA
LOS BAÑOS LAGUNA

09939343677

rowellbienes@gmail.com

DATE / PLACE OF BIRTH

10/01/2003

PINAMALAYAN, ORIENTAL MINDORO

NATIONALITY

FILIFINO

SKILLS

C#
HTML
PYTHON
JAVA SCRIPT
DIGITAL ARTS
PROBLEM SOLVING
LOGICAL THINKING
PROFICIENT IN BASIC
COMPUTER OPERATIONS

Profile

I am a Computer Science student with a strong interest in digital art and designing. I enjoy creating visual projects using digital art platforms and combining creativity with technical work. I have strong analytical skills and pay close attention to detail, especially when working on projects. While I am still developing my coding abilities, I am eager to learn and grow in the field, and I'm seeking opportunities to apply and expand my technical knowledge.

Education

LAGUNA STATE POLYTECHNIC UNIVERSITY LOS BANOS CAMPUS
Bachelor of Science in Computer Science
Major in Intelligence System
(2022-Present)

B.N. CALARA INTEGRATED NATIONAL HIGH SCHOOL
Accountancy, Business, and Management
(ABM)
(2020-2022)

LOS BAÑOS INTEGRATED SCHOOL (LBIS)
(2016-2020)

BAYOG ELEMENTARY SCHOOL
(2010-2016)

Hobbies

One of my main hobbies is creating visual projects using digital art platforms, where I enjoy combining creativity with technical skills. I also love playing NFT games, as they allow me to explore the connection between gaming, digital assets, and technology. In my free time, I enjoy working on creative projects that challenge my analytical skills and attention to detail. While I am still developing my coding abilities, I enjoy learning new tools and techniques that help me grow both creatively and technically.

Contact Person

Mother - Sally J. Bienes, 09944980701 (Address: 208 Purok 2 Malinta Los Baños Laguna)